MGM University

Chhatrapati Sambhajinagar

Name of Faculty – Engineering & Technology

Name of College – School of Engineering & Technology

Name of Department – Computer Science and Engineering

Name of Programme – B. Tech Integrated Computer Science and Engineering (IoT, Cyber Security Including Block Chain Technology)

CURRICULUM BOOKLET

(With effect from Academic year 2022-23)

MGM University

Chhatrapati Sambhajinagar

Published by –

Academic Section,
Registrar Office,
MGM University.

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MGM University

Vision

- To ensure sustainable human development which encourages self-reliant and self-content society.
- To promote activities related to community services, social welfare and also Indian heritage and culture.
- To inculcate the culture of non-violence and truthfulness through vipassanna meditation and Gandhian Philosophy.
- To develop the culture of simple living and high thinking

Mission

- To impart state of art education and technical expertise to students and give necessary training to teachers to create self-reliant society for future.
- To encourage students to participate in Indian and International activities in sports, literature, etc. so that future generation becomes base for free and liberal society
- To educate students in areas like Management, Finance, Human relations to inculcate philosophy of simple living and high thinking value of simple economic society.
- To inculcate culture of non-violence and truthfulness through Vipassana.

  To sustain activities of Indian culture (viz. classical dance, music and fine arts) through establishing institutes like Mahagami, Naturopathy, etc.

विद्यापीठगीत

अत्तदिपभवभवप्रदिपभव,

स्वरूपरूपभवहो

ज्ञानसब्बविज्ञानसब्बभव ,

सब्बदिपभवहो

अत्ताहिअत्तनोनाथो ,

अत्ताहिअत्तनोगति

अत्तमार्गपरअप्रमादसेहैतुझेचलना

सब्बकाकल्याणहो ,

वोकार्यकुशलकरना

सब्बकाउत्तममंगल , पथप्रदर्शकहो

अत्तदिपभवभवप्रदिपभव ,

स्वरूपरूपभवहो

ज्ञानसब्बविज्ञानसब्बभव ,

सब्बदिपभवहो

बुद्धमंशरनंगच्छामि :

धम्मंशरनंगच्छामि :

संघंशरनंगच्छामि :

School of Engineering and Technology (SOET) at a Glance

The School of Engineering and Technology (SoET), a self-aided, non-minority, conducted college of MGM University is the first institution to provide integrated B.Tech and M.Tech programs in the Marathwada region of Maharashtra. A sister institution of the illustrious Jawaharlal Nehru Engineering College (JNEC), Aurangabad, it was established as MGM Polytechnic College in 2010. However, recognizing the rising need of quality engineering education after schooling in the region and beyond, it was transformed into the School of Engineering and Technology (SOET) in 2022.

Inheriting a 39 years rich legacy of imparting world-class professional education from MGM Trust, SoET provides ultramodern research and innovation facilities to its students. Its integrated programs eliminate the need to appear for 12th standard and various engineering entrance tests, thus saving valuable time and energy of the students. It allows the students to utilize the same time to gain more practical knowledge of the field and a strong edge over the competition.

The college offers a wide variety of Undergraduate (Including Six Years Integrated Programs after 10th), Postgraduate (Including Five Years Integrated Programs after 12th) and PG Diploma programs to fulfil the dreams of the students. All these are advanced engineering programs founded on industry-oriented curriculum.

The Institution employs many modern teaching-learning methods like Project-based Learning, Research-based Learning and Collaborative Learning to develop comprehensive understanding and real-life skills in the students. The School has also signed several memoranda of understanding with renowned institutions, associations and industries to provide internship and project work opportunities to the students. It also joined hands with COURSERA, NPTEL, SWAYAM, ATAL and other MOOC platforms to facilitate blended learning facilities for the students.

Unique in its structure, methods and goals, the institution is strongly rooted in the philosophy of training and research that enhances the relationship between knowledge and its application and seeks to promote the creation of an ideal society through holistic education.

Vision

Enrich the students in technical education to work globally with human values.

Mission

- To inculcate technical skills by continuous assessment practices.
- To strengthen interaction between industry and institute.
- To involve the students in social activities with environmental awareness.

Programs offered at SOET

B.Tech (Integrated) UG Program 6 Years (after 10th)	**B.Tech UG Program 4 Years (after 12th)**
Civil Engineering	Computer Science and Engineering (IoT, Cyber Security Including Block Chain Technology)
Computer Science and Engineering (Data Science)	Computer Science and Engineering (Cyber Security)
Computer Science and Engineering (IoT, Cyber Security Including Block Chain Technology)	Electrical Engineering
Robotics and AI	Electronics Engineering (VLSI Design and Technology)
--	Robotics and Automation

Department of Computer Science and Engineering

The Bachelor of Technology (B. Tech.) in Computer Science and Engineering degree program started in the year 1983 and is approved by the All India Council for Technical Education (AICTE), New Delhi and offers Choice Based Credit System education. In addition to core courses, students can opt for discipline specific elective subjects, open elective subjects from different institutes of the University. In addition, this program is uniquely designed to increase the employability and to prepare students to work in a Multi-disciplinary work environment. The program offers Major degrees in ComputerScience and Engineering and is open to students opting for minor specializations as per their interests. Pedagogies concentrating on student's active participation are extensively used in the teaching-learning process.

Vision

To provide update technology in computer engneering to fulfill needs of industry and society.

Mission

- To impart technical knowledge and skills by involving students in educational activities.
- To enable students to be entrepreneurs enhancing interaction between institute and industry.
- To nurture moral and professional values to build up society and nation.

Name of Program – B.Tech (Integrated) Computer Science and Engineering (IoT, Cyber Security Including Block Chain Technology)

Duration – Six Years

Eligibility –

Admissions to First Year:-

(a) The Candidate should be an Indian National.

(b) Candidate should have passed 10th Std. examination or its equivalent examination with 50% and 45% (for reserve categories and Persons with Disability candidates belonging to Maharashtra State only) aggregate marks and the candidate should obtain non-zero positive score in MGMU-CET.

Lateral Entry Admissions (Direct Second Year):-

(a) The Candidate should be an Indian National.

(b) Passed 10+2 examination with Physics / Mathematics / Chemistry / Computer Science / Electronics / Information Technology / Biology / Informatics Practices Biotechnology/Technical Vocational Subject/Agriculture/ Engineering Graphics / Business Studies / Entrepreneurship (Any of the three)

OR

10th + (2 years ITI) with appropriate Trade in that order shall be eligible for admission to Second Year B.Tech (Integrated) course(s) of appropriate programme.

(c) Any other eligibility condition/qualification declared, from time to time, by the appropriate authority.

Name of Faculty: Engineering and Technology
Name of the College/Institute/Department/School: School of Engineering and Technology (SOET)
Name of the Programme: B. Tech Integrated Computer Science and Engineering (IoT, Cyber Security Including Block Chain Technology)
Programme Type (UG/PG): UG
Duration: Six Years

Program Name: B. Tech Integrated Computer Science and Engineering (IoT, Cyber Security Including Block Chain Technology)

(Semester: First)

Course code	Course Title	Type	Teaching Scheme			Evaluation Scheme (Marks)						Minimum Passing (Marks)						Credit	Remarks
						Internal			External		Total	Internal			External		Total		
			L	T	P	CA	MSE	TW	ESE	PR		CA	MSE	TW	ESE	PR			
22IUCC0101B	Mathematics-I	Theory	3	1	-	20	20	-	60	-	100	-	-	-	24	-	40	4	
22IUCC0102B	Physics-I	Theory	3	1	-	20	20	-	60	-	100	-	-	-	24	-	40	4	
22IUCC0103B	Chemistry-I	Theory	3	1	-	20	20	-	60	-	100	-	-	-	24	-	40	4	
22IUCC0104H	English-I	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3	
22IUCC0105T	Fundamentals of Computer System	Theory	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	Audit	
22IUCC0106T	Engineering Graphics	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3	
22IUCC0102L	Physics-ILab	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1	
22IUCC0103L	Chemistry-ILab	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1	
22IUCC0104L	LanguageLab-I	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1	
22IUCC0105L	Fundamentalsof Computer System Lab	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1	
22IUCC0106L	Engineering GraphicsLab	Practical	-	-	4	-	-	100	-	-	100	-	-	40	-	-	40	2	
Total			**17**	**3**	**12**	**100**	**100**	**300**	**300**	**0**	**800**	**0**	**0**	**120**	**120**	**0**	**320**	**24**	

(Semester: Second)

Course code	Course Title	Type	Teaching Scheme			Evaluation Scheme (Marks)						Minimum Passing (Marks)						Credit	Remarks
						Internal			External		Total	Internal			External		Total		
			L	T	P	CA	MSE	TW	ESE	PR		CA	MSE	TW	ESE	PR			
22IUCC0201B	Mathematics-II	Theory	3	1	-	20	20	-	60	-	100			-	24	-	40	4	
22IUCC0202B	Physics-II	Theory	3	1	-	20	20	-	60	-	100			-	24	-	40	4	
22IUCC0203B	Chemistry-II	Theory	3	1	-	20	20	-	60	-	100			-	24	-	40	4	
22IUCC0204H	Universal Human Value	Theory	2	-	-	-	-	-	50	-	50			-	20	-	20	2	
22IUCC0205T	Computer Workshop	Theory	2	-	-	-	-	-	-	-	-			-	-	-	-	Audit	
22IUCC0202L	Physics-II Lab	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1	
22IUCC0203L	Chemistry-II Lab	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1	
22IUCC0204L	Universal Human Value-Lab	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1	
22IUCC0205L	Computer Workshop Lab	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1	
22IUCC0206L	Workshop Practices Lab	Practical	-	-	4	-	-	100	-	-	100	-	-	40	-	-	40	2	
Total			**13**	**3**	**12**	**60**	**60**	**300**	**230**	**0**	**650**	**0**	**0**	**120**	**92**	**0**	**260**	**20**	

Program Name: B. Tech Integrated Computer Science and Engineering (IoT, Cyber Security Including Block Chain Technology)

(Semester: Third)

Course code	Course Title	Type	Teaching Scheme			Evaluation Scheme (Marks)						Minimum Passing (Marks)						Credit	Remarks
						Internal			External		Total	Internal			External		Total		
			L	T	P	CA	MSE	TW	ESE	PR		CA	MSE	TW	ESE	PR			
22IUCI0301T	Engineering Mathematics	Theory	3	1	-	20	20	-	60	-	100	-	-	-	24	-	40	4	
22IUCI0302T	Introduction to Biology	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3	
22IUCI0303T	Computer Programming	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3	
22IUCI0304T	Web Page Designing with HTML	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3	
22IUCI0305T	Basic Electronics	Theory	2	-	-	20	20	-	60	-	100	-	-	-	24	-	40	2	
22IUCI0306T	Effective Communication and soft skills	Theory	2	-	-	20	-	-	-	-	20	8	-	-	-	-	8	Audit	
ENV-C	Environmental Studies	Theory	2	-	-	20	20	-	60	-	100	-	-	-	24	-	40	Audit	
22IUCI0303L	Computer Programming Lab	Practical	-	-	4	-	-	60	-	40	100	-	-	24	-	16	40	2	
22IUCI0304L	Web Page Designing With HTML Lab	Practical	-	-	4	-	-	60	-	40	100	-	-	24	-	16	40	2	
22IUCI0305L	Basic Electronics Lab	Practical	-	-	2	-	-	30	-	20	50	-	-	12	-	8	20	1	
22IUCI0306L	Effective Communication and soft skills Lab	Practical	-	-	2	-	-	30	-	-	30	-	-	12	-	-	12	Audit	
Total			18	1	12	140	120	180	360	100	900	8	0	72	144	40	360	20	

(Semester: Fourth)

Coursecode	CourseTitle	Type	Teaching Scheme			Evaluation Scheme (Marks)						Minimum Passing (Marks)						Credit	Remarks
						Internal			External		Total	Internal			External		Total		
			L	T	P	CA	MSE	TW	ESE	PR		CA	MSE	TW	ESE	PR			
22IUCI0401T	Discrete Mathematics	Theory	3	1	-	20	20	-	60	-	100	-	-	-	24	-	40	4	
22IUCI0402T	Object Oriented Programming in C++	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3	
22IUCI0403T	Computer Graphics	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3	
22IUCI0404T	Digital Electronics	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3	
22IUCI0402L	Object Oriented Programming in C++ Lab	Practical	-	-	4	-	-	60	-	40	100	-	-	24	-	16	40	2	
22IUCI0403L	Computer Graphics Lab	Practical	-	-	2	-	-	30	-	20	50	-	-	12	-	8	20	1	
22IUCI0404L	Digital Electronics Lab	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1	
Total			**12**	**1**	**8**	**80**	**80**	**140**	**240**	**60**	**600**	**0**	**0**	**56**	**96**	**24**	**240**	**17**	

Program Name: B. Tech Integrated Computer Science and Engineering (IoT, Cyber Security Including BlockChain Technology)

(Semester: Fifth)

Coursecode	Course Title	Type	Teaching Scheme			Evaluation Scheme (Marks)						Minimum Passing (Marks)						Credit	Remarks
						Internal			External		Total	Internal			External		Total		
			L	T	P	CA	MSE	TW	ESE	PR		CA	MSE	TW	ESE	PR			
22IUCI0501T	Data Structure and Algorithm	heory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3	
22IUCI0502T	Java Programming	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3	
22IUCI0503T	oftware Engineering	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3	
22IUCI0504T	Programming in Python	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3	
22IUCI0505T	Microprocessor and Microcontroller	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3	
GS-01	Gandhian Studies	Theory	2	0	0	20	-	-	30	-	50	-	-		12		20	Audit	
22IUCI0501L	Data Structure and Algorithm	actical	-	-	2	-	-	30	-	20	50	-	-	12	-	**8**	20	1	
22IUCI0502L	Java Programming Lab	Practical	-	-	4	-	-	60	-	40	100	-	-	24	-	16	40	2	
22IUCI0503L	oftware Engineering Lab	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1	
22IUCI0504L	Programming in PythonLab	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1	
22IUCI0505L	Microprocessor and Microcontroller Lab	Practical	-	-	2	-	-	30	-	20	50	-	-	12	-	**8**	20	1	
Total			17	0	12	120	100	220	330	80	850	0	0	88	132	32	340	21	

(Semester: Sixth)

Course code	Course Title	Type	Teaching Scheme			Evaluation Scheme (Marks)						Minimum Passing (Marks)						Credit	Remarks
						Internal			External		Total	Internal			External		Total		
			L	T	P	CA	MSE	TW	ESE	PR		CA	MSE	TW	ESE	PR			
22IUCI0601T	Data Communication and Networking	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3	
22IUCI0602T	Advanced Java Programming	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3	
22IUCI0603T	Software Testing	heory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3	
22IUCI0604T	Database Management System	heory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3	
22IUCI0605T	Operating System	heory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3	
CSS	Communication Skill	Theory	2	-	-	10	10		30	-	50				12		20	2	
CSS-L	Communication Skill	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1	
22IUCI0601L	Data Communication and Networking Lab	ractical	-	-	2	-	-	30	-	20	50	-	-	12	-	8	20	1	
22IUCI0602L	Advanced Java Programming Lab	ractical	-	-	4	-	-	60	-	40	100	-	-	24	-	16	40	2	
22IUCI0603L	Software Testing Lab	ractical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1	
22IUCI0604L	Database Management system Lab	ractical	-	-	2	-	-	30	-	20	50	-	-	12	-	8	20	1	
22IUCI0605L	Operating System Lab	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1	
Total			17	0	14	110	110	270	300	80	900	0	0	108	132	32	360	24	

Program Name: B.Tech Integrated Computer Science and Engineering (IoT,Cyber Security Including BlockChain Technology)												
Semester VII												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No of Credits**	**Teaching (Contact hrs/ week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
PCC	CCI24PCL401	Artificial Intelligence	Lecture	3	3	-	60	40	100	-	16	40
PCC	CCI24PCL402	Human Computer Interaction	Lecture	2	2	-	60	40	100	-	16	40
PCC	CCI24PCL403	Network and Cyber Security	Lecture	2	2	-	60	40	100	-	16	40
PCC	CCI24PCP401	Artificial Intelligence Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	CCI24PCP402	Human Computer Interaction Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	CCI24PCP403	Network and Cyber Security Lab	Practical	1	-	2	30	20	50	-	8	20
MDM		Refer MGMU MDM Basket	Lecture	2	2	-	60	40	100	-	16	40
OE		Refer MGMU Open Elective Basket	Lecture	2	2	-	30	20	50		8	20
OE		Refer MGMU Open Elective Basket	Lecture	2	2	-	30	20	50		8	20
VEC	MGM56VEL102	Constitution of India	Lecture	2	2	-	30	20	50	-	8	20
EEMC	CCI24HSL404	Business Management & Financial Accounting	Lecture	2	2	-	60	40	100	-	16	40
FP	CCI24FPJ405	Field Project	Practical	2		4	50	-	50	20	-	20
		TOTAL		**22**	**17**	**10**	**530**	**320**	**850**	**-**	-	-

Semester - VIII

Course Category	Course Code	Course Title	Nature of Course	No. of Credits	Teaching (Contact Hrs / Week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
PCC	CCI24PCL451	Machine Learning	Lecture	2	2	-	60	40	100	-	16	40
PCC	CCI24PCL452	Cloud Computing	Lecture	2	2	-	60	40	100	-	16	40
PCC	CCI24PCL453	Ethical Hacking and Information Security	Lecture	3	3	-	60	40	100	-	16	40
PCC	CCI24PCP451	Machine Learning Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	CCI24PCP452	Cloud Computing Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	CCI24PCP453	Ethical Hacking and Information Security Lab	Practical	1	-	2	30	20	50	-	8	20
VSEC	CCI24VSP455	UI-UX Lab	Practical	2	-	4	60	40	100	-	16	40
EEMC	CCI24HSL456	Entrepreneurship Development	Lecture	2	2	-	60	40	100	-	16	40
AEC		Refer MGMU AEC Basket	Lecture	2	2	-	30	20	50	-	08	20
MDM		Refer MGMU MDM Basket	Lecture	2	2	-	60	40	100	-	16	40
OE	OE-III	Refer MGMU Open Elective Basket	Lecture	2	2	-	30	20	50		8	20
VEC	MGM24VEL201	Energy Conservation & Management System	Lecture	2	2	-	30	20	50	-	8	20
		TOTAL		**22**	**17**	**10**	**540**	**360**	**900**	**-**	**-**	**-**

Semester - IX

Course Category	Course Code	Course Title	Nature of Course	No. of Credits	Teaching (Contact Hrs / Week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
PCC	CCI24PCL501	Cyber and Digital Forensics	Lecture	2	2	-	60	40	100	-	16	40
PCC	CCI24PCL502	Cyber Law , Ethics and Security Policies	Lecture	2	2	-	60	40	100	-	16	40
PCC	CCI24PCL503	Block Chain Architecture	Lecture	2	2	-	60	40	100	-	16	40
PCC	CCI24PCL504	Big Data	Lecture	2	2	-	60	40	100	-	16	40
PCC	CCI24PCP501	Cyber and Digital Forensics Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	CCI24PCP503	Block Chain Architecture Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	CCI24PCP504	Big Data Lab	Practical	1	-	2	30	20	50	-	8	20
PEC	CCI24PEL505	Basic of CC	Lecture	2	2	-	60	40	100	-	16	40
	CCI24PEL506	DevOps-I										
PEC	CCI24PEP505	Basic of CC Lab	Practical	1	-	2	30	20	50	-	8	20
	CCI24PEP506	DevOps-I Lab										
MDM		Refer MGMU MDM Basket	Lecture	4	4	-	60	40	100	-	16	40
OE	OE-IV	Refer MGMU Open Elective Basket	Lecture	2	2	-	30	20	50		8	20
		TOTAL		**20**	**16**	**8**	**510**	**340**	**850**	**-**	**-**	**-**

Semester – X												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No. of Credits**	**Teaching (Contact Hrs / Week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
PCC	CCI24PCL551	Natural Language Processing	Lecture	3	3	-	60	40	100	-	16	40
PCC	CCI24PCL552	Block Chain Security	Lecture	2	2	-	60	40	100	-	16	40
PCC	CCI24PCP551	Natural Language Processing Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	CCI24PCP552	Block Chain Security Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	CCI24PCL553	Internet of Things	Lecture	2	2	-	60	40	100	-	16	40
PCC	CCI24PCP553	Internet of Things Lab	Practical	1	-	2	30	20	50	-	8	20
PEC	CCI24PEL554	CC using AWS	Lecture	2	2	-	60	40	100	-	16	40
	CCI24PEL555	Devops-II										
PEC	CCI24PEL556	Image Processing	Lecture	3	3	-	60	40	100	-	16	40
	CCI24PEL557	Quantum Computing										
PEC	CCI24PEP554	CC using AWS Lab	Practical	1	-	2	30	20	50	-	8	20
	CCI24PEP555	Devops-II Lab										
PEC	CCI24PEP556	Image Processing Lab	Practical	1	-	2	30	20	50	-	8	20
	CCI24PEP557	Quantum Computing Lab										
VSEC	CCI24VSP558	Mobile Application Development Lab	Practical	2	-	4	60	40	100	-	16	40
MDM		Refer MGMU MDM Basket	Lecture	2	2	-	60	40	100	-	16	40
		TOTAL		**21**	**14**	**14**	**570**	**380**	**950**	**-**	**-**	**-**

Semester - XI												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No. of Credits**	**Teaching (Contact Hrs / Week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
PCC	CCI24PCL601	Privacy & Security in IOT	Lecture	3	3	-	60	40	100	-	16	40
PCC	CCI 25PCL602	Artificial Neural Network	Lecture	2	2	-	60	40	100	-	16	40
PCC	CCI 25PCP602	Artificial Neural Network Lab	Practical	1	-	2	30	20	50	-	8	20
MDM		Refer MGMU MDM Basket	Lecture	2	2	-	60	40	100	-	16	40
PEC	CCI 25PEL603	CC using Azure IBM / DevOps- on AWS	Lecture	2	2	-	60	40	100	-	16	40
	CCI 25PEL604											
PEC	CCI 25PEP603	CC using Azure IBM Lab / DevOps- on AWS Lab	Practical	1	-	2	30	20	50	-	8	20
	CCI 25PEP604		Practical									
PEC	CCI 25PEL605	AR VR Technology	Lecture	2	2	-	60	40	100	-	16	40
	CCI 25PEL606	Quantum Algorithm										
PEC	CCI 25PEP605	AR VR Technology Lab	Practical	1	-	2	30	20	50	-	8	20
	CCI 25PEL606	Quantum Algorithm Lab										
RM	CCI 25RML606	Research Methodology	Lecture	4	4	-	60	40	100	-	16	40
RP	CCI 25RPJ607	Project	Practical	4	-	8	60	40	100	-	16	40
		TOTAL		**22**	**15**	**14**	**510**	**340**	**850**	**-**	**-**	**-**

Semester XII												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No of Credits**	**Teaching (Contact hrs/ week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
PCC	CCI24PCL651	MOOCS-1	Lecture	3	3	-	60	40	100	-	16	40
PEC	CCI24PEL652	MOOCS-2	Lecture	3	3	-	60	40	100	-	16	40
OJT	CCI24JTI653	Internship	Practical	14	-	28	100	50	150	-	20	60
		TOTAL		20	6	28	220	130	350	-	-	-

SEM-I

Course code: **22IUCC0101B**		**Course Name:** Mathematics-I
Credits: 4	**Teaching scheme:** L-3 T-1	**Evaluation scheme:** CA–40, ESE–60
Duration of End Sem. Exam: 3 hr.		
Pre-requisites:		
Course Objectives:		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Understand the basic concepts, principles, rules of algebra to solve the engineering problem in real world.		
CO2: Apply the basic concept of trigonometry to solve the engineering problem.		
CO3: Solve the engineering problem under the given geometrical condition.		
CO4: Utilize the basic concept of statistics to solve engineering problem.		
CO5: Understand the basic concept of system of complex numbers and illustrate simple problems.		
CO6: Understand the basic concept of system of complex numbers and illustrate simple problems.		

Contents:

Unit	Content	Teaching hours
1	**Algebra** 1.1 - Logarithm: Concept and Definition of Logarithm, Rules of Logarithms (Simple examples only) 1.2 -Permutation and Combination: Introduction, Fundamental Principle of counting, Permutations, Combinations. (Simple examples only) 1.3 -Binomial Theorem: Introduction, Binomial theorem for positive integral indices, General and middle terms. (Simple examples only) 1.4-Partial Fraction: Introduction, Definition and types of fraction, Resolve into partial based on nature of factors and related problems	8
2	**Determinant and Matrices** 2.1-Determinant :Definition and simple examples of determinant of order 2 & 3, Properties of Determinant, Minor and Cofactors, Crammer's Rule. 2.2-Matrices: Concept and definition of matrices Types of matrices, Algebra of matrices, Transpose, orthogonal, Adjoint of matrices, Inverse of matrix, Equivalent matrices, Matrix inversion method	8
3	**Trigonometry** Trigonometric ratios of compound, allied, multiple and sub multiple angles, Factorization and De- Factorization formulae, Inverse trigonometric ratios and related problems, Principle values and relation between trigonometric and inverse trigonometric ratios. Properties of triangle: sin, cosine, projection and tangent rule.(Without proof)	10
4	**Co-ordinate Geometry** Distance Formula, Section Formula, Different centers of triangle (without proof), Shifting of origin, Slope of line, Forms of Straight lines, Point of intersection of two lines, Angle between two lines (without proof) and Condition of parallel and perpendicular lines, Perpendicular Distance of a point from the line, Perpendicular Distance between two parallel lines. Conic Section: Circle, Parabola, Ellipse, Hyperbola.	10
5	**Data Analysis** 3.1 Measure of central tendency: Mean (AM, GM, HM,) Mode, Median. 3.2 Measure of dispersion: Range, MD, SD, and Test of consistency	6
6	**Complex Numbers** Definition of complex number, Algebra of Complex number, Argand diagram, Forms of Complex number, Properties of modulus and arguments, De Moivre's Theorem, Cube root of unity.	6

Reference Material:
1. Hall and Knight, Higher Algebra Arihant Publications India limited, **2.** Loney, SL (1912), Plane Trigonometry, Cambridge: University Press, JFM 25.0892.12 **3.** Loney, SL (1895), The elements of coordinate geometry, London: Macmillan and Co. **4.** S.C. Gupta & V.K. Kapoor, Fundamentals of Mathematical Statistics, 11th edition, Sultan Chand & Sons, New Delhi. **5.** David Freedman, Robert Pisani and Roger Purves, Statistics, 4th Edition. **6.** S.C. Gupta, Fundamental of Statistics, Himalaya. **7.** H.C. Saxena, Mathematical Statistics, Sultan Chand & Sons, New Delhi.
Online Resources:
1. www.scilab.org/-SCI Lab 2. www.mathworks.com/products/matlab/ -MATLAB 3. www.allmathcad.com/- MathCAD 4. www.wolfram.com/mathematica/ - Mathematica

Course code: 22IUCC0102B	**Course name: Physics-I**
Course category: PCC	
Credits: 4 **Teaching scheme:** L-3 T-1	**Evaluation scheme:** CA–40, ESE–60
Pre-requisites:	
Course Objectives:	
1.	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: dentify different systems of units and convert units from one system to another as well as conversant with Practical units.	
CO2:Understand equations of motion and their applications.	
CO3:Understand equations of motion and their applications.	
CO4:Acquire basic knowledge on semiconductor & application of P-N junction diode	
CO5:Use the properties of photoelectric effect, optical fiber &nano particle for various engineering application	
CO6: Understand concept of LIGHT	

Contents –

Unit	Content	Teaching hours
1	**Units & Measurements** 1.1Introduction of Unit, fundamental and derived quantities and their units, different system of Units (CGS, MKS, FPS and SI), 1.2Definition of accuracy and error, Estimation of Errors, absolute error, relative error and percentage error. Numerical problems. 1.3Explanation of Dimension of physical quantity Dimensional Equations of physical quantities and their uses with examples.	**8**
2	**Mechanics** .1 Introduction to scalar and vector Quantities, representation of vector, addition, subtraction and multiplication of vectors, parallelogram law of vector 2.2 Newton's Laws of motion: First law, Second law, &Third law, Statements & examples, Principle of conservation of linear momentum, statement and simple examples. 2.3Work, power and energy, its concept, units and dimension, Potential and Kinetic energy, its mathematical relations, Principle of conservation of energy, Recoil of gun.	**8**
3	**Circular Motion** 3.1 Circular motion, and angular velocity, relation between angular velocity and linear velocity, Rotational motion, 3.2 Simple Harmonic Motion, derivation of its equationsdefinition of amplitude, time period, frequency, phase etc., mathematical relations and units, Simple Pendulum	**6**

4	**Heat And Thermodynamics** 4.1 Concept of heat and temperature, Unit of heat, Joule and calorie, thermometer, its interval and fixed point, different scales of temperatures and their conversion formulae. 4.2 Transmission of heat, three modes of heat transfer, conduction, convection and Radiation, good and bad conductor of heat, coefficient of thermal conductivity, its S.I. unit and dimension. 1st law and 2nd law of thermodynamics, Joule's law & its derivation.	**8**
5	**Modern Physics** 5.1Photo electricity, Photon (quantum), Plank's hypothesis, energy of photon, properties of photons. 5.2 Photo electric effect: Circuit diagram, process of photoelectric emission, definitions:-threshold frequency, threshold wavelength, Stopping potential, characteristics of photoelectric effect, Application of photoelectric cell & LDR.	**6**

Text Books:
Reference Books: 1. Engineering Physics- R K Gour& S L Gupta, DhanpatRai Publication- New Delh 2.Engineering Physics- M N Avadhanulu& P G Kshirsagar, S Chand & Company Ltd.- New Delhi 3.Fundamental of Physics- Jearl Walker, Wiley India Edition 4.Engineering Physics- V Rajendran, Tata McGraw-Hill Publishing Company Ltd- New Delhi 5.Applied Physics- Manpreet Singh, Dr. Major Singh, Mrs.Hitashi Gupta, S K Kataria& Sons- New Delhi
Online Resources 1. http://nptel.ac.in/course.php?disciplineId=115 2.http://nptel.ac.in/course.php?disciplineId=104 3.http://hperphysics.phy-astr.gsu.edu/hbase/hph.html 4.www.physicsclassroom.com 5.www.physics.org 6.www.fearofphysics.com

Course code: 22IUCC0103B	**Course name:**Chemistry-I
Credits: 4 **Teaching scheme:** L-3 T-1	**Evaluation scheme:** CA–40, ESE–60
Duration of End Sem Exam: 3hr	
Pre-requisites:	
Course Objectives:	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Demonstrate an understanding of basic principles of organic chemistry and how they relate to everyday Experiences.	
CO2: Illustrate the Basic Concept of Structure of Atom	
CO3: Describe nature of bonding in molecular compounds.	
CO4: To know the common analysis techniques used in metallurgical industries.	
CO5: Students know how to approach the problem of choosing an alloy for a particular application.	
CO6: Use Relevant Electrolyte in Electrolysis for different Applications	

Contents:

Unit	Content	Teaching hours
1	**Basic principles of organic chemistry** 1.1. Introduction, 1.2 Structural representation of organic molecules-Condensed formula, structural formula, Bond line or Zigzag formula with examples. 1.3. Classification of organic compound. Acyclic/Open chain compound. Cyclic compound , Homocyclic,Heterocycliccompound,Alicyclic,Aromatic(Benzenoid,Nonbenzenoid),Heterocyclic (Aromatic . Nonaromatic),(Definition with example.) 1.4. Functional group in organic compounds. Halides,Cyanides,Isocyanides, Nitro compound,Alcohol,Phenol,Amines.Ether,Aldehydes,Ketones,Carboxylic acid, Ester, Amide, Sulphonic acid, Acyclic chloride,Acidunhydride,Alkyne,Alkene,Alkane.(Structure and Functional group with examples).	8
2	**Atomic Structure and Chemical bonding** 2.1 Introduction 2.2• Definition of atom, Bohr's atomic model, definition of atomic number, atomic mass number and their differences, 2.3 Isotopes and Isobars: Definitions, examples and distinction 2.4 Orbits and orbitals: s, p, d, f orbitals, shapes and description of s-orbital and p- orbital. Distribution of electrons in orbitals: 2.5 Definition of electronic configuration, Aufbau's principle, Hund's rule, orbital Electronic configurations (s, p, d, f) of elements having atomic number 1 to 30	6
3	**Chemical bonding** 3.1 Introduction 3.2 Valance Definitions of valence. Definition of electrovalence, positive and negative electrovalence, formation of Electrovalent compounds-MgO,CaCl2 3.3 Valance Definitions of valence. Definition of electrovalence, positive and negative electrovalence, formation of Electrovalent compounds-MgO,CaCl2 3.4 Definition of covalence, single, double and triple covalent bonds, formation of Covalent compounds H2O,CO2,N2	6
4	**Metals and Ores** 4.1 Introduction 4.2 Occurrence of metals in free and combined state,	8

	definitions- mineral, ore, Gangue, flux and slag, metallurgy. 4.3 Ores of Iron 4.4 Steps involved in the extraction of metals: Crushing, Concentration, Reduction, Refining 4.5 Mechanical properties of metals- Hardness, ductility, malleability, tensile strength, toughness, mach inability, weld-ability, forging, soldering, brazing, cast ability.	
5	**Alloys and its applications** 5.1 Introduction 5.2 Definition of alloy, purposes of making alloys with examples. 5.3 Preparation methods- Fusion, Compression 5.4 Classification of Alloys- Ferrous and nonferrous alloys with examples. 5.5 Examples of alloys-Composition, properties and application of duralumin, Woods metal. Brass, Solders. 4.7 Faraday's laws of electrolysis: Faraday's first and second law, relation between electrochemical equivalent and chemical equivalent. 4.8. pH and pOH: Definition of pH, pOH, pH Scale	6
6	**Electrochemistry** 6.1 Introduction 6.2 Basic concepts of electrolysis: Electrolyte, types of electrolyte- strong and weak electrolyte, their difference. 6.3 Ionization and electrolytic dissociation, Degree of ionization, factors affecting degree of ionization. .6.4 Applications of electrolysis- Electroplating of silver, electro refining of blister copper, 6.5 Faraday's laws of electrolysis: Faraday's first and second law, relation between electrochemical equivalent and chemical equivalent. 6.6. pH and pOH: Definition of pH, pOH, pH Scale	6

Text Books:

1. Barrow G.M., Physical Chemistry, McGraw-Hill Publication, New Delhi.
2. ShikhaAgrawal, Engineering Chemistry- Fundamentals and Applications, Cambridge Publishers.
3. Atkins, Physical Chemistry, OUP Oxford.
4. Engineering Chemistry, Wiley India Pvt, Ltd, First edition

Reference Books:

1. Jain P.C & Jain Monica, Engineering Chemistry, DhanpatRai& Sons, Delhi.
2. Bhal&Tuli, Text book of Physical Chemistry, S. Chand & Company, New Delhi.
3. O. G. Palanna , Engineering Chemistry, Tata McGraw-Hill Publication, New D
4. S. S. Dara, A textbook of Engineering Chemistry, McGraw-Hill Publication, New Delhi.
5. ShashiChawla, Engineering Chemistry, DhanpatRai and Co Education and Technical Publishers
6. V. R Gowarikar, Polymer Science, New Age International Ltd.
7. B. K. Sharma, Instrumental Methods of Chemical Analysis, Krishna Prakashan Media

Online Resources:

1. NPTEL Basic Courses Engineering Chemistry (https://nptel.ac.in/courses/122/101/122101001/)
2. https://www.ncertbooks.guru/engineering-chemistry/
3. Coursera Chemistry Courses (https://www.coursera.org/browse/physical-science-and-engineering/chemistry?languages=en&page=2)

Course Code: 22IUCC0104H		**Course Name:** English-I
Credits: 3	**Teaching Scheme:** L-3	**Evaluation Scheme:** CA–40, ESE–60
Duration of EndSemExam: 3hr		
Pre-requisites:		
Course Objectives:		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Student will be able to apply basic concept of grammar in day to day communication		
CO2: Student will be able to Summaries and comprehend the given passages		
CO3: Student will be able to Compose dialogue and paragraph for different situations		
CO4: Student will be able to use appropriate words as per the context		
CO5: Student will be able to Deliver prepared speeches to express ideas, emotions and thoughts.		

Contents –

Unit	Content	Teaching hours
1	**Basic Grammar** 1.1 Articles, 1.2 Preposition, 1.3 Conjunction, 1.4 Tenses, 1.5 Direct indirect speech, 1.6 Change the voice.	10
2	**Comprehension Skills** 2.1 Definition of Reading 2.2 Definition of Reading Comprehension 2.3 Kinds of Reading 2.4 Extensive Reading 2.5 Intensive Reading 2.6 Effective Strategies for Reading Comprehension	10
3	**Paragraph and dialogue writing** 3.1 Paragraph Writing: Elaborate and expand the ideas with cohesion, coherence and use of correct punctuation marks 3.2 Types of Paragraph: Narrative, Descriptive, Technical, Comparison and Contrast 3.3 Dialogue Writing: Based on various situations	10
4	**Vocabulary building** 4.1 Words Often Confused 4.2 Collocation 4.3 Prefix and Suffix 4.4 Synonyms and Antonyms	6
5	**Speeches** Speech Writing based on situations 5.1 Welcome Speech, 5.2 Farewell Speech, 5.3Vote of Thanks and Introducing a Guest	6
6	**Introduction to the sounds of English**	6

	6.1 English Phonetics 6.2 Phonetics symbols 6.3 Consonant and vowel sounds 6.4 Accent, rhythm and intonation. 6.5 The functions of intonation	

Reference Books:

1. English Grammar and Composition Book (Wren & Martin)
2. Oxford Modern English Grammar
3. Fundamentals of English Grammar (N.C.Sinha)

Online Resources:

1. https://www.britishcouncil.in/english/learn-online
2. http://learnenglish.britishcouncil.org/en/content
3. http://www.talkenglish.com
4. Languagelabsystem.com
5. www.wordsworthelt.com

Course Code: 22IUCC0105T		
Course Name: Fundamental of Computer System		
Credits: 2	**Teaching scheme:** L-2	**Evaluation Scheme:** Audit
Duration of End sem Exam: --		
Pre-requisites:		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Use Computer System and its Peripherals**.**		
CO2: Prepare Business Document using word processing tool.		
CO3: Interpret data and represent it graphically using spreadsheet.		
CO4: Prepare Professional Presentations.		
CO5: Identify the layout of wired and wireless LAN Environment		
CO6: Prepare and interpret database using csv file.		

Contents:

Unit	Content	Teaching hours
1	**Introduction to Computer System** Basic of computer system: Overview of Hardware and Software, block diagram of Computer System, Input/output unit CPU, Arithmetic logic Unit (ALU), System of software Memory Unit. 1.2 Generation of O.S.: Overview of Different Types and Generation of Operating system 1.3 Working with OS: Create and manage file and folders, Copy a file, renaming and deleting of files and folders, Searching files and folders, application installation, creating shortcut of application on the desktop. 1.4 Printer and Scanner: Overview of printer and scanner, Features of printer and Scanner, Difference between Printer and Scanner.	4
2	**Word Processing** Word processing: Overview of Word processor Basics of Font type, size, color, Effects like Bold, italic, underline, Subscript, superscript, Case changing options Previewing a document, Saving a document, Closing a document and exiting application. 2.2 Editing a document: Format and align text, Formatting Paragraphs, Line and paragraph spacing, using FIND and REPLACE, Setting line spacing, add bullet and numbers in lists, add borders and shading, document views, Page settings and margins, Spelling and Grammatical checks. 2.3 Inserting an image to word document: Insert and delete a page break, Insert page numbers, Insert the date and time, Insert special characters (symbols), Insert a picture from a file, Resize and reposition a picture 2.4 Working with table: Insert a table, Convert a table to text, Navigate and select text in a table, Resize table cells, Align text in a table, Format a table, Insert and delete columns and rows, Borders and shading, Repeat table headings on subsequent pages, Merge and split cells.	4
3	**Spreadsheet** Working with spread sheet: Overview of workbook and worksheet, Create Worksheet Save, Copy Worksheet, Delete Worksheet, Close and open Workbook 3.2 Editing worksheet: Insert and select data, adjust row height and column width, delete, move data, insert rows and columns, Copy and Paste, Find and	4

	Replace, Spell Check Zoom In-Out, Special Symbols, Insert Comments, Add Text Box, Undo Changes, Freeze Panes, hiding/unhanding rows and columns. 3.3 Formatting cells and sheet: Setting Cell Type, Setting Fonts, Text options, Rotate Cells, Setting Colors, Text Alignments, Merge and Wrap, apply Borders and Shades, Sheet Options, Adjust Margins, Page Orientation, Header and Footer, Insert Page Breaks, Set Background. 3.4 Working with formula: Setting Cell Type, Setting Fonts, Text options, Rotate Cells, Setting Colors, Text Alignments, Merge and Wrap, apply Borders and Shades, Sheet Options, Adjust Margins, Page Orientation, Header and Footer, Insert Page Breaks, Set Background 3.5 Advanced operation: Conditional Formatting, Data Filtering, Data Sorting, Using Ranges, Data Validation, Adding Graphics, Printing Worksheets, print area, margins, header, footer and other page setup options 3.6 Working with chart: Introduction to charts, overview of different types of charts, Bar, Pie, Line charts, creating and editing charts. Using chart options: chart title, axis title, legend, data labels, Axes, grid lines, moving chart in a separate sheet.	
4	**Presentation tool** Creating presentation : Outline of an effective presentation, Starting a Presentation Files, Creating a Basic Presentation, Working with textboxes, Apply Character Formats, Format Paragraphs View a Presentation, Saving work, creating new Slides, Applying a theme, Changing Colors, fonts and effects, apply custom Color and font theme, changing the background, Arrange Slide sequence. 4.2 Working with table: Insert a Table in a Slide, Format Tables, and Import Tables from Other Office Applications. 4.3 Animation: Adding and Modifying animation to a Presentation - Insert Images into a Presentation, insert audio clips, video/animation, Add Shapes, Add Visual Styles to Text in a Presentation, Edit Graphical Objects on a Slide, Format Graphical Objects on a Slide, Group Graphical Objects on a Slide, Apply an Animation Effect to a Graphical Object, Add Transitions, Add Speaker Notes, Print a Presentation. 4.4 Working with Charts: Insert Charts in a Slide, Modify a Chart, Import Charts from Other Office Applications.	4
5	**Basic of Networking** Introduction of networking Devices: Application of Network interface card (NIC), HUB, Switches, Routers, Modem 5.2 Concept of Network :- LAN,MAN,WAN Wireless network and devices; Wi-Fi, Access point, Repeaters, Bluetooth 5.3 Characteristics of Cable: CAT5,CAT6,fibre optic cable, use of crossover and straight cable,RJ-45 connectors, SC.ST,FC,LC type fiber Connectors 5.4 Types of Servers	4
6	**Databases** Create an appropriate database structure : Import data from existing files (including .csv, .txt) using specified field names to create tables. Create relationships between tables Create and use a data entry form . Types of database Characteristics, uses, advantages and disadvantages of a flat file and a relational database .Primary and foreign keys Characteristics of primary key and foreign keys. Form design Characteristics of good form design.Perform calculations Use arithmetic operations or numeric functions to perform calculations including calculated fields, calculated controls Use formulae and functions to perform calculations at run time including addition, subtraction, multiplication, division, sum, average, maximum, minimum, count .Sort data Use a single criterion, or multiple criteria to sort data into ascending or	6

	descending order.Search and select data Use a single criterion, or multiple criteria to select subsets of data using a query perform searches using a variety of operators including AND, OR, NOT, LIKE, >, =, <=, <> perform searches using wildcards.Display data Produce reports to display data including displaying all the required data and labels.	

Text Books:
1. B Ram –Computer Fundamental architecture and Organization, New age international Publications 2. Dr RajendraKawale ,"Computer Fundamental", Devraj Publications Dist. Solapur

Course Code: 22IUCC0106T	**Course Name:** Engineering Graphics
Credits: 3 **Teaching Scheme:** L-3	**Evaluation Scheme:** CA–40, ESE–60
Duration of EndSemExam: 3hr	
Pre-requisites:	
Course Objectives:	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Student will able to draw geometrical figures with proper dimensioning technique	
CO2: Student will able to draw the views of given object by using principles of orthographic projection	
CO3: Student will able to draw isometric views of given component or from orthographic projection	
CO4: Student will able to draw Engineering curves	
CO5: Student will able to draw free hand sketches of given engineering Elements.	
CO6: Student will able to create 2D computer drawings on CAD software	

Contents –

Unit	Content	Teaching hours
1	**Basics of Graphics:** Introduction to Engineering Graphics, Need of Engineering Drawing, Drawing Instruments, BIS code of practice for general engineering drawing, standard sizes of drawing sheets (ISO-A Series) , single stroke vertical Letters and Numbers, convention of lines and their applications, Scale- reduced, enlarged and full size, Dimensioning technique, and Geometrical construction	4
2	**Orthographic Projection:** Orthographic projection-first angle and third angle method their symbol, conversion of pictorial view into orthographic views- Orthographic projection of machine parts containing plain surfaces, slanting surfaces slots, ribs, cylindrical surfaces (First angle method is to be used)	7
3	**Isometric Projections:** Introduction to Pictorial views, Isometric scale and natural scale, Converting Orthographic Projections into Isometric Projections and Isometric views. Illustrative problems related to simple objects having plain, slanting , cylindrical surfaces and slots on slanting surfaces	7
4	**Engineering Curves** Curves used in Engineering Practice, Conic sections, Construction of conics by different methods, Rectangular- hyperbola, Cycloidal curves, Epi and hypo-cycloids, Involutes and Archimedean spiral, Loci of points on single slider crank mechanism with given specification	7
5	**Free Hand Sketches of Engineering objects** Free hand sketches of machine elements; Thread profiles, nuts bolts, studs set screws, washers, locking arrangments.	3
	Introduction to CAD Listing the computer technologies that impact on graphical communication, Demonstrating knowledge of the theory of CAD software [such as: The Menu System, Toolbars (Standard, Object Properties, Draw, Modify and	

6	Dimension), Drawing Area (Background, Crosshairs, Coordinate System), Dialog boxes and windows, Shortcut menus (Button Bars), The Command Line (where applicable), The Status Bar, Different methods of zoom as used in CAD, Select and erase objects	4

Reference Books:

1. N. D. Bhatt and Panchal V. M., "Engineering Drawing", Charotar Publishing House, Anand
2. P. J. Shah , " A Text Book of Engineering Drawing", S.Chand, New Delhi
3. Shah M.B. &RanaB.C,"Engineering Drawing & Computer Graphics", Pearson Publications,
4. .Agrawal B.&Agrawal C.M, "Engineering Graphics", TMH Publications
5. Narayana K.L. & P. Kannaiah, "Engineering Drawing", Scitech Publications
6. .P.I Vargese, "Engineering Graphics", McgrawHill Publications
7. D.A.Hindoliya," Engineering Graphics", B.S.Publications
8. Dabhade M. L., "Engineering Graphics", Vol.-I and Vol.-II, Vision Publications, Pune .
9. K.Venugopal, "Engineering Drawing and Graphics" ,New Age International Publishers

Online Resources:

1. https://www.machinedesignonline.com
2. https://www.me.umn.pdu/courses/me2018/handouts/engg%20graphics.pdf
3. https://www.youtube.com/watch?v=fvjk7PlxAuo
4. https://www.youtube.com/watch?v=3WXPanCq9LI

Course code: 22IUCC0102L	**Course name:** Physics-I Lab
Credits: 1 **Teaching scheme:** P-2	**Evaluation scheme:** TW–50, PR–0
Pre-requisites:	
Course Outcomes: At the end of the course, the students will be able to -	
CO1. Identify different systems of units and convert units from one system to another as well as conversant with Practical units.	
CO2. Understand equations of motion and their applications.	
CO3: Understand equations of motion and their applications.	
CO4. Acquire basic knowledge on semiconductor &applicatation of P-N Junction diode.	
CO5. Use the properties of photoelectric effect, optical fiber &nano particle for various engineering application	
CO6: Understand concept of LIGHT	

Contents –
Perform any Ten practical's from the list.

Practical	Content	Teaching hours
1	Measure the dimensions of given objects using Vernier caliper.	2
2	Measure the dimensions of given objects using micrometer screw gauge	2
3	Measure the dimensions of given objects using spherometer	2
4	To determine the valve of acceleration due to gravity [g] of a place with simple pendulum.	2
5	Determination of force constant using Helical Spring	2
6	Use Searle's thermal conductivity apparatus to find coefficient of thermal conductivity of a given material	2
7	Use Joule's calorimeter to determine Joule's Mechanical/ electrical equivalent of heat	2
8	To study the I-V characteristics of photoelectric effect	2
9	To determine the refractive index of the material of the glass slab by pin method.	2

Course code:22IUCC0103L		**Course name:**Chemistry-I Lab
Credits: 1	**Teaching scheme:** P-2	**Evaluation scheme:** TW–50, PR–0
Pre-requisites:		
Course Objectives:		
Course Outcomes: At the end of the course, the students will be able to		
CO1: Demonstrate an understanding of basic principles of organic chemistry and how they relate to everyday Experiences. **CO2:** Illustrate the Basic Concept of Structure of Atom **CO3:** Describe nature of bonding in molecular compounds. **CO4:** To know the common analysis techniques used in metallurgical industries. **CO5:** Students know how to approach the problem of choosing an alloy for a particular application. **CO6:** Use Relevant Electrolyte in Electrolysis for different Application		

Contents –

Unit	Content	Teaching hours
1	Do you know chemistry laboratory	2
2	Determine preliminary test and functional group of organic compound No.1	2
3	Determine preliminary test and functional group of organic compound No.2	2
4	Determine preliminary test and functional group of organic compound No.3	2
5	Determine preliminary test and functional group of organic compound No.4	2
6	Identify anion in the given ionic solution (non- metallic ions)	2
7	Identify cation in the given ionic solution (metallic ion)	2
8	Determine the percentage of iron in the given sample using redox titration	2
9	Determine the pH value of given solution using universal indicator	2
10	Determine the voltage generated from chemical reaction using Daniel cell.	2
11	Determine the electrode potential of Iron metal.	2
12	Determine the rate of corrosion on different temperatures for Aluminum	2

Course code: 22IUCC0104L		**Course name:** Language Lab-I
Credits: 1	**Teaching scheme:** P-02	**Evaluation scheme:** TW–50, PR–0
Pre-requisites:		
Course Outcomes: At the end of the course, the students will be able to –		
CO1: Student will be able to apply basic concept of grammar in day to day communication		
CO2: Student will be able to Summaries and comprehend the given passages.		
CO3: Student will be able to Compose dialogue and paragraph for different situations.		
CO4: Student will be able to use appropriate words as per the context		
CO5: Student will be able to Deliver prepared speeches to express ideas, emotions and thoughts		

Contents:

Unit	Content	Teaching hours
1	Make sentences using correct articles	2
2	Make sentences using correct Preposition	2
3	Make sentences using correct Conjunctions	2
4	Make sentences using correct active and passive voice.	2
5	Make sentences using correct direct and in direct speech.	2
6	Make sentences using correct tenses	2
7	Make oral presentations using correct grammar.	2
8	Write short paragraphs emphasizing on syntax and sentence structure.	2
9	Write different types of dialogues for role plays.	2
10	Construct sentences using various collocations.	2
11	Write different types of speeches using new vocabulary.	2
12	Deliver extempore short speeches of 3to5 minutes.	2

Course Code: 22IUCC0105L	**Course Name:**Fundamental of Computer System Lab	
Credits:1	**Teaching Scheme:** P-2	**Evaluation Scheme:** TW–50, PR–0
Duration of End Sem Exam:		
Pre-requisites:		
Course Outcomes: At the end of the course, the students will be able to –		
CO1: Use Computer System and its Peripherals**.**		
CO2: Prepare Business Document using word processing tool.		
CO3: Interpret data and represent it graphically using spreadsheet.		
CO4: Prepare Professional Presentations.		
CO5: Identify the layout of wired and wireless LAN Environment		
CO6: Prepare and interpret database using csv file.		

Contents –

Unit	Content	Teaching hours
1	Identify various Input/output devices. Connections and peripherals of computer system.	2
2	Install and work with wifi printer and scanner setting.	2
3	Create, edit and save documents : apply formatting features on the text, line, paragraph. Use bullets, numbering, page formatting.	2
4	Insert, edit images and shapes, sizing, cropping color, background, group/ungroup. Insert and apply various table formatting features on it.	2
5	Create, open and edit worksheet 1.Enter data and format it adjust row height and column width. 2.Insert and delete cells, rows and columns. 3.Apply wrap text, orientation feature on cell.	2
6	Insert formulas, ”IF” conditions, functions and named ranges in worksheet. Apply data sort, Filter and data validation features.	2
7	Create charts to apply various chart options.	2
8	Create slide presentation. 1.Apply design themes to given presentation. 2.Add new slides and insert pictures / images, shapes. 3.Add tables and charts in slides. 4.Run slides presentation in different modes. 5.Print slide presentations on handouts.	2
9	Apply animation effects to text and slides Add audio and video files in presentation.	2
10	Connect/Disconnect LAN cable, external hard disk modem. Connect desktop computers and laptops with LCD/DLP projectors.	2

11	Prepare and test crossover and straight cable, CAT5,CAT6 cable, using crimping Tools, splicer. Connect two switches/Hubs/switches/r outers physically.	2
12	Identify server by its type and verify its Specification	2
13	Create new database, table and build relationship between them, saving tables	2
14	Managing data in tables and performing calculation using functions, and sorting data from tables.	2

Course Code:22IUCC0106L	**Course Name:**Engineering Graphics Lab
Credits:2 **Teaching Scheme:** P-4	**Evaluation Scheme:** TW–100, PR–0
Duration of End Sem Exam:	
Pre-requisites:	
Course Outcomes: At the end of the course, the students will be able to –	
CO1: Student will able to draw geometrical figures with proper dimensioning technique	
CO2: Student will able to draw the views of given object by using principles of orthographic projection	
CO3: Student will able to draw isometric views of given component or from orthographic projection **CO4:** Student will able to draw Engineering curves	
CO5: Student will able to draw free hand sketches of given engineering Elements	
CO6: Student will able to create 2D computer drawings on CAD software	

Contents –

Unit	Content	Teaching hours
1	Draw different types of lines, dimensioning styles. Draw regular geometric construction	2
2	Draw one figure showing dimensioning techniques. Draw one problem on redraw the figure.	2
3	Draw two problem on orthographic projection using first angle method of projection having plain and slanting surface	2
4	Draw two problem on orthographic projection using first angle method of projection having cylindrical surface, rib	2
5	Draw two problems on Isometric view of simple object having plain and slanting surface using natural scale	2
6	Draw two problems on Isometric view of simple object having plain and slanting surface using natural scale	2
7	Draw two problem on Isometric projection of object having cylindrical surface by using isometric scale	2
8	Draw two problem on isometric projection of object having slot on slanting surface by using isometric scale	2
9	Draw Engineering curves generated from conics 1. Ellipse 2. Parabola 3. Hyperbola	2
10	Draw Involutes of circle and pentagon. Draw cycloid curve	2
11	Draw free hand sketches/conventional representation of machine elements such as thread profile, nuts, bolts, studs, set screws, washer, locking	2

	arrangement	
12	Draw free hand sketches/conventional representation of machine elements such as thread profile, nuts, bolts, studs, set screws, washer, locking arrangement	2
13	Draw basic 2D entities like: rectangle, polygon, circle, Arc, circular and rectangular array by using AutoCAD	2
14	Draw basic branch specific components in 2D using AutoCAD	2
15	Draw complex branch specific components in 2D using Autocad	2

SEM-II

Course Code: 22IUCC0201B		**Course Name:** Mathematics-II
Credits: 4	**Teaching scheme:** L-3 T-1	**Evaluation Scheme:** CA–40, ESE–60
Duration of End sem Exam: 3 hr		
Pre-requisites:		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Understand the concept to formulate a mathematical model using of function, and concept of Limits and continuity which rise to derivative.		
CO2: Estimate, to graph derivatives of the function and used rules of differentiation to solve problem involving rates of change, approximation of function and optimization problem		
CO3: Understand the concept of integration to find the arc length, area, volume and surface area using integrals.		
CO4: Solve ordinary differential equations and also apply these methods to solve engineering problems.		
CO5: Trace the Cartesian curve, parametric curve and Polar curve		

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Contents –

Unit	Content	Teaching hours
1	**Function limit and Continuity** Definition of Function, Ways to represent a function, Types of functions, Mathematical modeling, and transformation of functions. Limit: Concept and definition of limit using tangent and velocity problem, Laws of limits with simple problems. Concept of continuity	8
2	**Differential Calculus** Concept and definition of Derivatives through physical and geometrical problem. And Rates of Change, The Derivative as a Function. Rules of Differentiation, Derivative of inverse functions, Implicit Differentiation, Logarithmic Differentiation, Parametric Differentiation, Second order differentiation	8
3	**Application of Derivative** Rate of Change in the Natural and Social Sciences, Exponential Growth and Decay, Linear Approximations and Differentials, Maximum and Minimum Values, Indeterminate Forms and L'Hospitals Rule. Optimizing problems**.**	8
4	**Integral Calculus** Indefinite and Definite integrals. Methods and Properties, Applications to find arc length, Area, Volume and surface area.	10
5	**Differential Equation**	8

	Introduction to Initial Value problems. Linear and Bernoulli's equations, Exact equations and reducible to exact form. Applications of first order differential equations- Orthogonal trajectories, growth and decay problems, Electrical circuits, and falling bodies.	
6	**Curve Tracing** Tracing of standard curves in Cartesian form, Parametric form and Polar form	6

Reference Material:
1. Narayan, Shanti and Mittal, P.K. Differential Calculus, S Chand publication. 2. Narayan, Shanti and Mittal, P.K. Integral Calculus, S Chand publication 3. S.C. Gupta & V.K. Kapoor, Fundamentals of Mathematical Statistics, 11th edition, Sultan Chand & Sons, New Delhi. 4. Hu**g**hues- Hallett Gleason, Calculus Single and Multivariable, 4th edition, Wiley. 5. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons, New York. 6. B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, New Delhi. 7. P. N. Wartikar, J. N. Wartikar, Applied Mathematics (Vol I & II), Pune VidyarthiGrihaPrakashan, Pune. 8. H. K. Das and RajnishVerma, Higher Engineering Mathematics, S. Chand & CO. Pvt. Ltd., New Delhi.
Online Resources: 1.www.scilab.org/-SCI Lab 2.www.mathworks.com/products/matlab/ -MATLAB 3.www.allmathcad.com/-MathCAD 4.www.wolfram.com/mathematica/ - Mathematica 5.www.dplot.com/-DPlot

Course Code: 22IUCC0202B		**Course Name:** Physics-II
Credits: 4	**Teaching Scheme:** L-3 T-1	**Evaluation Scheme:** CA–40, ESE–60
Pre-requisites: Soil		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: To learn about the useful solid material in industry & understand its physical properties.		
CO2: To learn about the useful liquid material & understand its physical properties.		
CO3: Understand the concepts of sound wave		
CO4: To learn about the concepts of radioactive and ultrasonic principles.		
CO5: To learn about the concepts of electrostatic & X rays.		
CO6: To learn about the concepts of the principles of electricity , magnetism & semiconductors to solve engineering problems.		

Contents –

Unit	Content	Teaching hours
1	**Properties of Solid** 1.1 Deforming Force and Restoring Force, Elasticity, Plasticity, Rigidity 1.2 Stress and Strain and their types, Elastic limit and Hooke's law Stress - Strain diagram, Poisson's ratio, factors affecting elasticity, With numerical problem	6
2	**Properties of Liquid** 2.1 Fluid friction, pressure, pressure- depth relation, Pascal's law, Archimedes' principle 2.2 Viscosity, velocity gradient, Newton's law of viscosity. 2.3 Free fall of spherical body through viscous medium and Stokes' law, derivation of coefficient of viscosity 'ç' by Stokes' method, effect of temperature and adulteration on viscosity of liquids With numerical problem	8
3	**Wave Theory & Sound** 3.1 Wave , Longitudinal Wave, Transverse Wave, , wave length, amplitude, frequency, time-period and velocity of wave, relation between velocity, frequency and time - period of wave. 3.2 Sound waves, Newton's formula for velocity of sound, Musical sound & noise, Loudness, Pitch, Quality of sound (Timbre), Musical Scale.	8
4	**Radioactivity** 4.1 Radioactivity, α, β and γ rays and their properties. 4.2 Radioactive decay law, half-life period. Energy Mass Relation, Nuclear fission, Nuclear fusion. 4.3 Ultrasonic waves, properties of Ultrasonic waves, Applications of ultrasonic waves, Doppler Effect and its applications.	6
	Electrostatics & X Ray 5.1 Capacitors and capacitance, Parallel plate capacitor, effect of dielectric	

5	on capacitance, Series Combination of capacitors, Parallel Combination of capacitors, energy stored in a capacitor. Cells, emf of cell, internal resistance of cell 5.2 X ray Definition, Production of X-rays by modern Coolidge tube, Continuous & Characteristic X ray, Properties of X ray and Engineering , Medical, & Scientific Applications of X ray.	6
6	**Electricity , Magnetism & Semiconductor** 6.1Coulomb's law, Ampere's law Faraday's law motion of charges in electric & magnetic fields to determine e/m. Electrostatic focusing system. Magneto static focusing system. 6.1 Conductors, Insulators And Semiconductors, Energy bands, intrinsic and extrinsic semiconductors p-n junction diode, I-V characteristics of p-n junction, applications of p-n junction diode.	6

Reference Material :

1. Engineering Physics- R K Gour& S L Gupta, DhanpatRai Publication- New Delh

2. Engineering Physics- M N Avadhanulu& P G Kshirsagar, S Chand & Company Ltd.- New Delhi 3. Fundamental of Physics- Jearl Walker, Wiley India Edition

4. Engineering Physics- V Rajendran, Tata McGraw-Hill Publishing Company Ltd- New Delhi

5. Applied Physics- Manpreet Singh, Dr. Major Singh, Mrs.Hitashi Gupta, S K Kataria& Sons- New Delhi

E-Resources:

1. http://nptel.ac.in/course.php?disciplineId=115
2. http://nptel.ac.in/course.php?disciplineId=104
3. http://hperphysics.phy-astr.gsu.edu/hbase/hph.html
4. www.physicsclassroom.com
5. www.physics.org
6. www.fearofphysics.com

Course Code: 22IUCC0203B		**Course Name:** Chemistry-II
Credits: 4	**Teaching scheme:** L-3 P-0 T-1	**Evaluation Scheme:** CA–40, ESE–60
Duration of End sem Exam:3hr.		
Pre-requisites:		
Course Outcomes: At the end of the course, the students will be able to –		
CO1: Describe different processes for manufacturing metals and control of their corrosion		
CO2: Use relevant water treatment process to solve industrial problems		
CO3: Justify the selection of lubricants for different types of machines		
CO4: Describe different types of fuels and their properties and uses.		
CO5: Justify the selection of lubricants for different types of machines		
CO6: Used Relevant engineering materials in Industry		

Contents:

Unit	Content	Teaching hours
1	**Corrosion 1.1** Definition, causes of Corrosion Different types of Corrosion and factors affecting corrosion rate. **1.2 Protection of metal by**: Galvanising ,Tinning ,Electroplating ,colourizing 1.5 classification of plain carbon steel, low carbon, medium carbon,high carbon steel with their properties and Application. 1.3 **Paint:** Definition characteristics, constituent of paint and their function.	10
2	**Metallurgy** 2.1 Introduction 2.2 Definition of metallurgy Properties and application of commercial forms of iron- pig iron ,cast iron and wrought iron 2.3 Steel: Definition of steel ,Preparation of steel from pig iron using open hearth process, basic oxygen process. 2.4 classification of plain carbon steel, low carbon, medium carbon, high carbon steel with their properties and application.	10
3	**Water treatment:** 3.1 Introduction 3.2 Hardness : Classification, 3.3 Hard water in boilers and prevention: Boiler corrosion, Caustic embrittlement, priming and foaming, scales and sludge's 3.4 Water softening : lime soda process (hot lime soda and cold lime soda process) Zeolite process, Ion exchange process (Cation exchange process and anion exchange) 3.5 Potable water treatment: sedimentation, coagulation, filtration, and sterilization. Waste water treatment, BOD and COD of sewage water, Reverse Osmosis, recycling of waste water	10
4	**Fuels** 4.1 Introduction 4.2 Definition of fuel, calorific and ignition temperature. Characteristic of good fuel classification of fuels 4.3 Solid fuels: Coal classification and composition, proximate analysis, Ultimate analysis. Determination of gross calorific value by using Bomb calorimeter. 4.4 Liquid fuel: Fractional distillation of crude petroleum , boiling range, composition, properties, application of Bio- diesel 4.5 Gaseous fuel: Composition, properties ,application of Biogas, LPG, CNG	10

5	**Lubricants** 5.1 Introduction Definition of lubricants .Function of lubricants, Classification of lubricants. 5.2 Characteristic : Physical characteristic of lubricants : Viscosity, Viscosity index, oiliness, flash point, fire point, 5.3 chemical characteristic of lubricants- acid value, emulsification, saponification value.(only definition) 5.4 Selection of Lubricants for road rollers, steam engine, sewing machine, concrete mixture, I.C. engine, cutting tools, gears.	10
6	**Non -metallic Engineering material: Polymers (Plastics, Rubber) :** 6.1Introduction 6.2 Plastics: Definition of plastic, polymer, polymerization, types of Polymerization with examples. Types of plastic- thermo softening plastics and thermosetting plastics and their difference, properties and applications of plastics. 6.3 Rubber: Types of rubber. Natural Rubber- definition, drawbacks of natural rubber, vulcanization of rubber with chemical reaction, applications of vulcanized rubber Synthetic rubber- definition, difference between natural and synthetic rubber, examples of synthetic rubber, properties of synthetic rubber like -elasticity, tack, and abrasion resistance, their definition and related application Thermal Insulators 6.4 Thermal Insulators -Definition, characteristics of thermal insulators, classification- organic and inorganic thermal insulators, their examples, preparation, properties and applications of polystyrene and Distilled or pure water	10

Reference Materials

1. Jain P.C & Jain Monica, Engineering Chemistry, DhanpatRai& Sons, Delhi.
2. Bhal&Tuli, Text book of Physical Chemistry, S. Chand & Company, New Delhi.
3. O. G. Palanna , Engineering Chemistry, Tata McGraw-Hill Publication, New Delhi.
4. S. S. Dara, A textbook of Engineering Chemistry, McGraw-Hill Publication, New Delhi.
5. ShashiChawla, Engineering Chemistry, DhanpatRai and Co Education and Technica Publishers
6. V. R Gowarikar, Polymer Science, New Age International Ltd.
7. B. K. Sharma, Instrumental Methods of Chemical Analysis, Krishna Prakashan Media
8. Barrow G.M., Physical Chemistry, McGraw-Hill Publication, New Delhi.
9. ShikhaAgrawal, Engineering Chemistry- Fundamentals and Applications, Cambridge Publishers.
10. Atkins, Physical Chemistry, OUP Oxford.
11. Engineering Chemistry, Wiley India Pvt, Ltd, First edition.

E-Resources:

1. NPTEL Basic Courses Engineering Chemistry (https://nptel.ac.in/courses/122/101/122101001/
2. https://www.ncertbooks.guru/engineering-chemistry/
3. Coursera Chemistry Courses (https://www.coursera.org/browse/physical-science-andengineering/chemistry?languages=en&page=2
4. “Introduction to Corrosion” NPTEL Course (https://www.youtube.com/playlist?list=PL8lylDWRkaW8BXestE4baRnN0699S11Lq
5. “Tribology &LubricationNPTEL Course (https://www.youtube.com/playlist?list=PLLwnvFqJAltJvWAfEVU4gcUK27hEA7FD)

Course Code: 22IUCC0204H	**Course Name:** Universal Human Values
Credits: 2**Teaching Scheme:** L-2 P-0	**Evaluation Scheme:** ESE–50
Duration of End Sem. Exam: 1 hr.	
Pre-requisites:	

Course Objectives:
Course Outcomes: At the end of the course, the students will be able to –
CO-1 To create an awareness on Engineering Ethics and Human Values.
CO-2 Understand the significance of value inputs in a classroom and start applying them in their life and profession
CO-3 Distinguish between values and skills, happiness and accumulation of physical facilities, the Self and the Body.**C**
O-4 Understand the role of a human being in ensuring harmony in society and nature.
CO-5 Distinguish between ethical and unethical practices, and start working out the strategy to actualize a harmonious environment wherever they work.

Contents –

Unit	**Content**	**Teaching hours**
1	**HUV Introduction** 1.1 Need, Basic Guidelines Content and Process for Value Education, 1.2 Continuous Happiness and Prosperity, 1.3 Self-Exploration, Continuous Happiness and Prosperity 1.4 Right Understanding, Relationship and Physical Facilities 1.5 Understanding Happiness and Prosperity Correctly.	8
2	**Harmony of Human Being** 2.1 Harmony in Myself 2.2 Understanding Human Being as a co-existence of the Sentient 'I' and the material 'Body' 2.3 Programs to ensure Sanyam and Swasthya.	8
3	**Harmony in Human-Human Relationship** 3.1 Understanding Harmony in the Family 3.2Understanding the meaning of Vishwas, 3.3Understanding the meaning of Samman, 3.4Understanding the Harmony in the Society, 3.5Visualizing a Universal Harmonious order in Society.	8
4	**Harmony in the Nature and Existence** 4.1 Understanding the Harmony in the Nature 4.2 Interconnectedness and Mutual Fulfilment among the Four Orders of Nature 4.3 Understanding Existence as Co-existence.	8
5	**Harmony on Professional Ethics** 5.1 Natural Acceptance of Human Values, 5.2 Definitiveness of Ethical Human Conduct, 5.3 Basis for Humanistic Education, 5.4 Humanistic Constitution and Humanistic Universal Order, 5.5 Competence in Professional Ethics, Case Studies of Typical Holistic Technologies.	8
6	**SELF DEVELOPMENT** 6.1 Goal setting and its importance. 6.2 Characteristics of Goal setting (SMART- Specific, Measurable, Attainable, Realistic, Time bound) 6.3 Time Management - Importance, prioritization of work, time matrix, time	8

	savers, and time wasters. 6.4 Stress Management - Definition, types of stress, causes of stress, managing stress, and stress busters.	

Reference Material:
1 Corliss Lamont, Philosophy of Humanism 2.Gaur. R.R. ,Sangal. R, Bagaria. G.P, A Foundation Course in Value Education, Excel Books, 2009. 3.Gaur. R.R. ,Sangal. R ,Bagaria. G.P, Teachers Manual Excel Books, 2009. 4.I.C. Sharma . Ethical Philosophy of India Nagin& co Julundhar 5.Mortimer. J. Adler, – Whatman has made of man 6.William Lilly Introduction to Ethic Allied Publisher
Online Resources: 1. https://www.britishcouncil.in/english/learn-online 2. http://learnenglish.britishcouncil.org/en/content 3. http://www.talkenglish.com 4. Languagelabsystem.com 5. www.wordsworthelt.com

Course Code: 22IUCC0205T	**Course Name:**Computer Workshop
Credits: 2 **Teaching Scheme:** L-2	**Evaluation Scheme:** Audit
Duration of End Sem Exam:	
Pre-requisites:	
Course Objectives:	
Course Outcomes: At the end of the course, the students will be able to –	
CO1: Troubleshoot common motherboard problems.	
CO2: Select processors required for relevant systems.	
CO3: Partition/Format hard disk drives.	
CO4: Troubleshot peripherals and networks.	
CO5: Test Power supplies	
CO6: Networking and Maintenance	

Contents:

Unit	Content	Teaching hours
1	**Motherboard** Motherboard: Components Layout Connections Motherboard Types: Types and features Enhancing features of motherboard: Adding and or replacing components. Troubleshooting: problems of motherboard.	4
2	**Processor and BIOS** Processor: Common features, Types, Basic structure of CPU,Different levels of cache, system bus, clock speed, packaging. Multiple core processors: Description, Two core processor architecture and multicourse architecture. Co-Processors: Graphics, math. BIOS: Basic I/O system services, Bios interaction, date and time, Boot device priority, Boot setting configuration, password security.	4
3	**Hard Disk** Hard disk Interfaces:EIDE,Serial ATA,SCSI,USB and IEEE 1394,RAID,Solid state Drive. Disk Structures: Heads, tracks, sectors, cylinders, clusters, landingzone, MBR,Zone bit recording. Disk Performance parameters characteristics: Seeks and latency, Data transfer rate. File system: FAT16,FAT32,NTFS,Unix file system,EXT2/EXT3,RAID	4
4	**I/O and Modem** Troubleshoot I/O devices: Keyboard, Switches, Mouse, Scanners, webcam,monitors,printers,speaker and Mike,LCD projector. I/O cables: Specifications of I/O cables,Types of I/O cables,Types of I/O ports,Internal and external modem, Block diagram and specification. Network Interfaces: Definition of network interface, Types of network interface, troubleshooting of network connectivity, Antivirus.	4
	Power supply Purpose and features of SMPS,Working of SMPS Fault finding in power supply	4

5	Uninterrupted Power supply: Characteristics of UPS,Types of UPS,online and offline. Preventive Maintenance of power supply.	
6	**Networking and Maintenance** Share and control resource and Internet connection through network. Implement Network Security to protect from various attacks on networking.MAC Address and its features. Firewall : Technology, implementaion	6

Reference Material:

1. James. K. L. "The Computer hardware installation, interfacing, troubleshooting and maintenance", PHI Learning ,New Delhi,2014 ISBN:978-81-203-4798-4
2. Gupta, Vikas ,"Comdex: Hardware and Networking course kit" Dreamtech Press, New Delhi ISBN:978-93-5119-265-7
3. Minasi,Mark, "The Complete Pc Upgrade and maintenance Guide" BPB Publication, Mumbai, New Delhi ISBN:978-81-265-0627

Online Resources:

1. https://www.tutorialspoint.com/computer_fundamentals/computer_motherboard.htm
2. https://www.javatpoint.com/what-is-a-multicore-processor
3. https://www.geeksforgeeks.org/co-processor-computer-architecture/
4. https://www.digitalocean.com/community/tutorials/an-introduction-to-networking terminology-interfaces-and-protocols
5. https://www.tutorialspoint.com/information_security_cyber_law/network_security.htm
6. https://www.tutorialspoint.com/what-is-a-proxy-server

Course Code: 22IUCC0202L	**Course Name:** Physics-II Lab
Credits: 1 **Teaching scheme:** P-2	**Evaluation scheme:** TW–50
Pre-requisites:	

Course Outcomes: At the end of the course, the students will be able to –
CO 1: To learn about the useful solid material in industry & understand its physical properties.
CO 2: To learn about the useful liquid material & understand its physical properties.
CO 3: Understand the concepts of sound wave
CO 4: To learn about the concepts of radioactive and ultrasonic principles.
CO 5: To learn about the concepts of electrostatic & X rays
CO 6: To learn about the concepts of the principles of electricity , magnetism & semiconductors to solve engineering problems.

Contents:

Unit	Content	Teaching hours
1	Use Searle's method to determine the Young's modulus of given wire	2
2	Determine the coefficient of viscosity of given liquid by Stoker's materials	2
3	Find the downward force, along an inclined plane, acting on a roller due to gravity and its relationship with the angle of inclination.	2
4	Determine the refractive index of glass slab	2
5	Determine the refractive index of prism using Pin Method	2
6	Use resonance tube to determine velocity of sound.	2
7	Determine the specific resistance of given wire.	2
8	Use magnetic compass to draw the magnetic lines of forces of magnet of different shapes.	2
9	Use p-n junction diode to draw forward bias and reverse bias I-V characteristics.	2

Course Code:22IUCC0203L	**Course Name:** Chemistry-II Lab
Credits: 1 Teaching scheme: P-2	**Evaluation scheme: TW–50**
Pre-requisites:	
Course Outcomes: At the end of the course, the students will be able to	
CO1: Describe different processes for manufacturing metals and control of their corrosion **CO2:** Use relevant water treatment process to solve industrial problems **CO3:** Justify the selection of lubricants for different types of machines **CO4:** Describe different types of fuels and their properties and uses. **CO5:** Used Relevant engineering materials in Industry	

Contents –

Unit	Content	Teaching hours
1	Prepare the corrosive acidic medium for Aluminum	2
2	Determine the rate of corrosion on different temperatures for Aluminum	2
3	Determine the acid base titration neutralization reaction	2
4	Determine the percentage of iron present in given Hematite ore by KMnO4 solution	2
5	Determine the percentage of copper in given copper ore	2
6	Determine Chloride content in given water sample by Mohr's method.	2
7	Determine total hardness, temporary hardness and permanent hardness of watersample by EDTA method	2
8	Determine the alkalinity of given water sample	2
9	Determine the effect of temperature on viscosity for given lubricating oil using Redwood Viscometer	2
10	Determine the moisture and ash content in given coal sample by using proximate analysis	2
11	Prepare phenol formaldehyde resin used in manufacturing of Bakelite plastic	2
12	Prepare Urea formaldehyde resin and understand the structure and properties for its applications in engineering	2

Course code: 22IUCC0205L **Course name: Computer Workshop Lab** **Course category:** PCC
Credits: 01 **Teaching scheme:** P-2Hrs/Week **Evaluation scheme:** CA–0, MSE–0, ESE–0, TW–30, PR–20
Pre-requisites: 1
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Troubleshoot common motherboard problems.
LO2: Select processors required for relevant systems.
LO3: Partition/Format hard disk drives.
LO4: Troubleshot peripherals and networks.
LO5: Test Power supplies
LO6: Networking and Maintenance

List of Experiments–

Sr.No.	Name of Experiment
1	Identify different hardware components onmotherboard.
2	Troubleshoot common problems of motherboard.
3	Configure BIOS setting.
4	Partition and manage hard disk: Format harddrives with different file systems
5	Install operating system –Windows family
6	Install operating system –Unix family
7	Troubleshoot hard disk problems
8	Set Keyboard, Mouse, Monitor, Speaker, Webcam, Microphone and LCD Projector ect.
9	Install SMPS,measure voltage levels in mainconnectors of SMPS connecting various subsystems.
10	Troubleshoot computer system by diagnosing the problem.
11	Use diagnostic software for fault finding andviruses
12	Undertake preventive maintenance of PC using vacuum cleaner and simple tools.
13	Configure internet connection to the PC usingwireless technology and troubleshoot various connection related problems
14	Install Proxy Server and configure it.
15	Set up basic protection using public keys and MAC address filters.
16	Practice on firewall technologies to secure the network perimeter

Reference Books:

1. B Ram –Computer Fundamental architecture and Organization, New age international Publications
2. DrRajendraKawale ,"Computer Fundamental", Devraj Publications Dist. Solapur

Online Resources: 1. https://www.tutorialspoint.com

Course Code: 22IUCC0204L	**Course Name:** Universal Human Values Lab
Credits: 1 Teaching scheme: L-0 P-2 T-0	**Evaluation scheme: TW–50**
Course Objectives:	

Course Outcomes: At the end of the course, the students will be able to –
CO-1 To create awareness on Engineering Ethics and Human Values.
CO-2 Understand the significance of value inputs in a classroom
and start applying them in their life and profession
CO-3 Distinguish between values and skills, happiness and accumulation of physical facilities, the Self and the Body.
CO-4 Understand the role of a human being in ensuring harmony in society and nature.
CO-5 Distinguish between ethical and unethical practices, and start working out the strategy to actualize a harmonious environment wherever they work.

Contents –

Unit	Content	Teaching hours
1	Introduce yourself in detail	2
2	Explain your ideas of success/progress/development.	2
3	Write down 5 needs of your self and 5 needs of your body	2
4	Explain self- esteem with two examples	2
5	Chalk out programs to ensure that you are responsible to your body- for the nurturing, protection and right utilization of the body.	2
6	Form small groups in the class and in that group initiate dialogue and ask the eight questions related to trust	2
7	Write a note in the form of story, poem, skit, essay, narration, dialogue to educate a child. Evaluate it in a group.	2
8	Choose any one subject being taught today. Evaluate it and suggest suitable modifications to make it appropriate and holistic.	2
9	Explain harmony between body and mind in your own words.	2
10	Suggest some ideas to be a happy human being.	2
11	Collect information on any two great humanitarians.	2
12	Explain importance of goal setting .	2

Course Code:22IUCC0206L	**Course Name:**Workshop practice Lab
Credits: 2 Teaching scheme: P-4	**Evaluation scheme: TW–100**
Pre-requisites: Basic knowledge of computer, Computer Aided Drawing I lab	
Course Outcomes: At the end of the course, the students will be able to –	
CO 1: Select tool and machinery according to job.	
CO 2: Use hand tools in different shops for performing different operation.	
CO 3: Opera equipment and machinery in different shop.	
CO 4: Prepare composite or utility job according to drawing.	
CO5: Maintain workshop tools and equipment's	

Contents –

Unit	Content	Teaching hours
1	Prepare fitting job (male and female type) as per given drawing or job with following operation marking operation punching operation drilling operation cutting operation filling operation fitting operation (male and female assembly) checking correctness of fit of mating parts	8
2	Prepare bill of material along with estimate cost according given drawing of job such as repairing of class room furniture / book shelves / metallic doors / motor sari guard / battery stand with locking device considering the following applicable operation. a) marking operation as per drawing . b)cutting operation as per drawing . c) cleaning operation as per drawing d) edge preparation operation as per drawing e) filling operation as per drawing f) bending operation as per drawing g) welding operation as per drawing	8
3	Prepare steel frame / structure of utility job (like stool, benches, tables, drawing desk, window grill, ventilator, door frame or similar job) involving arc welding joints as per given drawing or job.	8
4	Prepare simple turning job with facing/taper turning/knurling/grooving/threading operation as per drawing	8
5	Prepare sheet metal utility job using following operation - Part I a. Cutting and Bending b. Edging c. End Curling d. Lancing e. Soldering f. Riveting	8

SEM-III

Course code: 22IUCI0301T **Course name: Engineering Mathematics** **Course category:** PCC
Credits: 4 **Teaching scheme:** L-3 T-1 **Evaluation scheme:** CA–40, ESE–60
Pre-requisites:
Course Objectives:
Course Outcomes: At the end of the course, the students will be able to -
CO.1: Find the partial derivatives of functions and apply the concepts of partial differentiation to find maxima and minima of functions containing two variables
CO.2: Evaluate the double and triple integrals.
CO.3: Calculate area, volume, surface area using double and Triple integral.
CO.4: Solve problems on directional derivatives, gradient, Divergence, Curl of vector field.
CO.5: Solve the problems on fundamental theorems of vector calculus viz. Green's, Stokes and Divergence theorems.
CO.6: Find the solution of linear differential equations having their applications in mechanical and electrical systems.

Contents –

Unit	Content	Teaching hours
1	**Partial Differentiation**: Function of several variables, Partial derivatives, Chain rule, Total Derivative, Change of variables, Euler's theorem on Homogeneous function. Level curves. Applications of Partial differentiation: Jacobian, Tangent planes, Errors and Approximations, Differentials. Extreme value problems. Lagrange's multipliers.	**10**
2	**Multiple Integrals:** Double integration in Cartesian &polar coordinates, Evaluation of double integrals by changing the order of integration and changing to polar form, triple integration.	**08**
3	**Applications of Multiple integrals:**To find Area by double integration, Surface Area, Volume of Solid of Revolution, Volume by triple integration.	**08**
4	**Vector differential Calculus**:Vector fields, Gradient and directional derivatives. Independence of path and potential functions. Divergence of vector field, Curl of vector field, vector identities, Applications.	**10**
5	**Vector Integral Calculus:** Line and Surface integrals, Green's theorem, Divergence theorem, Stokes theorem. Applications.	**08**
6	**Differential equations of higher orders: Linear**:Differential equations of second and higher order with constant coefficients, The method of Variation of parameters. Linear differential equations with variable coefficients – Cauchy's Linear Equation and Legendre's Linear Equation, Initial and boundary value problems. Applications of second order differential equations- vibrating of spring, electrical circuits, and Bending of Beam problems.	**08**

Text Books:1. James Stewart, Calculus Early Transcendental, 7th edition, Cengage. 2. George B. Thomas, Ross L. Finney, Calculus and Analytical Geometry, 9th edition, Pearson. 3. Shanti Narayan, Differential Calculus, S. Chand & Co.
Reference Books: 1. Hughues- Hallett Gleason, Calculus Single and Multivariable, 4th edition, Wiley. 2. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons, NewYork. 3. B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, NewDelhi. 4. P. N. Wartikar, J. N. Wartikar, Applied Mathematics (Vol I & II), Pune VidyarthiGrihaPrakashan,Pune. 5. H. K. Das and RajnishVerma, Higher Engineering Mathematics, S. Chand & CO. Pvt. Ltd., New Delhi. 6. RudraPratap, Getting Started with MATLAB: A Quick Introduction for Scientists and Engineers, Oxford University Press, 2010.
Online Resources: 1. @ NPTEL lecture on Calculus of several variable. 2. @NPTEL lectures on Multivariable calculus variable. 3. @ Khanacademy lectures on Multivariable calculus variable.

Course code: 22IUCI0302T	**Course name: Introduction to Biology**	
Course category: PCC		
Credits: 3	**Teaching scheme:** L-3	**Evaluation scheme:** CA–40, ESE–60
Pre-requisites:		
Course Objectives:		
1.		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Evaluate the Basic Biological Concepts viarel event Industrial Applications and Case Studies.		
CO2: Evaluate the Principles of Designand Development for Exploring Novel Bioengineering Projects.		
CO3: Corroborate the Concepts of Biomimetic for Specific Requirements.		
CO4: Think Critically towards Exploring Innovative Biobased Solutions for Socially Relevant Problems.		

Contents –

Unit	Content	Teaching hours
1	**BiomoleculesandtheirApplications** Carbohydrates(CellulosebasedWaterFilters,PHAandPLAasBioplastics) NucleicAcids(DNAVaccineforRabiesandRNAVaccinesforCovidl9,Forensics,DNAFingerprinting) Proteins(ProteinsasFood, WheyProteinand MeatAnalogs, Plantbased Proteins) Lipids(Biodiesel,CleaningAgents/Detergents) Enzymes(Glucose-oxidaseinBiosensors,LignolyticEnzymeinBio-Bleaching)	12
2	**HumanOrganSystemsandBioDesigns-I** BrainasaCPUSystem(Architecture,CNSandPeripheralNervousSystem,SignalTransmission,EEG,RoboticArmsforProsthetics,EngineeringSolutionsforParkinson's Disease) EyeasaCameraSystem(ArchitectureofRodandConeCells,OpticalCorrections,Cataract,LensMaterials,BionicEye) HeartasaPumpSystem(Architecture,ElectricalSignalling-ECGMonitoringandHeartrelatedIssues,ReasonsforBlockagesofBloodVessels,DesignofStents,PaceMakers,Defibrillators)	12
3	**HumanOrganSystemssndBioDesigns-II** LungsasPurificationSystem (Architecture,GasExchange Mechanisms, Spirometry,AbnormalLungPhysiology- COPD,Ventilators,Heart-LungMachine) KidneyasaFiltrationSystem(Architecture,Mechanism ofFiltration,CKD,DialysisSystems) Muscular and Skeletal Systems as Scaffolds (Architecture, Mechanisms,	11

	Bioengineering solutions for MuscularDystrophyandOsteoporosis)	
4	**TrendsInBioengineering** BioprintingTechniques and Materials 3D printing of Ear,Bone,Skinand Foods Electrical Tongue and Electrical Nose in Food Science DNA Origami and Biocomputing Bioimaging and Artificial Intelligence for Disease Diagnosis Self-Healing Bioconcrete (Based on Bacillus Spores,Calcium Lactate Nutrients and Biornineralization Processes)	**10**

Text Books:

1. "BiologyforEngineers"byThyagarajanS., SelvamuruganN.,RajeshM.P.,Nazeer R.A.,ThilagarajW.,BarathiS.,andJaganthanM.K.ofTataMcGraw-Hill, New Delhi,2012.

2. "BiologyforEngineers"bySohiniSinghandTanuAllenof VayuEducationofIndia,NewDelhi,2014.

Reference Books:

1. "HumanPhysiology"byStuartFox,KristaRompolskiofMcGraw-HilleBook,16thEdition, 2022.

2."BiomedicalInstrumentation"byLeslie CromwellofPrentice Hall2011.

3.BiologyforEngineers"byArthurT. Johnson,CRCPressofTaylorand Francis,2011.

4."Biomimetics: Nature-Based Innovation"byYosephBar-CohenofCRC Press1stEdition, 2012.

5."BioInspiredArtificialIntelligence"byTheories,MethodsandTechnologies,D.Floreanoand C.MattiussiofMIT Press,2008.

6."3DBioprinting: Fundamentals,PrinciplesandApplications"byIbrahimOzbolatofAcademicPress, 2016.

7."Electronic Nosesand TonguesinFood Science"byMariaRodriguezMendeofAcademicPress, 2016.

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: 22IUCI0303T		**Course name: Computer Programming**
Course category: PCC		
Credits: 3	**Teaching scheme:** L-3	**Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Basic Computer Knowledge		
Course Objectives: Understand fundamental programming concepts such as variables, data types, operators, control structures, and functions.		
Develop logical thinking and problem-solving skills to analyze problems and design appropriate algorithms.		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Develop flowchart and algorithm to solve problem logically.		
CO2: Write simple C programs using arithmetic expressions		
CO3: Develop C programs using control structures.		
CO4: Develop C programs using arrays and structures.		
CO5: Develop C programs using Pointers.		

Contents –

Unit	Content	Teaching hours
1	**Fundamentals of C Language :** Introduction to C Language, Importance of C Language, Use of C Language, Application of C Language, .Features of C Language	**4**
2	**Overview of C language :** Introduction to Algorithm, Introduction to flowchart, Basic Structure of C Programming, Tokens in C, Keywords, Identifiers, Format Specifies	**8**
3	**C Language Fundamentals :** Character set, Identifiers, Keywords, Operators, Constants, Variables, Data Types, Statements, Decision Making of C language, If Statement, If else Statement, Nested if Statement, if else if ladder ,.Switch case,, Looping, while loop, do while loop, For Loop	**10**
4	**Arrays and String :** Introduction to Array, Array types, Introduction to string, String functions in C.	**8**
5	**Functions & Pointers:** Function: Concept of Function, User defined Function, System Defined Function, and Types of parameter passing in function. **Pointers:** Need of Pointers, Types of Pointers, Pointer Expression, Arrays of Pointers, Pointers and Function.	**10**
6	**Structure and Unions :** Structure in C, Nested structure in C, Array of structure in C, Introduction To Unions, Difference between Structure and Unions.	**5**

Text Books:

1.C in Depth – Deepali Srivastava & S. K. Srivastava
2.Programming in C — Ashok Kamthane & Amit Kamthane

Reference Books:

1. Title of Book: Programming in ANSI C Author: Balagurusamy. E.
2. Title of Book: Let us C Author: Kanetkar, Yashwant

Online Resources:

1. https://www.tutorialspoint.com
2. https://javatpoint.com
3.https://www.w3schools.com

Course code: 22IUCI0304T **Course name: Web Page Designing with HTML** **Course category:** PCC
Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–40, ESE–60
Pre-requisites: 1.Basics of HTML 2. Web Page Creation 3.Basic Tags of HTML
Course Objectives:
1.To understand basic concepts of website design and web page content.
2. To create lists and apply hyperlinks on text and images.
3. To use HTML block-level and text-level formatting tags effectively.
4. To organize web content using tables and frames.
5. To apply CSS for styling and presentation of web pages.
6. To publish websites on Internet or Intranet.7. Develop a basic static or dynamic website following web design principles and responsive design concepts.
Course Outcomes: At the end of the course, the students will be able to -
CO1: Use of website and content of website designing in web page.
CO2: Use block level and text level formatting tags to present content in web page.
CO3: Arrange list and apply hyperlink on images.
CO4: Organize and arrange content using table and frames.
CO5: Apply presentation schemes on content of web page using CSS.
CO6: Publish website on Internet or Intranet.

Contents –

Unit	Content	Teaching hours
1	**Introduction to Web Design** Introduction of Internet, WWW, Website, Working of Websites, Webpages, Front End, Back End, Client and Server Scripting Languages, Responsive Web Designing, Types of Websites (Static and Dynamic Websites). Editors: Downloading free Editors like Notepad++, Sublime Text Editor, Making use of Editors, File creation and editing, saving. Why to create Web site: use of Website, History & Feature of HTML What is HTML : HTML Documents , Basic structure of an HTML document ,Creating an HTML document	**8**
2	**Introduction to HTML Tag** What is HTML Tag : Use of Tags, Types of Tags, Cascading of tags Block Level tags and Horizontal Rules: Headings. Paragraphs. Line Breaks. Divisions. Text Level Tags And Special Characters: Centered Text, Block Quotes. Preformatted text, types of Address, HR tag Bold Italic. Teletype ,Underline, Strikethrough, Superscript Subscript DIV tag, Displaying special characters	**8**
3	**Lists, URL & Images** Lists: Ordered Lists Unordered Lists. Definition Lists, Nested Lists URL: URL And Anchor Tag:URL Types of URL. Absolute URL, Relative URLS, pros and cons of relative and absolute URLs, Anchor Tag Linking	**8**

	various documents for internal and external links Images, Colors And Backgrounds: Inserting Images, formatting image for sizing, alignment Border and using other attributes with IMG tag Inserting image as page background Creating solid color page background.	
4	**Tables and Frames** Table Table tag with attributes TABLE, TR. TH, TD tags. Border, cell spacing, cell padding, width, align, bgcolor attributes. Frames: Types of Frames with their attributes, Creating frames: FRAMESET tag - rows, cols attributes.	**7**
5	**Cascading Style Sheets** CSS Introduction to CSS, Types of CSS, CSS Selectors: Universal Selector, ID selector, Tag Selector, Class Selector, Sub Selector, Attribute Selector, Group Selector CSS Properties: Back Ground properties, Block Properties, Box properties, List properties, Border Properties, Positioning Properties Data base connectivity, Form designing with database	**7**
6	**Hosting of Website** Concept of Internet and Intranet Publishing website on Intranet. Installing and configuring web server, uploading files on intranet site, access intranet-based website; Publishing website site on Internet, hiring Web space, uploading files using FTP, Virtual Hosting, access internet-based website	**7**

Text Books:
Reference Books: 1. HTML and XHTML- the complete reference – Powell, Thomas – Tata McGraw Hill, New Delhi,2014, ISBN: 9780070701946 2. HTML and CSS – The Complete Reference -- Tata McGraw Hill, New Delhi
Online Resources: 1. https://www.tutorialspoint.com 2. https://javatpoint.com **3.** https://www.w3schools.com

Course code: 22IUCI0305T **Course category:** PCC		**Course name: Basic Electronics**
Credits: 3	**Teaching scheme:** L-3	**Evaluation scheme:** CA–40, ESE–60
Pre-requisites:		
Course Objectives:		
1.		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Identify Electronics componants in Electronics Group.		
CO2: Apply Electronics componants safety		
CO3: Use Relevant Electrical or Electronic Protective Device Safety.		
CO4: Use Various Sensors AndTransducsersCurriculum		

Contents –

Unit	Content	Teaching hours
1	**Electronics Componants & Siganls** A brief history of Electronics, Defination of Semiconductor , Types of Semiconductor Active and Passive Componants Resistor , Inductor , Capacitor Symbol , Colour Code , and Specification Voltage & Current sources, Ideal and Practical Voltage Source and Current Source Signal , Types of Signal (Analog and Digital)	**10**
2	**Diode Circuits** P N Junction Diode : Symbol , Construction, Working and Applications Zener Diode : Symbol, Working ,appliaction , Zener Diode as A Voltage Regulator Rectifier Circuit : Half Wave Rectifier , Full Wave Rectifier, Bridge Wave Rectifier and Related performance parameter Diffeence between HWR & FWR Introduction to Filter : Working of Inductor (L) & Capacitor (C) Filter , Application of Various Filter.	**12**
3	**Electronics Componants & Siganls** A brief history of Electronics, Defination of Semiconductor , Types of Semiconductor Active and Passive Componants Resistor , Inductor , Capacitor Symbol , Colour Code , and Specification Voltage & Current sources, Ideal and Practical Voltage Source and Current Source Signal , Types of Signal (Analog and Digital)	**12**
4	**Diode Circuits** P N Junction Diode : Symbol , Construction, Working and Applications Zener Diode : Symbol, Working ,appliaction , Zener Diode as A Voltage Regulator Rectifier Circuit : Half Wave Rectifier , Full Wave Rectifier,	**11**

	Bridge Wave Rectifier and Related performance parameter Diffeence between HWR & FWR Introduction to Filter : Working of Inductor (L) & Capacitor (C) Filter , Application of Various Filter.	

Electronics Componants & Siganls A brief history of Electronics, Defination of Semiconductor , Types of Semiconductor Active and Passive Componants Resistor , Inductor , Capacitor Symbol , Colour Code , and Specification Voltage & Current sources, Ideal and Practical Voltage Source and Current Source Signal , Types of Signal (Analog and Digital)
Diode Circuits P N Junction Diode : Symbol , Construction, Working and Applications Zener Diode : Symbol, Working ,appliaction , Zener Diode as A Voltage Regulator Rectifier Circuit : Half Wave Rectifier , Full Wave Rectifier, Bridge Wave Rectifier and Related performance parameter Diffeence between HWR & FWR Introduction to Filter : Working of Inductor (L) & Capacitor (C) Filter , Application of Various Filter.

Course code: 22IUCI0306T Course name: Effective Communication and Soft skills **Course category:** PCC
Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–20, ESE–00
Pre-requisites:
Course Objectives:
1.
Course Outcomes: At the end of the course, the students will be able to -
CO1: Student will be able to utilizeeffectivecommunication skills.
CO2: Student will be able to select and apply the appropriate methods of communication in various situations
CO3: Student will be able to develop presentation skills and body language.
CO4: Student will be able touse appropriate words as per the context
CO5: Student will be able todraft correct format of emails and business letters

Contents –

Unit	Content	Teaching hours
1	**Introduction to communication** Definition of communication Process of communication Types of communication (Formal,Informal,Verbal,Nonverbal,Vertical, Horizontal, Diagonal) Principles of communication- seven C's of communication Barriers to communication-Physical Barrier, Mechanical, Psychological, Language.	**06**
2	**Non verbal communication** Introduction to nonverbal communication Aspects of body language (Kinesics) (**Facial expression, eye contact, vocalic, paralanguage,gesture,posture, dress & appearance, haptics**) Non-verbal codes- Proxemics, Chronemics, Artefacts Body language and confidence Use of body language at workplace.	**06**
3	Formal Written Communication Office Drafting- Notice, memo & e-mail. Job application with resume. Business correspondence-(Enquiry letter, order letter ,complaint letter, adjustment letter.) Report writing- Accident report, fall in production, investigation report.	**06**
4	Presentation skills What is Presentation Importanceand types of Presentation. Elements of effective presentation.	**06**

	Tips for effective Presentation	
5	**Listening skills:** Importance of listening skill, Listening Process. Active and Passive listening Barriers to listening Strategies of effective listening.	**04**
6	**Soft skills:** Public Speaking, Interview skills. Group discussion Team work and interpersonal skills	**04**

Text Books:
Reference Books: 1. Text book of Communication Skills. (Sanjay Kumar, PushpaLata) Oxford University Press. 2.Communication Skills (URMILA RAI) Himalaya Publications 3. English for success (E.Sureshkumar, P. Sreehari, J.Savithri) Foundation Books. 4.Effective communication skills (M.AshrafRizvi)
Online Resources: 1. https://in.indeed.com/career-advice/resumes-cover-letters/communication-skills 2. https://www.skillsyouneed.com/ips/communication-skills.html 3.https://corporatefinanceinstitute.com/resources/management/communication/ 4. https://www.coursera.org/articles/communication-skills

Course code: ENV-C **Course category:** PCC		**Course name: Environmental Studies**
Credits: 3	**Teaching scheme:** L-3	**Evaluation scheme:** CA–40, ESE–60
Pre-requisites:		
Course Objectives:		
1.		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Understand the need of Environment Study		
CO2: Analyze the use of resources		
CO3: Understand the concept of an ecosystem		
CO4: Understand the concept ofBiodiversity and its conservation		
CO5: Analysis the concept of Pollution		
CO6: Analysis is the Social Issues and the Environment		

Contents –

Unit	Content	Teaching hours
1	**The multidisciplinary nature of environmental studies** Definition, scope and importance Need for public awareness	**2**
2	**Natural Resources:** Natural resources and associated problems. (a) Forest resources: Use and over-exploitation, deforestation, case studies. Timber extraction, mining, dams and their effects on forests and tribal people. (b) Water resources: Use and over-utilization of surface and ground water, floods, drought, conflicts over water, dams-benefits and problems. (c) Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies. (d) Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies. (e) Energy resources: Growing energy needs, renewable and non-renewable energy sources, use of alternate energy sources, case studies. (f) Land resources: Land as a resource, land degradation, man induced landslides, soil erosion and desertification. · Role of an individual in conservation of natural resources. · Equitable use of resources for sustainable lifestyles.	**8**

3	**Ecosystems** Concept of an ecosystem · Structure and function of an ecosystem · Producers, consumers and decomposers · Energy flow in the ecosystem · Ecological succession · Food chains, food webs and ecological pyramids · Introduction, types, characteristic features, structure and function of the following ecosystem: a. Forest ecosystem b. Grassland ecosystem c. Desert ecosystem d. Aquatic ecosystems (ponds, streams, lakes, rivers, ocean estuaries	**6**
4	**Biodiversity and its conservation** Introduction – Definition: genetic, species and ecosystem diversity · Biogeographical classification of India · Value of biodiversity: consumptive use, productive use, social, ethical aesthetic and option values · Biodiversity at global, national and local levels · India as a mega-diversity nation · Hot-spots of biodiversity · Threats to biodiversity: habitat loss, poaching of wildlife, man wildlife conflicts · Endangered and endemic species of India · Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity	**8**
5	**Environmental Pollution** Definition · Causes, effects and control measures of: a. Air pollution b. Water pollution c. Soil pollution d. Marine pollution e. Noise pollution f. Thermal pollution g. Nuclear pollution · Solid waste management: Causes, effects and control measures of urban and industrial wastes. · Role of an individual in prevention of pollution · Pollution case studies · Disaster management: floods, earthquake, cyclone and landslides	**8**

6	**Social Issues and the Environment** From unsustainable to sustainable development · Urban problems and related to energy ·Water conservation, rain water harvesting, watershed management · Resettlement and rehabilitation of people; its problems and concerns. Case studies. · Environmental ethics: Issues and possible solutions · Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Case studies. · Wasteland reclamation · Consumerism and waste products · Environmental Protection Act · Air (Prevention and Control of Pollution) Act · Water (Prevention and control of Pollution) Act · Wildlife Protection Act · Forest Conservation Act · Issues involved in enforcement of environmental legislation · Public awareness	8

Text Books:
Reference Books: 1.Basic Environmental Sciences- Michael Allaby 2. Environmental Sciences- Y.K. Singh 3. Environmental Studies- Rajgopalan.
Online Resources: www.cpcb.nic.in

Course code: 22IUCI0303L **Course name: Computer Programming Lab** **Course category:** PCC	
Credits: 02 **Teaching scheme:** P-4	**Evaluation scheme:** TW–60, PR–40
Pre-requisites:1	
Laboratory Outcomes: At the end of the session, the students will be able to -	
LO1: Develop Flowchart and Algorithm to solve problems logically.	
LO2: Write simple C programs using arithmetic Expressions.	
LO3: Develop C Programs using control Structures.	
LO4: Use functions in C Program.	
LO5: Develop Programs using Pointers.	

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1	To understand the concept of Algorithm and Flowchart.	2
2	Develop C Program using operators.	2
3	Develop program using increment/decrement operator.	2
4	Develop C Program to print ASCII value of character.	2
5	Develop C program to find whether the entered number is positive negative or zero using Else if	2
6	Write a program to find whether the given year is leap year or not.	2
7	C Program to Find the greatest of 3 numbers.	2
8	Develop C program to print Half Pyramid of Number.	2
9	Using switch statement print day of week by taking number from 1 to 7.	2
10	Write a program to find sum of digits of given number.	2
11	Develop a C Program using While loop	2
12	Prepare a program using concept of structure.	2
13	Develop a program using the concepts of pointers.	2
14	Write a program to Print student grade by accepting student's percentage.	2
15	Develop a program to sort 10 numbers	2
16	Program to find largest element in array	2
17	C Program to calculate average of all elements present in array	2
18	C Program to find length of String	2

Reference Books 1. Title of Book: Programming in ANSI C Author: Balagurusamy. E.
2. Title of Book: Let us C Author: Kanetkar, Yashwant

Online Resources: https://www.tutorialspoint.com
https://javatpoint.com
https://www.w3schools.com

Course code: 22IUCI0304L	**Course name: Web Page Designing with HTML Lab**	
Course category:PCC		
Credits: 02	**Teaching scheme:** P-4	**Evaluation scheme:** TW–60, PR–40
Pre-requisites:1		
Laboratory Outcomes: At the end of the session, the students will be able to -		
LO1: Use of website and content of website designing in web page..		
LO2: Use block level and text level formatting tags to present content in web page.		
LO3: Arrange list and apply hyperlink on images.		
LO4: Organize and arrange content using table and frames.		
LO5: Apply presentation schemes on content of web page using CSS.		
LO6: Publish website on Internet or Intranet		

List of Experiments–

Sr.No.	Name of Experiment	Pratical Hours
1	Install any one editor and display hello message on screen.	2
2	Create a web page using Block Level tags and HR tags.	2
3	Create a web page using Text Level tags and display special character in web page.	2
4	Create a web page for implementation of different types of list.	2
5	Create a web page for inserting image in web page and change attribute of image.	2
6	Create a web page to link: a) Different web page of same web site b) Different web page of different web site **c)** Images	2
7	Create a web page for implementation of table tags	2
8	Create a web page for implementation of frame tags	2
9	Create a web page by implementation of different selector	2
10	Create a web page by using different CSS properties	2
11	Create a web page to create different forms: a) Login form b) Student Entry From **c)** Forms with linking of different tags	2
12	Create above web page and connect with database	2
13	Create a web page for multimedia a) Audio b) Viedo c) Source	2
14	Install a web server for publishing website	2
15	Publish a web site on internet by using free hosting site.	2
16	Mini Project	2

Reference Books:

1. HTML and XHTML- the complete reference – Powell, Thomas – Tata McGraw Hill, New Delhi, 2014, ISBN: 9780070701946
2.HTML and CSS – The Complete Reference -- Tata McGraw Hill, New Delhi

Online Resources:

1.https://www.tutorialspoint.com
2.https://javatpoint.com
3.https://www.w3schools.com

ourse code: 22IUCI0305L	**Course name:Basic Electronics Lab**
Course category:PCC	
Credits: 01	**Teaching scheme:** P-2 **Evaluation scheme:** TW–30, PR–20
Pre-requisites:1	
Laboratory Outcomes: At the end of the session, the students will be able to -	
LO1: Test Various Active and Passive Componants.	
LO2: Build Simple and Functional Circuits on Breadboard.	
LO3: Build Transistor and Amplifier Circuit.	
LO4: Use Various Sensors And Transducsers.	

List of Experiments–

Sr.No.	Name of Experiment	Pratical Hours
1	Introduction to Various Equipment in the Laboratory	2
2	Measure Voltage And Frequency With Cathod Ray Oscilloscope (CRO)	2
3	Use of Analog and Digital Multimeter to measure Voltage and Current.	2
4	Identify Various Active and Passive componants in given circuit.	2
5	Use of Multimeter to measure value of Various componants.	2
6	Use of LCR-Q Meter to measure value of Inductor (L) and Capacotor .	2
7	Test P N Junction Diode using DMM (Digital Multi Meter)	2
8	Test Performane of N-P-N Transistor.	2
9	Identify Three Terminals of Transistor using Digital Multi Meter.	2
10	Build Half Wave Rectifier on Bread Board.	2

Text Books:

1 ."Basic Electronics " by Dr. V K Mehta Books.
2."Digital Electronics and Logic Design" by Dr. Adsul A.P. and Jagdale R.H. of Nirali Publication.
3."Digital Electronics" by J.S. Katre of Tech-Max Publication, Pune.

Reference Books: 1. "Modern Digital Electronics" by Jain R.P. of Tata McGraw Hill Publication, New Delhi.
2. www. Electrical & Electronics for u
3. www.Electronics-Tutorials
4. www.wikipedia.com

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: 22IUCI0306L	**Course name: Effective Communication and Soft skills Lab**

Course category: PCC		
Credits: 01	**Teaching scheme:** P-2	**Evaluation scheme:** TW–30, PR–00
Pre-requisites:1		
Laboratory Outcomes: At the end of the session, the students will be able to -		
LO1: Use of website and content of website designing in web page.		
LO2: Use block level and text level formatting tags to present content in web page.		
LO3: Arrange list and apply hyperlink on images.		
LO4: Organize and arrange content using table and frames.		
LO5: Apply presentation schemes on content of web page using CSS.		
LO6: Publish website on Internet or Intranet		

List of Experiments–

Sr.No.	Name of Experiment	Practial Hours
1	Explain the importance of the business communication.	2
2	Explain the importance of body language at workplace.	2
3	Case study on nonverbal and verbal communication	2
4	Prepare guidelines for presentation.	2
5	Design power point presentation on any topic.	2
6	Prepare interview questionnaire.	2
7	Organize debate on given topic.	2
8	Deliver speech on the given topic. (Welcome speech and farewell speech)	2
9	Prepare job application with resume.	2
10	Draft a complaint letter on the given topic.	2

Reference Books:

1. Text book of Communication Skills. (Sanjay Kumar, PushpaLata) Oxford University Press.
2. Communication Skills (URMILA RAI) Himalaya Publications
3. English for success (E.Sureshkumar, P. Sreehari, J.Savithri) Foundation Books.

Online Resources:

1. https://in.indeed.com/career-advice/resumes-cover-letters/communication-skills
2. https://www.skillsyouneed.com/ips/communication-skills.html
3.https://corporatefinanceinstitute.com/resources/management/communication/
4. https://www.coursera.org/articles/communication-skills

SEM-IV

Course code: 22IUCI0401T **Course name: Discrete Mathematics** **Course category:** PCC
Credits: 3 **Teaching scheme:** L-3 T-1 **Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Basic Mathematics
Course Objective:
1.To develop a strong foundation in logical reasoning and mathematical thinking
2. To develop a clear understanding of relations functions, recurrence relations their properties and types.
3.To develop an understanding of number system
4. To develop the ability to apply fundamental counting techniquesto solve combinatorial and mathematical problems effectively.
5. To develop the ability to understand and apply fundamental concepts of graphs and trees.
6. To develop understanding of algebraic system to solve mathematical problems.
Course Outcomes: At the end of the course, the students will be able to -
CO1: apply principles of logic, set theory, and proof techniques, to solve problems systematically.
CO2 : analyze mathematical structures by applying concepts of relations, functions, and recurrence relations
CO3 : apply number theory concepts to solve mathematical and computational problems**.**
CO4 : solve combinatorial and probability-based problems by applying counting techniques
CO5 : analyze and apply graph and tree conceptsto solve problems and implement algorithms
CO6 : apply the concepts of algebraic systems to solve mathematical

Contents –

Unit	Content	Teaching hours
1	**Set Theory, Logic and Proofs** Set Theory, Logic and Proofs: Propositions, Conditional Propositions, Logical Connectivity, Propositional calculus, Universal and Existential Quantifiers, First order logic, Proofs: Proof Techniques, MathematicalInduction. Set, Combination of sets, Finite and Infinite sets, Un- countably infinite sets, Principle of inclusion and exclusion, strong Induction	**10**
2	**Relations, Functions, Recurrence Relations** Definitions, Properties of Binary Relations, Equivalence Relations and partitions, Partial ordering relations and lattices, Chains and Anti chains. Theorem on chain, Warshall's Algorithm & transitive closure, Recurrence relations. Functions: Definition, Domain, Range, Image, etc. Types of functions: Surjection, Injection, Bijection, Inverse, Identity, Composition of Functions.	**10**
3	**Number Theory** Basics of Modulo Arithmetic, Basic Prime Number Theory, GCD, LCM, Divisibility, Euclid's algorithm, Factorization, Chinese Remainder Theorem Fields: Naturals, Integers, Rationales, Real's, Complex Numbers Properties of operations: associative, commutative, distributive, identity, inverse	**10**
4	**Counting**	**10**

	Basic Counting Techniques (sum, product, subtraction, division, exponent), Pigeonhole and Generalized Pigeonhole Principle with many examples, Permutations and Combinations and numerical problems ,Binomial Coefficients Pascal's, Identity and Triangle, Generating Permutations and Combinations.	
5	**Graphs& Trees** Basic terminology, multi graphs and weighted graphs, paths and circuits, Shortest path Problems, Euler and Hamiltonian paths and circuits, factors of a graph, planar graph and Kuratowski's graph and theorem, independent sets, graph coloring. Trees, rooted trees, path length in rooted trees, binary search trees, spanning trees and cut set, theorems on spanning trees, cut sets ,circuits, minimal spanning trees, Kruskal's and Prim's algorithms for minimal spanning tree.	**12**
	Graphs& Trees Basic terminology, multi graphs and weighted graphs, paths and circuits, Shortest path Problems, Euler and Hamiltonian paths and circuits, factors of a graph, planar graph and Kuratowski's graph and theorem, independent sets, graph coloring. Trees, rooted trees, path length in rooted trees, binary search trees, spanning trees and cut set, theorems on spanning trees, cut sets ,circuits, minimal spanning trees, Kruskal's and Prim's algorithms for minimal spanning tree.	**12**

Text Books: 1. Schaum's Outlines Series, Seymour Lipschutz, MarcLipson, 2. C. L. Liu and D P Mohapatra, "Elements of Discrete Mathematics A Computer Oriented Approach," 3rd Edition by, Tata McGraw –Hill.
Reference Books: 1. Kenneth H. Rosen, "Discrete Mathematics and Its application", Seventh Edition by Tata McGraw –Hill.
Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: 22IUCI0402T Course name: Object Oriented Programming using C++ **Course category:** PCC
Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Basics of programming concept, Basics of DS, Fundamental of Procedure Oriented , Basics of DMS
Course Objectives:
1. To develop a foundational understanding of object-oriented programming concepts using C++
2. To develop the understanding for designing and implementation of classes and objects
3. To develop the ability to understand reusability concept using inheritance in OOP
4. To develop the ability to use pointers and implement polymorphism for designing efficient object-oriented programs.
5. To develop the ability to perform file operations in C++
6. To develop the ability to handle runtime errors using exception handling techniques
Course Outcomes: At the end of the course, the students will be able to -
CO1: understand and apply object-oriented programming concepts using C++ fundamentals
CO2: design, implement, and manipulate classes and objects
CO3: analyse, implement inheritance to design reusable and hierarchical object-oriented programs.
CO4: apply pointers and implement polymorphism to design efficient object-oriented programs.
CO5: use C++ stream classes to open, read, write, and manage files efficiently in object-oriented programs.
CO6: apply exception handling concept to detect, manage, and recover from runtime errors effectively.

Contents –

Unit	Content	Teaching hours
1	**Principles of Object oriented Programming :** Procedure Oriented Programming verses Object Oriented Programming (OOP), basic Concept of OOP, Application of OOP,C verses C++, Structure of C++ Program, Tokens, Keywords, variables, Constants, Built in data Types, User defined data Types, Typecasting, Operators, Expressions, Control Structures: Decision making statements and Loops, scope resolution operator, memory management operator, Array, Strings and Structures in C++,	**6**
2	**Classes and Object :** Class , Object, access specifies, Defining member Functions, Creating Objects, memory allocation for Objects, Static data members & function , friend function, Array of Objects, Object as function arguments, Concepts of Constructors. Types of constructors, Destructors	**8**
3	**Inheritance :** Introduction to Inheritance, defining derived class, visibility modes and effects, Types of Inheritance: Single ,multilevel, multiple, hierarchical, hybrid. Virtual base class, abstract class. Constructors in derived class. Function overriding.	**8**

4	**Polymorphism & Pointers :** Introduction of Polymorphism, Types of Polymorphism, Compile time Polymorphism Function overloading. Operator overloading, overloading of unary and binary operators, Rules for operator overloading. Run time polymorphism, Virtual functions, rules for virtual functions, pure virtual function. Concepts of Pointer: Pointer Declaration, Printer operator, address operator, Pointer arithmetic, Pointer to Object: Pointer to Object. this pointer, Pointer to derived class	**8**
5	**Working with Files :** C++ stream classes, Classes for file stream operations, Opening files, closing files reading from and writing to files, Detection of end of file, file modes	**8**
6	**Exception Handling and Templates :** Exception Handling: Fundamentals, other error handling techniques, simple exception handling- Divide by Zero, Multiple catching, re-throwing an exception, exception specifications, user defined exceptions, processing unexpected exceptions, constructor, destructor and exception handling, exception and inheritance, Templates: The Power of Templates, Function template, overloading Function templates, and class template, class template and Non type parameters, template and friends Generic Functions, The type name and export keywords.	**7**

Text Books:

1. Deitel, "C++ How to Program", 4th Edition, Pearson Education, ISBN:81-297-0276-
2. Robert Lafore, "Object-Oriented Programming in C++", fourth edition, Sams Publishing, ISBN: 0672323087 (ISBN 13: 9780672323089

Reference Books:

1. Herbert Schildt, "C++-The complete reference", Eighth Edition, McGraw Hill Professional, 2011, ISBN:978-00-72226805
2. Matt Weisfeld, "The Object-Oriented Thought Process", Third Edition Pearson ISBN-13:075-2063330166
3. E.Balagurusamy, "Object-Oriented Programming with C++", 7th edition, Graw-Hill Publication, ISBN 10: 9352607996 ISBN 13: 9789352607990
4. Cox Brad, Andrew J. Novobilski, "Object-Oriented Programming: An Evolutionary Approach", Second Edition, Addison-Wesley, ISBN: 13:978-020-1548341

Online Resources:

1.https://nptel.ac.in/courses/106/105/106105151/
2. https://swayam.gov.in/nd1 noc20 cs07/preview
3. https://www.classcentral.com/course/swayam-programming-in-c-6704.

Course code: 22IUCI0403T **Course category:** PCC		**Course name: Computer Graphics**
Credits: 3	**Teaching scheme:** L-3	**Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Basic Programming Concepts, Computer Fundamentals		
Course Objectives: **Introduce the fundamental concepts of computer graphics**, including graphics systems, display devices, and graphics software		
Understand mathematical foundations required for computer graphics such as 2D and 3D coordinate systems, vectors, and matrix transformations.		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Manipulate visual and geometric information of images.		
CO2: Implement standard algorithms to draw various graphics objects using C program		
CO3: Develop programs for 2D and 3D Transformation		
CO4: Implement various clipping algorithm		
CO5: Develop programs to create curves.		

Contents –

Unit	Content	Teaching hours
1	**Basics of computer Graphics** Images and Objects, pixel and resolution, Text mode, Graphics mode, Basic graphic pipeline. Bitmap and vector based graphics, Application of computer graphics. Display devices: Raster scan displays, random scan display, flat panel display, LED, LCD display, plasma, touch screen. Output Primitives Graphics functions and standards. Latest trends in computer Graphics.	**8**
2	**Raster Scan graphics** Basic concept in line drawing Circle generating Algorithm Polygons-types of Polygon Scan conversions, Frame Buffer Character generation methods.	**6**
3	**Overview of transformations** Two Dimensional Transformations: Translation, Scaling, Rotation. Reflection, Shearing Matrix representations and homogeneous coordinates: Translation Scaling, Rotation, Reflection, shearing. Composite Transformations-Rotation about arbitary point. Three Dimensional transformation: Translation, Scaling, Rotation Types of Projections: Perspective and Parallel Projection	**8**

4	**Windowing and clipping** Windowing and clipping concepts: Window to viewport transformation Line clipping: Cohen Sutherland clipping algorithm. Cyrus Beck. Liang Barsky, Midpoint subdivision Polygon clipping: Sutherland –Hodgman Text clipping	**8**
5	**Introduction to curves** Curve generation: Arc generation using DDA algorithm, Interpolation Types of curves: Hilbert's Curve, Koch curve, B-Spline, Bezier curves.	**8**
6	**Animation and virtual reality** Basic Principles of Animation and Types of Animation Introduction to the flash interface Layers & Views Animation -Principles , Frame by frame animation, tweening, masks Introduction to virtual reality	**7**

Text Books: Computer Graphics: Donald Hearn, Baker M. Pauline Computer Graphics: Mourya Rajesh K.
Reference Books: Computer Graphics: Dr.Chopra Rajiv
Online Resources: **1.** https://www.tutorialspoint.com 2. https://javatpoint.com 3. https://www.w3schools.com

Course code: 22IUCI0404T **category:** PCC		**Course name: Digital Electronics Course**
Credits: 3	**Teaching scheme:** L-3	**Evaluation scheme:** CA–40, ESE–60
Pre-requisites:		
Course Objectives:		
1.		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Understand Basics of Digital System.		
CO2: Use Number System and Codes for Problem Solving of Digital Electronic System.		
CO3: Use for Logic Gates and Families to Construct Logic Circuits.		
CO4: Build Simple and Functional Combinational Logic Circuits.		
CO5: Build Simple and Functional Sequential Logic Circuits.		
CO6: Test Data Convertors and PLD's in Digital Electronics Systems.		

Contents –

Unit	Content	Teaching hours
1	**Introduction to Digital Techniques** Signal - Overview of Analog and Digital Signal System - Defination, Characteristics, Classification of Various Systems Digital Systems - Positive and Negative Logic, Representation, Characteristics, Advantages, Disadvantages and Applications of Digital Systems, Digital Computers	6
2	**Number System and Code** Number System - Classification - Binary, Octal, Decimal, Hexadecimal Number System, Conversion of Number Systems, 1's Complement and 2's Complement Binary Arithmetic - Addition, Subtraction, Multiplication, Division Codes - BCD Code, BCD Addition, BCD Subtraction using 9's and 10's complement, Gray Code, Excess-3, ASCII Code, Hamming Code	6
3	**Logic Gates and Families** Basic Gates, Universal Gates, Derived Gates - Logical Symbol, Logical Expression and Truth Table of Gates- Deriving all gates using Universal Gates, Multiple Input Gates (3 - Inputs**)** **Boolean Algebra: Fundamentals of Boolean Laws** **Duality Theorem, De Morgan's Theorem** Logic Families - Characteristics of Logic Families, Noise Margin, Power Dissipation, Figure of Merit, Fan In, Fan Out, Speed of Operation, Operating Temperatures, Power Supply Requirement& Comparison between Different Logic Families, Overview	6

4	**Combinational Logic Circuits** Standard Boolean Representation - Sum of Product (SOP) and Product of Sum (POS), Min-Term and Max-Term, Conversion between SOP and POS Forms, Realization using NAND / NOR Gates K-Map Reduction Technique for the Boolean Expression - Minimization of Boolean Functions up to 4 Variables (SOP and POS Form) Design of Arithmetic Circuits and Code Converter using K-Map - Half and Full Adder, Half and Full Subtractor, Gray to Binary and Binary to Gray (up to 4 bits) Arithmetic Circuits - (IC 7483) Adder and Subtractor, BCD Adder Encoder / Decoder: Basics of Encoder, Decoder, Comparison, (IC 7447) BCD to 7 Segment Decoder / Driver Multiplexer and Demultiplexer - Working , Truth Table and Applications of Multiplexers and Demultiplexures, MUX Tree, IC 74151 as MUX; DEMUX Tree, DEMUX as Decoder, IC 74155 as DEMUX Buffer - Tristate Logic, Unidirectional and Bidirectional Buffer (IC741,S244,74LS245)	**8**
5	**Sequential Logic Circuits** Basic Memory Cell - RS Latch using NAND and NOR Triggering Methods - Edge Trigger and Level Trigger SR Flip Flops - SR Flip Flop, Clocked SR Flip Flop with Preset and Clear, Drawbacks of SR Flip Flop JR Flip Flops - Clocked JK Flip Flop with Preset and Clear, Race Around Condition in JK Flip Flop, Master Slave JK Flip Flop, D and T Type Flip Flop Excitation Table of Flip Flops, Block Schematic and Function Table of IC-7474, 7475 Shift Register - Logic Diagram of 4 Bit Shift Registers, Serial Input Serial Output, Serial Input Parallel Output, Parallel Input Serial Output, Parallel Input Parallel Output, 4 Bit Universal Shift Register Counters - Asynchronous Counter: 4 Bit Ripple Counter, 4 Bit Up / Down Counter, Modulus of Counter, Synchronous Counter: Design of 4 Bit Synchronous Up / Down Counter, Decade Counter: Block Schematic of IC 7490 Decade Counter, IC 7490 as MOD-N Counter, Ring Counter, Twisted Ring Counter	**8**

6	**Data Convertors and PLD's** Data Converter - DAC: Types, Weighted Resistor Circuit and R-2R Ladder Circuit, DAC IC 0808 Specifications, ADC: Block Diagram, Types and Working of Dual Slope ADC, SAR ADC, ADC IC 0808 / 0809, Specification Memory - RAM and ROM Basic Building Blocks, Read and Write Operation ,Types of Semiconductor Memories PLD: Basic Building Blocks and Types of PLDs, PLA, PAL, GAL CPLD: Basic Building Blocks, Functionality	**6**

Text Books:
1. "Digital Electronics and Logic Design" by Dr. Sanjay Sharma of KATSON Books.
2. "Digital Electronics and Logic Design" by Dr. Adsul A.P. and Jagdale R.H. of Nirali Publication.
3. "Digital Electronics" by J.S. Katre of Tech-Max Publication, Pune.

Reference Books:
1."Modern Digital Electronics" by Jain R.P. of Tata McGraw Hill Publication, New Delhi.
2. "Digital Circuits and Design" by Salivahan S. and Arivazhagan S. of Vikas Publication House, Delhi.
3. "Digital Principles" by Malvino A.P. and Leach D.P. of Tata McGraw Hill Publication, New Delhi.
4.”Digital Electronics, Principles and Integrated Circuits" by Anil K. Maini of Wiely India Edition.
5. "Digital Fundamentals" by Thomas Floyd of Pearson Publication

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: 22IUCI0402L Course name: Object Oriented Programming using C++ Lab **Course category: PCC**		
Credits: 02	**Teaching scheme:** P-4	**Evaluation scheme:**, TW–60, PR–40
Pre-requisites:1		
Laboratory Outcomes: At the end of the session, the students will be able to -		
LO1: Develop C++ Programs to solve problem using Procedure Oriented Approach.		
LO2: Develop C++ Programs using classes and Objects.		
LO3: Implement Inheritance in C++ Program		
LO4: Use Polymorphism in C++ Program		
LO5: Develop C++ Programs to perform file Operations.		
LO6: Use Exception Handling and Templates in C++ Program		

List of Experiments–

Sr.No.	Name of Experiment
1	Develop minimum Two Programs using constants, variables, arithmetic expression, operators, exhibiting data type conversion.
2	Develop a program to implement decision making statements(if-else, switch)
3	Develop program to implement to demonstrate control structures(for, while, do-while)
4	Develop program to implement 1-Dimensional array
5	Develop program to perform matrix operations using multi-Dimensional array
6	Develop programs that implements a class and use it with Objects
7	Develop programs that implements a class and create array of Objects.
8	Write a Program to implement friend Function.
9	Write a Program to implement inline Function
10	Write a Program to implement all types of constructors (constructor overloading) with destructor
11	Write a Program for implementing single inheritance.
12	Write a Program for implementing Multi-level inheritance.
13	Write a Program for implementing multiple inheritances.
14	Develop minimum 1 Program to Demonstrate pointer to Object.
15	Develop minimum 1 Program to Demonstrate pointer to Derived Class.
16	Write a Program to Demonstrate operator overloading for Unary operator
17	Write a Program to Demonstrate operator overloading for Binary operator.
18	Write a Program to Demonstrate function Overloading.
19	Write a Program to read and write data to and from file

Text Books: 1. Deitel, "C++ How to Program", 4th Edition, Pearson Education, ISBN:81-297-0276-2
2. Robert Lafore, "Object-Oriented Programming in C++", fourth edition, Sams Publishing, ISBN:0672323087 (ISBN 13: 9780672323089

Reference Books: 1. Herbert Schildt, "C++-The complete reference", Eighth Edition, McGraw Hill Professional, 2011, ISBN:978-00-72226805
2. Matt Weisfeld, "The Object-Oriented Thought Process", Third Edition Pearson ISBN-13:075-2063330166
3. E.Balagurusamy, "Object-Oriented Programming with C++", 7th edition, Graw-Hill Publication, ISBN 10: 9352607996 ISBN 13: 9789352607990
4. Cox Brad, Andrew J. Novobilski, "Object-Oriented Programming: An Evolutionary Approach", Second Edition, Addison-Wesley, ISBN:13:978-020-1548341

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: 22IUCI0403L **Course name: Computer Graphics Lab** **Course category:**Program Core Course
Credits: 01 **Teaching scheme:** P-2Hrs/Week **Evaluation scheme:**CA–0, MSE–0, ESE–0, TW–30, PR–20
Pre-requisites:1
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Manipulate visual and geometric information of images.
LO2: Implement standard algorithms to draw various graphics objects using C program
LO3: Develop programs for 2D and 3D Transformation
LO4: Implement various clipping algorithm
LO5: Develop programs to create curves

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1	Write a program to draw following graphics object using build in c function: i)pixel ii) Lines iii)circle iv)Rectangle v) Ellipse	2
2	Implement DDA algorithm to draw line.	2
3	Implement Bresennham's algorithm to draw line.	2
4	Writhe a program to fill polygon using Flood fill method	2
5	Writhe a program to fill polygon using Boundary fill method	2
6	Write a program in C to perform two dimensional transformation i)Translation ii)Scaling	2
7	Write a program in C to perform two dimensional transformation(Rotation)	2
8	Write a program to clip line using Cohen-Sutherland algorithm.	2
9	Write a program to draw Hilbert's Curve	2
10	Write a program to draw Koch curve and Bezier curves.	2

Text Books: 1. Computer Graphics: Donald Hearn, Baker M. Pauline 2. Computer Graphics: Mourya Rajesh K.
Reference Books: 1.Computer Graphics: Dr.Chopra Rajiv
Online Resources: 1. NPTEL / SWAYAM lectures.

Course code: 22IUCI0404L **Course category: PCC**		**Course name: Digital Electronics Lab**
Credits: 01	**Teaching scheme:** P-2	**Evaluation scheme:**TW–50, PR–00
Pre-requisites:1		
Laboratory Outcomes: At the end of the session, the students will be able to -		
LO1: Understand Basics of Digital System.		
LO2: Use Number System and Codes for Problem Solving of Digital Electronic System.		
LO3: Use for Logic Gates and Families to Construct Logic Circuits.		
LO4: Build Simple and Functional Combinational Logic Circuits.		
LO5: Build Simple and Functional Sequential Logic Circuits.		
LO6: Test Data Convertors and PLD's in Digital Electronics Systems.		

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1	Introduction to Digital Techniques and Its Applications.	2
2	Test the Functionality of Basic Logic Gates.	2
3	Test the Functionality of Universal and Derived Logic Gates.	2
4	Verify Duality Theorem and De Morgan's Theorem.	2
5	Design and Verification of the Truth Tables of Half and Full Adder Circuits.	2
6	Design and Verification of the Truth Tables of Half and Full Subtractor Circuits	2
7	Test the Function of Multiplexer IC 74150/74151 and Demultiplexer IC 74154/74155 or any other equivalent.	2
8	Realize and Verify SR Flip Flop.	2
9	Realize and Verify Master Slave JK Flip-Flop.	2
10	Implement 4 Bit Ripple Counter.	2
11	Implement 4 Bit Universal Shift Register.	2
12	Implement 4 Bit R-2R D / A Converter.	2

Text Books: 1. "Digital Electronics and Logic Design" by Dr. Sanjay Sharma of KATSON Books.
2. "Digital Electronics and Logic Design" by Dr. Adsul A.P. and Jagdale R.H. of Nirali Publication.
3. "Digital Electronics" by J.S. Katre of Tech-Max Publication, Pune.

Reference Books: 1. "Modern Digital Electronics" by Jain R.P. of Tata McGraw Hill Publication, New Delhi.
2. "Digital Circuits and Design" by Salivahan S. and Arivazhagan S. of Vikas Publication House, Delhi.

Online Resources: 1. www.cse.yorku.ca/~mack/1011/01.NumberSystems.ppt
2.www.people.sju.edu/~ggrevera/arch/slides/binary-arithmetic.ppt
3.www.mathsisfun.com/binary-number-system.html

SEM-V

Course code:22IUCI0501T **Course category:** PCC		**Course name: Data Structure and Algorithm**
Credits: 3	**Teaching scheme:** L-3	**Evaluation scheme:** CA–40, ESE–60
Pre-requisites:		
Programming Language such as: C, C++		
Course Objectives:		
1. To assess how the choice of data structures and algorithm design methods affects the performance of programs.		
2. To choose the appropriate data structure and algorithm design method for a specified application.		
3. To solve problems using data structures such as linear lists, stacks, queues, hash tables, binary trees, heap s, tournament trees, binary search trees, and graphs and writing programs for these solutions.		
4. To solve problems using algorithm design methods such as the greedy method, divide and conquer, dynamic programming, backtracking, branch and bound and writing programs for these solutions.		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Apply knowledge of basic programming constructs to design algorithms and programs.		
CO2: Analyze clearly different possible solutions to a program and select the most efficient one. (for eg, Sorting and Searching)		
CO3: Demonstrate functioning of linear data structures (stack, queue and linked list) and apply them in required applications		
CO4: Write neat code by selecting appropriate data structure and demonstrate a working solution for a given problem		
CO5: Distinguish between linear and non-linear data structures and determine their applicability		
CO6: Analyze performance of algorithms by determining time and space complexity.		

Contents –

Unit	Content	Teaching hours
1	**Introduction to Data Structures** Introduction to Data & Data Structures, types of Data Structure -Primitive & Non- Primitive, Arrays, Types of Arrays, Memory representation of Array, traversing linear array, inserting and deleting elements in array. Structures, Functions, Recursive Functions, Introduction to time & space complexity.	**8**
2	**Searching, Sorting and Hashing** Searching – Linear Search and Binary Search.Sorting – Bubble sort, Selection sort, Insertion sort, Hashing: Hashing (Hash Function), Hash Collision, Collision resolution: Chaining Open Addressing, Linear Probing, Quadratic Probing, Double Hashing.	**6**
3	**Stacks and Queues** Stacks: Definition, Array Representation of stacks. Evaluation of a Postfix Expression, Transforming Infix into Postfix expression. Queues: Definition, Array Representation of Queues, Types of Queues: Circular Queues, Double ended Queue, Priority Queue, Stack using two queues, Queue using two stack.	**8**

4	**Linked List** Linked List: Representation of Linked Lists in memory, Insertion & deletion from Linked List, Traversing & Searching a Linked List, Types of Linked List: Circular Linked List and Doubly Linked Lists, Stack and Queue Implementation using Linked List, Addition of two Polynomials using Linked List.	**8**
5	**Tree Data Structures** Introduction to Tree and Terminologies , Binary Trees, Properties of Binary Trees, Array and liked representation of Binary Trees, Traversals-In order, Post order, Preorder, Binary Search trees- Definition, Insertion, Deletion, Traversal, and Searching. Evaluation of Expression, AVL Trees.	**8**
6	**Graph** Definitions, Terminologies, Matrix and adjacency list representation of graphs. Traversal Methods: Breadth First Search and Depth First Search, Minimum Spanning Tree, Prim's & Krushal Algorithm.	**7**

Text Books: 1. "Data Structures with C", S. Lipschutz, Mc Graw Hill.2010
2. " Schaum's Outline of Data Structures", Lipschutz, S., McGraw-Hill
3. "Fundamentals of Data Structures in C++", E. Horowitz, Sartaj Sahni, D.Mehta, Universities Press.1995
4. "Data Structures using C", E. Balaguruswamy, McGraw Hill 2013
Reference Books: 1. "Data Structures using C" Aarron M. Tenenbaum, Yedidyah Langsam,Moshe J. Augensten , PHP
Online Resources: 1. https://archive.nptel.ac.in/courses/106/106/106106127/
2. https://nptel.ac.in/courses/106102064

Course code:22IUCI0502T **Course name:** Java Programming **Course category:** PCC
Credits: 3 Teaching scheme: L-3 Evaluation scheme: CA–40, ESE–60
Pre-requisites: Computer Fundamentals, Programming fundamentals, Basics of problem solving and algorithms, Basics of OOP, Data Structure.
Course Objectives:
1**.** To understand the fundamentals of Java programming and its programming environment.
2. To apply object-oriented programming concepts for developing Java applications.
3. To design Java programs using decision-making, looping, and control structures
4. To implement exception handling, multithreading, and file input/output operations in Java.
5. To develop Java applications using packages, interfaces, applets, and graphics programming
Course Outcomes: At the end of the course, the students will be able to -
CO1: Understand and develop program using Object Oriented methodology in Java.
CO2: Apply and design inheritance relationships in Java classes to promote code reuse.
CO3: Apply multithreading concepts in Java programs to achieve concurrent execution.
CO4: Understand & apply Exception Handling.
CO5: Create Java programs that involve graphics and applets.
CO6: Understand and work with file streams.

Contents –

Unit	Content	Teaching hours
1	**Basic Syntactical construct sin Java** Java Features and the Java Programming Environment, Object Oriented, Compiled, Interpreted, Platform independent, Portable, Robust and Secure, Dynamic. Defining a class, creating object, accessing class members Java Tokens and Data types, Constants and Symbolic Constants, variables, dynamic initialization, data types, array and string, scope of variable, typecasting, and standard default values. Operators and Expressions, Arithmetic Operators, Relational Operators, Logical Operators, Increment and Decrement, Conditional Operators, Bit wise Operators, Instance of Operators, Dot Operators, Operator precedence and associativity , Evaluation of Expressions, Type conversions in expressions, Mathematical Functions - min(),max(), sqrt(), pow(), exp(), round(), abs().Decision making and looping: If statement, if else statement, nested if else statement, if else if ladder, the switch statement, nested switch statement, The ?:operator, The while statement, the Do while statement ,the 'for' statement, break, continue and return statement, nested loops, labelled loops, for-each version of the for loop.	**7**
2	**Derived Syntactical Construct sin Java** Constructors and methods, types of constructors, nesting of methods, argument passing the 'this' keyword, command liner arguments, varargs: variable-length arguments, garbage collection, finalize() method, the object class. Visibility Control Public, Priv. Protected, default, friendly private Protected access. Arrays and Strings: Types of arrays, creating an array. strings. string classes and string buffer. vectors, -wrapper classes. enumerated types.	**8**

3	**Inheritance , Interface and Package** Inheritance: concept of inheritance Types of Inheritance Single Inheritance. multilevel Inheritance, Hierarchical Inheritance. method and constructor overloading and overriding, dynamic method dispatch, final variables, final methods. Use of super. abstract methods and classes. static members. Interfaces: Define Interface, implementing interface ,accessing interface, variables and methods. extending interfaces, interface references, nested interfaces Package: Define package, type of package naming and creating packages, accessing package, import statement, static import ,adding class and interfaces to a package.	**8**
4	**Exception Handling and Multithreading** Errors and Exception: Types of errors ,exceptions, try and catch statement, nested try statement, throws and Finally statement, build-in exceptions. Chained exceptions, creating own exception (throw clause), subclasses. Multithreaded Programming Creating a Thread: By extending to thread class and by implementing runnable Interface. Life cycle of thread: Thread Methods: wait(), sleep(), notify(), resume(), suspend(), stop().Thread exceptions. thread priority and methods ,synchronization, inter-thread communication. deadlock.	**8**
5	**Java Applets and Graphics Programming** Introduction to applets: Applet, Applet life cycle (skeleton), Applet tag, Adding Applet to HTML file, passing parameter to applet, embedding <applet> tags in java code, adding controls to applets. Graphics Programming: Graphics classes, lines, rectangles, ellipse, circle, arcs, polygons, color and fonts, setColor(), getColor(), setForeGround(), setBackGround(), font class, variable defined by font class: name. pointsizesize, style, font methods: getFamily(). getFont(), getFontname (), getSize(). getStyle(), getAllFonts() and get available font family name()of the graphics environment tclass.	**8**
6	**Managing Input/Output Files in Java** Introduction and Concept of Streams. Stream Classes. Byte Stream Classes: Input Stream Classes, Output Stream Classes. Character Stream Classes, Using streams Using File Class: I/O Exceptions, Creation of Files, Reading/Writing characters, Reading/Writing Bytes, Handling Primitive Data types.	**6**

Text Books: 1. PatrickNaughton-HerbertScheldt,TheCompleteReferencJava2(ThirdEdition)TMH 2.Deitel&Deitel: “How to Program JAVA”, PearsonEducation 3.E.Balagurusamy, Programming with Java, BPBPublication
Reference Books:
Online Resources: http://www.nptel.ac.in/courses/106105084/25 https://docs.oracle.com/javase/8/docs

Course code: 22IUCI0503T **Course Category:** PCC		**Course name: Software Engineering**
Credits: 3	**Teaching scheme:** L-3	**Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Basic programming Concept		
Course Objectives:		
1**.** To understand and select appropriate software process models.		
2. To develop and document software requirements using SRS.		
3. To apply software modeling techniques for data and system design.		
4. To estimate software size and cost effectively.		
5. To apply project management principles in software development.		
6. To implement quality assurance and security practices in software projects.		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Understand suitable Software Process model for software development.		
CO2: Create software requirement specifications.		
CO3: Apply Software modelling to create data designs		
CO4: Evaluate size and cost of software product.		
CO5: Apply Projectmanagement principlesinsoftwaredevelopment.		
CO6: Apply Quality Assurance and security principles in software development.		

Contents –

Unit	Content	Teaching hours
1	**Software Development Process** Software, Software Engineering as layered approach and its characteristics, Types of software. Software development framework. Software Process Framework, Process models: Perspective Process Models, Specialized Process Models. Agile Software development: Agile Process and its importance, Extreme Programming, Adaptive Software Development, Scrum, Dynamic Systems Development Method (DSDM), Crystal.Selection criteria for software process model.	8
2	**Software Requirement Engineering** Software Engineering Practices importance, Core principles. Communication Practices, Planning Practices, Modeling practices, Construction practices, software deployment (Statement and meaning of each principle for each practice). Requirement Engineering: Requirement Gathering and Analysis, Types of requirement (Functional, Product, Organizational, External Requirements), Eliciting Requirements, Developing Use-cases, Building requirement models, Requirement Negotiation, Validation. Software Requirement Specification: Need of SRS, Format, and its Characteristics.	9

3	**SoftwareModelling and Design** Translating Requirement model into design model: Data Modelling. Analysis Modelling: Elements of Analysis model. Design modelling: Fundamental Design Concepts (Abstraction, Information hiding, Structure, Modularity, Concurrency, Verification, Aesthetics). Design notations: Data Flow Diagram (DFD), Structured Flowcharts, and Decision Tables. Testing — Meaning and purpose, testing methods — Black-box and White-box, Level of testing — Unit testing. Test Documentation — Test Case Template, Test plan, Introduction to defect report, test summary report.	10
4	**Software Project Estimation** The management spectrum—4P's Metrics for' Size Estimation: Line of Code (Loc), Function Points (FP). Project Cost Estimation Approaches: Overview of Heuristic, Analytical, and Empirical Estimation. COCOMO (Constructive Cost Model), COCOMO II. Risk Management: Risk Identification, Risk Assessment ,Risk Containment, RMMM strategy.	8
5	**Software Project Management** Project Scheduling: Basic Principles Work breakdown structure Activity network and critical path Method, Scheduling techniques (CPM. PERT). Project Tracking: Timeline charts. Earned Value Analysis, Gantt Charts.	5
6	**Software Quality Assurance and Security** Software Quality Management vs. Software Quality Assurance. Phases of Software Quality Assurance: Planning, Activities, audit, and review Quality Evaluation standards: Six Sigma. ISO for software. CMMI: Levels, Process areas. Software Security ,Introduction to DevOps, Secure software engineering	5

Text Books:

1. Pressman, Roger , "Software Engineering: A Practitioner Approach", Mc Graw Hill Higher Education, New Delhi,(Seventh Edition) ISBN 978-0-07-337597-7
2.Fairly Richard,"Software Engineering Concepts",Mc Graw Hill Education,New Delhi-2001,ISBN-13:9780074631218

Reference Books:

1.Jain,Deepak,"Software Engineering:Principles and practices"Oxford University Press,New Delhi ISBN 9780195694840

Online Resources:

1.http://www.rspa.com/spi/
2.www.tutorialspoint.com//software_engineering/
3.www.versionone.com/agile-101/
4.www.sei.cmu.edu
5.www.nptel.ac.in/courses/

Course code: 22IUCI0504T **Course name: Programming in Python** **Course category:** PCC
Credits: 3 **Teaching scheme:** L-3 P-0 T-0 **Evaluation scheme:** CA–40, ESE–60, TW–0, PR–0
Pre-requisites: Basic knowledge of programming
Course Objectives:
1. Learn the basics of Python and how to write simple programs.
2. Use loops and conditional statements to control program flow.
3. Work with data structures like lists, tuples, sets, and dictionaries.
4. Apply functions, classes, and packages to build modular programs.
5. Explore Python libraries for data science such as NumPy, Pandas, and Matplotlib.
6. Perform file handling and exception handling to make programs reliable.
Course Outcomes: At the end of the course, the students will be able to -
CO1: Display message on screen using Python Script or IDE
CO2: Develop python demonstrate euse of operators.
CO3: Perform operations on Data Structures in Python.
CO4: Develop functions forgivenproblem.
CO5: Study about the various Packages in Python for data science
CO6: Handle Exception

Contents –

Unit	Content	Teaching hours
1	**Python foreverybody** Why Program, Hardware Overview, Python as aLanguage,Features of Python, Why Python, Installation Python Jupyter notebook, Using the Python Playground ,how to write programandcompileinJupyternotebook.WritinginputandoutputStatementsinPython,outputformatting ,Writing Comments, keywords in Python, variables, indentation and comments	**6**
2	**Condition alLoops and Iteration** BasicOperators,writingexpressionsConditional Statements,inPython,LoopsandIteration,Definite Loops,range function, String Function	**8**

3	**Data Structures in Python** Lists: Defininglists, accessing values inlist, deletingvaluesinlist, updating lists.b) Basic List Operations c) Built-in List functions Tuples:a) Accessing values in Tuples, deleting values in Tuples, and updating Tuples. b) Basic Tuple operations.c) Built-in Tuple Functions Sets: a) Accessing values in Set, deleting values in Set and updating Sets. b) Basic Set operations.c) Built-in Set functions Dictionaries: a) Accessing values in Dictionary, deleting values in DictionaryandupdatingDictionary.b) Basic Dictionary operations. c)Built-inDictionaries functions	**8**
4	**Functions Classes,Modules&Packages** Functions, Types of functions, String Functions, Lamda functions, Recursivefunctions, modulesandpackages,Classes: creatingclasses,objects.DataHiding&Inheritance	**8**
5	**Packages inPython forData Science** Numpy introduction, Numerical operations on Numpy,IntroductionofMatplotlib,gettingstartedwithPandas,Dataframes basics in Pandas, key operations on dataframes, Binarysearch,FindingelementsincommoninlistsusingHash,Findinglargeste lements	**8**
6	**File I/o Handling and Exception Handling** I/O Operations: Reading keyboard input, Printing to screen File Handling: Opening file in different modes, accessing file Contents using standard library functions, Reading and writing files, Closing a file, Renaming and deleting files, Directorie sin Python, File and directory related standard functions Exception Handling: Introduction, Exception handling-lary: except:" statement, 'raise' statement, User defined exceptions	**7**

Text Books: 1.KNageshwara"PythonProgramming", ScitechPublicationsPvt. Ltd. 2.KentD.Lee,"PythonProgrammingFundamentals",SecondEdition,SpringerPublication. 3.WesMcKinney,"PythonforDataAnalysis"O'ReillyPublication.
Reference Books:

Online Resources:

1.ThePythonLanguageReference:http://docs.python.org/2/reference/index.html
2.ThePythonStandardLibrary:http://docs.python.org/2/library/
3.https://docs.scipy.org/doc/scipy/reference/tutorial/stats.html
4.http://matplotlib.org/api/mlab_api.html#module-matplotlib.mlab
5.http://conference.scipy.org/proceedings/scipy2010/pdfs/seabold.pdf
6.http://seaborn.pydata.org
7.https://www.datacamp.com/community/data-science-cheatsheets
8.PEP20--The ZenofPython :https://www.python.org/dev/peps/pep-0020/
9.https://docs.scipy.org/doc/numpy-dev/user/numpy-for-matlab-users.html

Course code: 22IUCI0505T **Course name: Microprocessor & Microcontroller** **Course category:** PCC		
Credits: 3	**Teaching scheme:** L-3	**Evaluation scheme:** CA–40, ESE–60
Pre-requisites:		
Course Objectives:		
1.		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Understand the Basic Concepts of 8085 Microprocessor.		
CO2: Understand the Basic Concepts of 8086 Microprocessor.		
CO3: Understand and Analyze the Concept of Programming's.		
CO4: Understand the Basic Concepts of Microcontroller.		
CO5: Apply Knowledge of Memory and Interfacing I/O for Programming.		
CO6: Understand the Concepts of Advance Processors.		

Contents –

Unit	Content	Teaching hours
1	**Introductionto Microprocessors** Microprocessor Architecture and Its Operations, Memory, Input & Output Devices,RISC and CISC Processors, Harvard and Von Neumann Architecture 8085 - Pin Diagram, Architecture, Functional Diagram,TimingDiagrams,LogicDevicesfor Interfacing, Memory Interfacing, Interfacing Output Displays, Interfacing Input Devices VariousInterfacingTechniques,MemoryOrganization &DifferentInterfacings.	**8**
2	**8086 Microprocessor** 8086 - Pin Diagram, Architecture, Functional Diagram, Register Organization, Memory Segmentation, Signal Descriptions of 8086, Interrupts of 8086 8086 Instruction Formats, Addressing Modes, Instruction Set, Assembler Directives, Macros PeripheralDevices-8237DMAController,8255 Programmable Peripheral Interface, 8253 / 8254 Programmable Timer / Counter, 8259 Programmable Interrupt Controller, 8251 USART and RS232C	**8**
3	**Basic Programming & Concepts** Flow Chart Symbols, Data Transfer Operations, Arithmetic Operations, Logic Operations, Branch Operation, Writing Assembly Language Programs Programming Techniques - Looping, Counting and Indexing, Additional Data Transfer and 16 Bit Arithmetic Instruction Logic Operation - Rotate, Compare, Counter and Time Delays, 8085 & 8086 Interrupts	**8**

4	**Introduction to Microcontroller 8051** Introduction, Evolution, Comparison Between Microprocessor and Microcontroller, 8051 Microcontroller - Overview, Block Diagram, Architecture, Key Features, Pin Diagram, I/O Ports, Memory Organization, Addressing Modes and Instruction Set of 8051 Programming Model, Data and Program Memory, Internal RAM Organization, Special Function Registers, Control Registers, Counters and Timers, Interrupt Structures. PSW and Flag Bits, Register Banks and Stack, Typesof Special Function Registers and their uses in 8051, MemoryAddressDecoding,8031/51Interfacingwith ExternalROMAndRAM	**8**
5	**8051 Programming** Introduction to 8051 Assembly Programming, Assembling and Running an 8051 Program, Data Typesand Assembler Directives Arithmetic, Logic Instructions and Programs, Jump, Loop and Call Instructions, I/O Port Programming,Timers,SerialPortProgramming,InterruptsProgramming ExternalMemoryInterfacing,InterfacingofLED, LCDADC,DAC,StepperMotor,KeyboardusingAssembly / C Programming	**6**
6	**Advance Processors** ARM Processor Fundamentals, ARM Architecture, Register, CPSR, Pipeline, Exceptions and Interrupts Interrupt Vector Table ARM Instruction Set - Data Processing, Branch Instructions, Load Store Instructions, Software Interrupt Instructions, Program Status Register Instructions, Loading Constants, Conditional Execution Introduction to Thumb instructions, Introduction to CORTEX Processor and Its Architecture, OMAP Processor and Its Architecture.	**7**

Text Books:

“1. "MicroprocessorArchitecture, Programming and Application"byRameshGaonkarofWilleyEastern Ltd.2. "Fundamentals of Microprocessor and Microcontroller" by B.Ram of Dhanpat Rai and Sons.

2."FundamentalsofSystem-on-ChipDesignonARMCortex-MMicrocontrollers"byReneBeuchat, Andrea Guerrieri, Sahand Kashani of ARM Education Media.

3."ARMSystemDevelopersGuide"byAndrewNSloss,Dominic Symes,ChrisWrightofElsevier,2012.

Reference Books:

1. "MicroprocessorsandInterfacing"byHall,DouglasV.ofTataMcGrawHill.
2. "Advanced Microprocessors and Peripherals" by A. K. Ray and K.M. Bhurchandani of MHE, 2nd Edition.
3. "The 8051 Microcontrollers Architecture, Programming & Applications" by Kenneth. J. Ayala of Cengage Learning.
4. "The 8051 Microcontrollers, Architecture and Programming and Applications" by K.Uma Rao, Andhe Pallavi of Pearson Publication.
5. "ARMSystem-On-ChipArchitecture"byFurberofPearsonEducation.

Online Resources:

1. https://www.tutorialspoint.com
2. https://javatpoint.comhttps://
3. www.w3schools.com

Course Code:22IUCI0501L Course Name: Data Structure and Algorithm Lab Course Category: PCC
Credits: 01 **Teaching scheme:** P-2 k **Evaluation Scheme:** TW– 30, PR–20
Pre-requisites: Programming Language such as: C, C++
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Implement programs using arrays and structures to solve computational problems.
LO2: Design and demonstrate searching, sorting, and hashing algorithms using appropriate data structures.
LO3: Implement and test stack and queue operations using both arrays and linked list representations.
LO4: Develop programs to create, manipulate, and demonstrate various linked list structures and their operations.
LO5: Implement and evaluate different tree data structures such as binary trees, binary search trees.
LO6: Develop programs to represent and analyze graphs using adjacency matrix and adjacency list, and design efficient graph algorithms.

List of Experiments—

Sr. No.	Name of Experiment	Practical Hours
1	Write a ‘C’ program to perform following Operations on Array: Create, Insert, Delete,Display.	2
2	Write a ‘C’ Program to Search a particular data from the given Array of Strings using Linear Search Method.	2
3	Write a ‘C’ program to Search a particular data from the given Array of numbers using Binary Search Method.	2
4	Write a ‘C’ Program to Sort an Array of numbers using Bubble Sort Method.	2
5	Write a ‘C’ Program to Sort an Array of numbers using Selection Sort Method	2
6	Write a ‘C’ Program to Sort an Array of numbers using Insertion Sort Method.	2
7	Write a 'C' Program to perform PUSH and POP Operations on Stack using an Array.	2
8	Write a 'C' Program to perform INSERT and DELETE Operations on Linear Queue using an Array.	2
9	Write a 'C' Program to perform INSERT and DELETE operations on Circular Queue using an Array.	2
10	Write a 'C' Program to Implement Singly Linked List with Operations: (i) Insert at beginning, (ii) Search, (iii) Display	2
11	Write a 'C' Program to Implement BST (Binary Search Tree) and Traverse in In-Order.	2
12	Write a program to implement Graph operations.	2

Text Books: 1. “Data Structures with C”, S. Lipschutz, Mc Graw Hill.2010

2. ” Schaum's Outline of Data Structures”, Lipschutz, S., McGraw-Hill

3. “Fundamentals of Data Structures in C++”, E. Horowitz, Sartaj Sahni, D.Mehta, Universities Press.1995

4. “Data Structures using C”, E. Balaguruswamy, McGraw Hill 2013
Reference Books: 1. “Data Structures using C” Aarron M. Tenenbaum, Yedidyah Langsam,Moshe J. Augensten , PHP
Online Resources: 1. https://archive.nptel.ac.in/courses/106/106/106106127/
2. https://nptel.ac.in/courses/106102064

Course code: 22IUCI0502L **Course category: PCC**		**Course name: Java Programming Lab**
Credits: 02	**Teaching scheme:** P-4	**Evaluation scheme:** TW–60, PR–40
Pre-requisites: Computer Fundamentals, Programming Fundamentals, Problem solving and algorithms, OOP, Data Structures, Web Technolgies.		
Laboratory Outcomes: At the end of the session, the students will be able to -		
LO1: Understand and developprogramusing Object OrientedmethodologyinJava.		
LO2: Implement inheritance relationships in Java classes to promote code reuse.		
LO3: Apply multithreading concepts in Java programs to achieve concurrent execution.		
LO4: Implement ExceptionHandling.		
LO5: Create Java programs that involve graphics and applets.		
LO6: Create Java programs for handling I/O and filestreams.		

List of Experiments–

Sr.No.	Name of Experiment
1	Set up a Java Programming development environment and test by implementing a small program.
2	Develop programs to demonstrate use of if statements and its different forms.
3	Develop programs to demonstrate use of - a) Switch—Case statement Conditional if(?:)
4	Develop programs to demonstrate use of Looping Statement 'for'
5	Develop programs to demonstrate use of 'while'. - do- while
6	Develop a program for implementation of explicit type conversion in Java.
7	Develop a program for implementation of Constructor.
8	Develop a program for implementation of different functions of String Class.
9	Develop a program for implementation of Arrays in Java.
10	Develop a program for implementation of Vectors in Java.
11	Develop a program for implementation of Wrapper Class to a) convert primitive into object. convert object into primitive
12	Develop programs which implement the concept of overriding.
13	Develop a program for implementation of Single and Multilevel inheritance.
14	Develop a program for implementation of interface
15	Develop a program to import different classes in package.
16	Develop a program for implementation of try, catch and finally block.
17	Develop programs for implementation of throw, throws clause.
18	Develop a program for implementation of multithreading operation.
19	Write a program to create animated shape using graphics and applets. You may use the following shapes: a) Lines and Rectangles. b) Circles and Ellipses.

	c) Arcs Polygons with fillPolygon method.
20	Develop a program for implementation of I/O stream classes.

Text Books: 1.PatrickNaughton-HerbertScheldt,TheCompleteReference-Java2(ThirdEdition)TMH 2.Deitel&Deitel:"HowtoProgramJAVA", PearsonEducation 3.E.Balagurusamy, ProgrammingwithJava, BPBPublication
Reference Books:
Online Resources: http://www.nptel.ac.in/courses/106105084/25https://docs.oracle.com/javase/8/docs

Course code: 22IUCI0503L **Course name: Software Engineering Lab** **Course category:** PCC
Credits: 01 **Teaching scheme:** P-2 **Evaluation scheme:** TW–50, PR–00
Pre-requisites: Basic programming Concept
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Understand suitable Software Process model for software development.
LO2: Create software requirement specifications.
LO3: Apply Software modelling to create datad esigns
LO4: Evaluate size and cost of software product.
LO5: Apply Project management principl esin software development.
LO6: Apply Quality Assurance and security principl esin software development.

List of Experiments–

Sr.No.	Name of Experiment
1	Write problem statement to define the project title with bounded scope of the project.
2	Select relevant process model to define activities and related tasks set for assigned project.
3	Gather application specific requirements for assimilate into RE (Requirements Engineering) model.
4	Prepare broad SRS (software requirement specification) for the above selected project.
5	Prepare use-cases and draw use-case diagram using Software Modeling Tool.
6	Develop data designs using DFDs (data flow diagram), Decision tables, and E-R (entity-relationship) diagram.
7	Draw class diagram, Sequence diagram, Collaboration diagram, State Transition Diagram for the assigned project.
8	Write test cases to validate requirements of assigned project from SRS document.
9	Identify risks involved in the project and prepare RMMM (Risk Management, Mitigation, and Monitoring) plan.
10	Estimate cost of the project using COCOMO (Constructive Cost Model) / COCOMO II approach for the assigned project.
11	Prepare SQA plan that facilitates various attributes of quality of process and product.
12	Write Test Cases to validate Requirement of Assigned project from SRS document.

Text Books:

1. Pressman, Roger , "Software Engineering: A Practitioner Approach", Mc Graw Hill Higher Education, New Delhi,(Seventh Edition) ISBN 978-0-07-337597-7

2.Fairly Richard,“Software Engineering Concepts”,Mc Graw Hill Education,New Delhi-2001,ISBN- 13:9780074631218

Reference Books:

1.Jain,Deepak,”Software Engineering:Principles and practices”Oxford University Press,New Delhi ISBN 9780195694840

Online Resources:

1.http://www.rspa.com/spi/
2.www.tutorialspoint.com//software_engineering/
3.www.versionone.com/agile-101/
4.www.sei.cmu.edu
5.www.nptel.ac.in/courses/

Course code: 22IUCI0504L **Course category:** PCC	**Course name: Programming in Python Lab**
Credits: 01 **Teaching scheme:** P-2	**Evaluation scheme:** TW–50, PR–00
Pre-requisites:1	
Laboratory Outcomes: At the end of the session, the students will be able to -	
LO1: Display message on screen using Python Script or IDE	
LO2: Develop python program to demonstrate euse of operators.	
LO3: Perform operations on Data Structures in Python.	
LO4: Develop functions for given problem.	
LO5: Study about the various Package sin Python for data science	
LO6: Handle Exception	

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1	Write Program to perform input/output operations Write a program to take input (integer, float, string) and print it.	2
2	**Program based on operators** 1. Write a program to simulate a simple calculator (+ - / * %) that takes two operands as input and displays the result. 2. Write a program to find area and perimeter of geometric objects. 3. The distance between two cities (in km.) is input through the keyboard. Write a program to convert and print this distance in meters, feet, inches, and centimeters. 4. Write a C program to interchange two numbers. **5.** Write a program to compute Fahrenheit from Centigrade.	2
	Programs based on Decision making. 1. Write a program to read marks from keyboard and your program should display equivalent grade according to conditions : (100-80) Distinction , (79-60) – First Class , (59-40) - Second Class, <40 - Fail 2. Write a program to input basic salary of an employee and calculate gross salary according to given conditions. a)Basic Salary <= 10000 : HRA = 20%,DA=80% b)Basic Salary is between 10001to20000 : HRA=25%, DA=90% c)Basic Salary>=20001:HRA=30%, DA = 95% 3. If the ages of three brothers are input through the keyboard, write a C Program to determine the youngest and oldest of the three. 4. Write a program to calculate overtime pay of employee. Overtime is paid at the rate of Rs. 12.00 per hour for every hour worked above 40 hours. Assume that employee do not work for fractional part of an hour. 5. Write a program for checking the speed of drivers. If speed is less than 70, it should print “Ok”. Otherwise, for every 5kmabove the speed limit	2

3	(70), it should give the driver one demerit point and print the total number of demerit points. For example, if the speed is 80, it should print: “Points: 2”. If the driver gets more than 12 points, the function should print: “License suspended”	
4	**Programs using while and for loops** 2. WAP to find factorial of given number 3. WAP to check whether given number is Palindrome or not 4. WAP to check whether given number is Armstrong or not 5. WAP to print Fibonacci series 6. Write a Python program which iterates the integersfrom1to 50.Formultiplesof three print "Fizz" instead of the number and for the multiples of five print "Buzz". For numbers which are multiples of both three and five print "FizzBuzz". 7. WAP to check whether given number is Perfect number or not 8. WAP to check whether given number is Prime number or not 9. Write C program to print given star and number patterns and reverse it. *** 1** **** 12** ***** 123** ****** 1234**	2
5	**Programs on string** 2. Write Python Program to find length of string without using len() function. 3. Count all letters, digits, and special symbols from a given string. 4. Python Program to Count the Number of Vowels in a String. 5. Python Program to Calculate the Number of Upper Case Letters and Lower Case Letters in a String. 6. Python Program to Check whether given string is palindrome or not	2
6	**Programs on List and Tuple** 1. Write a Python program to sum all the items in a list. 2. Write a Python program to multiply all the items in a list 3. Write a Python program to get the largest number from a list. 4. Write a Python program to get the smallest number from a list 5. Write a Python program to count all element sin list and count Occurrences Of A List Item In Python 6. Write a Python program to create a tuple with different data types 7. Write a Python program to check whether an element exists within a tuple 8. Write a Python program to reverse atuple Write a Python program calculate the product of all the numbers given intuple. Original Tuple:(2,4,8,8,3,2,9) Product - multiplying all the numbers of the said tuple: 27648	2
	Programs on set and dictionary 1. Write a Python program to concatenate following dictionaries to create a new one. Sample Dictionary : dic1={1:10, 2:20} dic2={3:30, 4:40}	2

7	2. dic3={5:50,6:60} Expected Result:{1:10,2:20,3:30,4: 40,5:50,6:60} 3. Write a Python program to check whether a given key already exists in a dictionary 4. Write a Python script to generate and print a dictionary that contains a number (between 1 and n) in the form (x, x*x) Sample Dictionary(n=5):Expected Output : {1: 1,2: 4,3: 9, 4: 16, 5:25} 5. Write a Python program to merge two Python dictionaries Write a Python program to get the maximum andminimum value in a dictionary 7. Write a Python program to create set difference, Union and intersection 8. Write a Python program to check if two given sets have no elements in common	
8	**Programs using function** 1. Write Functions to calculate your trip's costs: Define a function called hotel _cost with one argument nights as input Define a function called plane _ride_ cost that takes a string, city, as input. Define a function called rental_ car _cost with an argument called days. Define a function called trip _cost that takes two arguments, city and days. Like the example above, have your function return the sum of calling the rental_car_cost(days), hotel_cost(days),and plane_ride_cost(city) functions. 2. Write a program into check a given number is even or odd using the function. 3. Write a function Exchange to interchange the values of two variables, say x and y. illustrate the use of this function in a calling function. 4. Write a program to find Sum of natural number using recursion. 5.Write a program to print Fibonacciseries Number using recursion	2
9	**Program using NumPy, Matplotli band Pandas library** 1. Write a program to perform matrix addition, substraction, multiplication. 2. Plot all types of graph using Matplotlib. 3. Write a program which performs basic operation of Pandas.	2
10	**Program on Binary Search** Write a program to find given element using Binary search	2
11	Write a program to find given element using Binary search	2
12	**Program on I/o Handling and Exception Handling** Write a program to handle user defined Exception for given problem.	2

Text Books:

1. K Nageshwara“PythonProgramming”, ScitechPublicationsPvt.Ltd.
2. Kent D.Lee, “PythonProgrammingFundamentals”, SecondEdition, Springer Publication.
3. Wes Mc Kinney, “PythonforDataAnalysis”O’ReillyPublication.

Reference Books:

1. Dr Nageshwar Rao ,”Core Python Programming” -Wiley

Online Resources:

1.ThePythonLanguageReference: http://docs.python.org/2/reference/index.html
2.ThePythonStandardLibrary: http://docs.python.org/2/library/
3.https://docs.scipy.org/doc/scipy/reference/tutorial/stats.html
4.http://matplotlib.org/api/mlab_api.html#module-matplotlib.mlab
5.http://conference.scipy.org/proceedings/scipy2010/pdfs/seabold.pdf
6.http://seaborn.pydata.org
7.https://www.datacamp.com/community/data-science-cheatsheets
8.PEP20--The ZenofPython :https://www.python.org/dev/peps/pep-0020/
9.https://docs.scipy.org/doc/numpy-dev/user/numpy-for-matlab-users.html

Course code: 22IUCI0505L	**Course name: Microprocessor & Microcontroller Lab**	
Course category: PCC		
Credits: 01	**Teaching scheme:** P-2	**Evaluation scheme:** TW–30, PR–20
Pre-requisites:1		
Laboratory Outcomes: At the end of the session, the students will be able to -		
LO1: Understand the Basic Concepts of 8085 Microprocessor.		
LO2: Understand the Basic Concepts of 8086 Microprocessor.		
LO3: Understand and Analyze the Concept of Programming's.		
LO4: Understand the Basic Concepts of Microcontroller.		
LO5: Apply Knowledge of Memory andInterfacingI/OforProgramming.		
LO5: Understand the Concepts of Advance Processors.		

List of Experiments–

Sr.No.	Name of Experiment
1	Study of 8085&8086 Microprocessor Kit
2	Programming for DataTransfer Operations
3	Programming for Arithmetic/Logical Operations
4	Programming for String Operations
5	Studyof Different Interfacing Cards
6	Study of 8255
7	Introduction to 8051S imulator
8	Arithmetic Operations using 8051
9	Logical Operations using 8051
10	DataTransfer Operations using 8051
11	Bit Manipulation Operations using 8051
12	Study of ARM & CORTEX Processors

Text Books:

1. "MicroprocessorArchitecture,ProgrammingandApplication"byRameshGaonkarofWilleyEasternLtd.2.
2. "FundamentalsofMicroprocessorandMicrocontroller"by B.RamofDhanpatRaiandSons.
3. "Fundamentals of System-on-Chip Design on ARM Cortex - M Microcontrollers" by Rene Beuchat, AndreaGuerrieri, Sahand Kashani of ARM Education Media.
4. "ARMSystemDevelopersGuide"by AndrewNSloss,DominicSymes,ChrisWrightofElsevier,2012.

Reference Books:

1. "MicroprocessorsandInterfacing"byHall,DouglasV.ofTataMcGrawHill.
2. "AdvancedMicroprocessors and Peripherals" by A. K. Ray and K.M. Bhurchandani of MHE, 2nd Edition.
3. "The 8051 Microcontrollers Architecture, Programming & Applications" by Kenneth. J. Ayala of Cengage Learning.
4. "The 8051 Microcontrollers, Architecture and Programming and Applications" by K.Uma Rao, Andhe Pallavi of Pearson Publication.
5. "ARMSystem-On-ChipArchitecture"byFurberofPearsonEducation.

Online Resources:

1. AssemblerTASM/MASM,LinkerLINK/TLINK,DebuggerOSDebuge/TD
2. www.intel.com
3. www.pcguide.com/ref/CPU
4. www.CPU-World.com/Arch/
5. www.techsource.com/engineering-parts/microprocessor.html
6. https://www.elprocus.com/8051-assembly-language-programming/
7. https://www.tutorialspoint.com/assembly_programming/
8. http://www.slideshare.net/search/slideshow?searchfrom=headerandq assembly+languagetprogrammingtof+8086andud=anyandft=allan dlang=**andsort=

SEM-VI

Course code: 22IUCI0601T Course name: Data Communication and Networking Course category: PCC
Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Understanding of computer hardware (CPU, memory, storage), Programming Fundamentals, Basic computer organization
Course Objectives: Understand Fundamentals of Data Communication, Learn Networking Models and Standards, Study Transmission Media and Devices
1.
Course Outcomes: At the end of the course, the students will be able to -
CO1: Analyse the function of data communication and computerNetwork
CO2: Select relevant transmission media and switching Techniques.
CO3: Analyse the transmission errors with respect to IEEE standards
CO4: Configure various networking devices
CO5: Configure different TCP/IPservices

Contents –

Unit	Content	Teaching hours
1	**Introduction** Data Communications, Networks, Network Types, Internet History. Types of Network, the OSI model. Introduction to Physical Layer-1: Data and Signals, Digital Signals, Transmission Impairment, Data Rate limits, Performance, Digital Transmission: Digital to digital conversion.	**8**
2	**Fundamentals ofdata communication and Computer Network** Process of Data Communication and its Components. Protocols, standards, Standard organization, Bandwidth, Data Transmission Rate, Baud rate and bits per second. Modes of Communication Analog signal and Digital signal Fundamental of Computer network Classification of network Network Architecture	**6**
3	**Transmission media and switching** Communication media 3.2.Unguidedtransmission media Lines of Sight Transmission Point to Point Multiplexing Switching	**8**
4	**Error detection, correction and wireless communication** Types of error Error detection IEEE Standards 4.4.Wireless LANs BluetoothArhitecture Mobilegenerations	**8**

5	**Network Topologies And networkingDevices** Network Topologies Network Connecting Devices: Hub,Switch,Router,repeater,Bridge,Gateway, Modem, Wireless infrastructure Components	**8**
6	**Reference Models** OSI Reference Model TCP/IP Model: Layered Architecture, Data link Layer Introduction:Adddressingmechanismin Internet IP addressing IPv4 and IPv6 OSI and TCP/IP Network Model	**7**

Text Books: 1. Data communication and networking Author: Forouzan Behrouz
Reference Books: 1. Computer Networks Author: Tanenbaum Andrew S
Online Resources: https://www.tutorialspoint.com https://www.w3schools.com

Course code: 22IUCI0602T	**Course name: Advanced Java Programming**
Course category: PCC	
Credits: 3 **Teaching scheme:** L-3	**Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Core Java, Object-Oriented Programming (OOP), Java AWT (GUI Programming Basics), Java Collections & Data Structures, Basic Networking Concepts, Basic Database/SQL Concepts	
Course Objectives:	
1. Understand and apply GUI programming concepts using AWT and Swing components, layout managers, and event-handling mechanisms to develop interactive Java applications.	
2. Implement advanced event. handling in Java using listener interfaces, adapter classes, and inner classes for robust user-interface functionality.	
3. Develop networked Java applications using sockets, TCP/IP, UDP, and internet addressing concepts.	
4. Perform database connectivity and operations using JDBC, including connections, statements, prepared statements, result sets, and transaction management.	
5. Utilize Java Collections Framework and generics to manage and manipulate data efficiently, including sets, lists, maps, and custom objects for real-world applications.	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Create Javaprograms usingGUIframeworks suchasAWT or Swing.	
CO2: Apply knowledge of event handling mechanisms in AWT and Swingcomponents	
CO3: Write Java code to handle various events in aprogram.	
CO4:Apply networking concepts to communicate over networks. Also, handle databaseoperations	
CO5: Utilize Javacollections tomanageandmanipulate data efficiently.	
CO6: Create Java to handle HTTP requests and generate dynamic web content.	

Contents –

Unit	Content	Teaching hours
1	**Abstract Windowing Toolkit (AWT)** Component, container, window, frame, panel. Creating windowed programs and applets. AWT controls and layout managers: use of AWT controls: labels, buttons, checkbox, checkbox, group, scroll bars, text field text area. Use of layout managers: flow Layout, border Layout, grid Layout ,card Layout,GridbagLayout,menu bars,menus,dialogboxes,file dialog.	7
2	**Swings** Introduction to swing: Swing features, Difference between AWT and Swing. Swing Components JApplet, Icons and Labels, Text Fields, Combo Boxes .Buttons: The JButton, Check Boxes, RadioButtons. Advanced Swing Components :Tabbed Panes ,Scroll Panes, Trees, Tables, Progress bar, tooltips .MVC Architecture. Key Listener Interface, Mouse Listener interface, Mouse Motion Interface, Text Listener Interface, Windows Listener Interface.	8

3	**Event Handling** ThedelegationEventModel:Eventsources,EventlistenersEventclasses: The Action Event class, the Item Event class, the Key Event class, the Mouse Event class, the Text Event class, the Window Event class .Adapter classes. Inner classes .Event listener Interfaces :Action Listener Interface, Item Listener Interface,	**8**
4	**Networking Basicsand Database interaction** Socket Overview: Client/Server, Reserved Sockets, Proxy Servers, Internet Addressing. Java and the Net:The Networking Classes and interfaces.InetAddress: Factory Methods, Instance Methods. TCP/IP Client Sockets: Who is URL: Format, The URI Class URL Connection: TCP/IP Server Sockets. Datagrams : Datagram Packet , Datagram Server and Client Introduction to JDBC DDBC JDBC Architecture: Two tier models and three tier models Types of JDBC Drivers Driver Interfaces andDriver Manager class: Connection Interfacen Statement lnterface, Prepared StatementInterface Result Set Interface The essential JDBC Program	**8**
5	**Java Collections andGenerics** The Collections Framework, The Set Interface, Set Interface Methods, Set Implementation Classes, The List Interface, List ImplementationClasses, TheMap Interfaces, Map Implementation Classes, Utility classes, Generics in Java, Primitive wrapper classes--{Collection Framework: Array List class, Linked List class, List Iterator interface, Hash Set class, Linked Hash Set class ,Tree Set class, Priority Queue class,ArrayDequeclass,Mapinterface,HashMapclass,LinkedHashMapclass ,TreeMapclass,Hashtableclass, Comparable and Comparator, Properties class}	**8**
6	**Servlets** The Life Cycle of a Servlet Creating simple Servlet: The Servlet API, javax. servlet Package, Servlet Interface, Servlet ConfigInterface, Servlet Context Interface, Servlet Request Interface, Servlet Response Interface, Generic Servlet Class The javax. Servlet .http Package: Http Servlet Request Interface, Http Servlet Response Interface, Http Session Interface ,Cookie Class, Http Servlet Class, Http Session Event Class , Http Session Binding Event Class Handling HTTP Requests and Responses Handling HTTPGET Requests Handling HTTPPO STRequests. Cookies and Session Tracking	**6**

Text Books: 1.Java:TheCompleteReferenceySchildt,Herbert,Publisher:McGrawHillEducationIndia 2.Java2ProgrammingBlackBook–byHolzner,Stevenetal:DreamtechPress,Newdelhi 3.Java Server Programming Tutorial JAVA EE6 Black Book- by KogentLearnig Solutions: Dreamtech Press, New delhi
Reference Books:

Online Resources:
1.https://nptel.ac.in/courses/106105084/30
2.https://www.tutorialspoint.com/java
3.https://www.javatpoint.com/servlet-tutorial

Course code: 22IUCI0603T **Course category:** PCC		**Course name: Software Testing**
Credits: 3	**Teaching scheme:** L-3	**Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Basic Programming Knowledge, **Software Engineering**		
Course Objectives:		
1. To introduce students to the fundamental concepts and principles of software testing.		
2. To enable students to understand and apply various software testing methods and techniques.		
3. To develop the ability to design and prepare test cases for different types and levels of software testing.		
4. To train students to design and prepare a comprehensive test plan for a software application.		
5. To equip students with the skills required to analyze software applications, identify defects, and prepare defect reports.		
6. To provide hands-on experience in evaluating software performance using automated testing tools.		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Apply fundamental concepts and principles of software testing in practical scenarios.		
CO2: Apply appropriate software testing methods and techniques for different types of software systems.		
CO3: Design and develop test cases for various levels and types of software testing.		
CO4: Design and prepare a comprehensive test plan for a given software application.		
CO5: Analyze software behavior to identify defects and prepare detailed defect reports.		
CO6: Evaluate software performance by executing automated testing tools and interpreting performance metrics.		

Contents –

Unit	Content	Teaching hours
1	**Basics of Software Testing** Software Testing, Objectives of Testing Failure, Error, Fault, Defect, Bug Terminology. Test Case, When to Start and Stop Testing of Software (Entry and Exit Criteria).Verificationand Validation (V Model), QualityAssurance, Quality Control.	**8**
2	**Software Testing Methods** Methods of Testing: Static and dynamic Testing The box approach: White Box Testing: Inspections, Walkthroughs, Technical Reviews, Functional Testing, Code Coverage Testing, Code Complexity Testing. Black Box Testing: Requirement Based Testing, Boundary Value Analysis, Equivalence Partitioning,	**9**

3	**Types and Levels of Testing** Levels of testing: Unit Testing: Driver, Stub Integration Testing: Top-Down Integration, Bottom-Up Integration, Bi-Directional Integration Testing on web Application: Performance Testing: Load Testing, Stress Testing, Security Testing. Client-Server Testing. Acceptance Testing: Alpha Testing and Beta Testing, Special Tests: Regression Testing, GUI Testing,	**10**
4	**Test Management** Test Planning : Preparing a Test Plan, Deciding Test Approach, Setting Up Criteria for Testing, Identifying Responsibilities, Staffing, Resource Requirements, Test Deliverables, Testing Tasks Test Management: Test Infrastructure Management, Test People Management. Test Process: Base Lining a Test Plan, Test Case Specification. Test Reporting: Executing Test Cases, Preparing Test Summary Report.	**8**
5	**Defect Management** Defect Classification, Defect Management Process. Defect Life Cycle, Defect Template Estimate Expected Impact of a Defect, Techniques for Finding Defects, Reporting a Defect.	**5**
6	**Testing Tools and Measurements** Manual Testing and Need for Automated Testing Tools Advantages and Disadvantages of Using Tools Selecting a Testing Tool When to Use Automated Test Tools, Testing Using Automated Tools. Metrics and Measurement: Types of Metrics. Product Metrics and Process Metrics, Object oriented metric sin testing.	**5**

Text Books:

1. Sriniwasan Desikan, Gopalswamy Ramesh, *Software Testing: Principles and Practices*, Pearson Education, Pearson India, 2005.
   ISBN: 978-81-7758-121-8.
2. M. G. Limaye, *Software Testing: Principles, Techniques and Tools*, Tata McGraw-Hill Education, New Delhi, 2007.
   ISBN: 978-0-07-013990-9.

Reference Books:

1. Naresh Chauhan, *Software Testing: Principles and Practices*, Oxford University Press, Noida.
2. Yogesh Singh, *Software Testing*, Cambridge University Press, Bengaluru.
   ISBN: 978-1-107-65278-1.

Online Resources:

1.http://www.selenium.com
2.http://en.wikipedia.org/wiki/Test_automation
3.http://en.wikipedia.org/wiki/Software_testing#Testing_tools
4.http://www.softwaretestingsoftware.com
5.www.nptel.ac.in/courses/
6.www.toolsqa.com

Course code: 22IUCI0604T **Course name: Database Management System** **Course category:** PCC
Credits: 3 **Teaching scheme:** L-3 P-0 T-0 **Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Understand fundamental concepts of data, databases, and DBMS, Gain awareness of emerging database technologies
Course Objectives:
1. Understand fundamental concepts of data, databases, and DBMS, and explain the advantages of DBMS over traditional file processing systems.
2. Describe the applications of databases and explain the three-level architecture and overall structure of a DBMS.
3. Apply data abstraction and data modeling techniques using the Entity–Relationship (E-R) model.
4. Explain the relational data model and key concepts of RDBMS, including normalization and its various forms.
5. Use Structured Query Language (SQL) effectively to define, manipulate, control, and query data using DDL, DML, DCL, and DQL commands. Apply integrity constraints, SQL operators, built-in functions, joins, subqueries, views, indexes, sequences, and synonyms to solve database problems.
6. Develop database programs using PL/SQL, including procedures, functions, cursors, triggers, and exception handling.
Course Outcomes: At the end of the course, the students will be able to -
CO1: Recallsand understandsfundamentalconceptsrelatedtodatabases.
CO2: Applynormalizationrulestoensuredata integrity
CO3: ApplySQLcommandstocreateand managedatabases.
CO4: ApplyPL/SQLcodetoimplementbusiness logicandmanipulatedatawithinthe database.
CO5: Createstoredproceduresandfunctionstoencapsulatereusablelogic.
CO6: Implement security measures(e.g.,accesscontrols,encryption)toprotectthedatabaseandensure confidentiality.

Contents –

Unit	Content	Teaching hours
1	**Database System Concep**t Concept of Data, Database, DBMS, advantages of DBMS over file processing system, application of database. Three-level Architecture for Database system. Data abstraction. Overall Structure of DBMS. Data Modelling. Data Modelling Using the E-R Model.	**8**
2	**Relational Data Model** Fundamentals of RDBMS: Key concept, Normalization and its types. Introduction to Structured Query Language: DDL commands Data Integrity Constraint DML commands DCL commands DQL commands SQL operators	**6**

3	**Interactive SQL and Advance SQL** In-built Functions: String Arithmetic Date and time Aggregate functions Queries using: Group by, Having, and Order by clause Joins: Inner and Outer JoinSub queriesViews and Joins. Sequences, Indexes, Synonyms.	**8**
4	**PL/SQL Programming** Introduction of PL/SQL, execution environment, PL/SQL data Types, Variables, Constants. Control Structure, Exception handling Cursors Procedures, Functions. Database Triggers	**8**
5	**Database security and TransactionProcessing** DatabasesecurityProtectingthedatawithindatabase-Databaseprivileges Transaction Database backup Database recovery	**8**
6	**Emerging DatabaseTechnologies** Object oriented and object relational databases Web databases Distributed Databases Dataware housing and datamining.	**7**

Text Books:

1.Database System Concepts Mc Graw Hill in ANSIC Korth Mc Graw Hill Education,NewDelhi,2015
2.Introduction to Database Management Systems 1 SRD Group McGraw Hill Education,New Delhi, 2015

Reference Books:

1.Introduction to Relational databases & SQL programming Allen McGraw HillEducation, New Delhi, 2015
2.Complete Reference:Mysql Vikram Vaswani McGraw HillEducation,NewDelhi.2015

Online Resources: 1.https://nptel.ac.in/course.html

Course code: 22IUCI0605T	**Course name: Operating System**
Course category: PCC	
Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–40, ESE–60	
Pre-requisites:	
Course Objectives:	
1.	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Use operating system tools to perform various functions	
CO2: Apply scheduling algorithms to calculate turnaroundtimeandaveragewaiting time.	
CO3: Learn sharing of resources without interference using mutual exclusion.	
CO4: Calculate efficiency of different memory management techniques.	
CO5: Apply file management techniques.	
CO6: Improve Linux as a technical plat form.	

Contents –

Unit	Content	Teaching hours
1	**Introduction Operating System** Operating System overview objectives and functions, Types of Operating system, The Evolution of Operating System, Operating System Structure and Operations- System Services, Systems Calls, System Programs, Systemstructure,SystemBoot,VirtualMachinesSystem DesignandImplementation.	6
2	**Process&CPUScheduling** ProcessConcept,ProcessScheduling,andOperationson Processes,InterprocessCommunication,Threads,Inter- process Communication, Scheduling criteria	8
3	**Process Synchronization** Critical Section Problem, Scheduling algorithms and performance evaluation.Deadlock- Principles ofDeadlock, DeadlockPrevention,DeadlockAvoidance,Deadlock Detection,RecoveryfromDeadlock.	8

4	**MemoryManagement** Basic concept, Logical and Physical address map, Memory Partitioning: Fixed Partitioning, Dynamic Partitioning, Buddy System, Relocation, Paging, Segmentation, Virtual Memory - Basics of Virtual Memory—Hardwareandcontrolstructures—Localityof reference,Pagefault,WorkingSet,Dirtypage/Dirtybit—Demand paging, Page Replacement Algorithms: Optimal, First in First Out (FIFO), Second Chance (SC),Notrecentlyused(NRU)andLeastRecently used(LRU).	**8**
5	**Input/Output and File Management** **I/Omanagement:I/0** devices, Device controllers, Direct memory access, Device drivers, Secondary **Storage Structure:** Disk structure, Disk scheduling algorithms. **File Management:** Overview, File Organization and Access,FileDirectories,FileSystemstructure,Allocation methods (contiguous, linked, indexed), Free-space management (bit vector, linked list, grouping), directory implementation(linearlist,hashtable),efficiencyand Performance.	**8**
6	**CaseStudy:UNIX Operating System** Introduction,OverviewofUNIX,StructureofUNIXOS, Booting,FileSystemOfUNIX,UNIXandLINUX Comparison.LinuxDesignPrinciples,LinuxBooting Process,KernelModules,ProcessManagement,Scheduling, MemoryManagement,File Systems, Input and Output, Inter-processCommunication.	**7**

Text Books:

1.OperatingsystemConcepts - SilberschatzGalvin/JohnWileyandSons
2.Operating System– GodboleAchyut,TataMCGrowHillEducation,2015, ISBN:9780070591134

Reference Books:

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: CSS		**Course name: Communication skills**
Course category: PCC		
Credits: 3	**Teaching scheme:** L-3	**Evaluation scheme:** CA–20, ESE–30
Pre-requisites:		
Course Objectives:		
1.		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Student will be able to utilize communication skills in different formal settings.		
CO2: Student will be able to select and apply the appropriate methods of communication in various situations.		
CO3: Student will be able to develop presentation skills and body language.		
CO4: Student will be able to apply learned interview skills .		
CO5: Student will be able to draft correct format of emails and business letters.		

Contents –

Unit	Content	Teaching hours
2	**Non verbal communication** Introduction to nonverbal communication Aspects of body language (Kinesics) (Facial expression, eye contact, vocalic, paralanguage, gesture, posture, dress & appearance, haptics) **Tips for effective use of Non-verbal communication** Body language and confidence Use of body language at workplace.	**5**
3	**Interview Skills.** Introducton,Principles of Interview,Types of Interview,Preparing for Interview,Success in Interview.	**5**
4	**Group Discussion and Team Presentation.** Introduction to Group Discusssion. Methodology of Group Discussion. Guidelines for Group Discussion. Elements of effective presentation. Tips for effective team Presentation	**5**

5	**Formal Written Communication** **Office Drafting-** Notice, memo & e-mail. Job application with resume. Writing a Case Study. Report Writing	**5**
6	**Introduction to communication** Nature and scope of communication Process of communication Function of Communication Cross cultural communication Corporate communication-Effectiveness in Managerial communication	**5**

Text Books:

1. *Sanjay Kumar, Pushpa Lata* (Oxford University Press) ,Communication Skills ,Core textbook covering communication fundamentals and practice.
2. *Urmila Rai* (Himalaya Publications), Communication Skills Standard student textbook with explanations and exercises.
3. *E. Sureshkumar, P. Sreehari, J. Savithri,* English for Success(Foundation Books) ,Student-oriented book blending communication and English skill practice.

Reference Books:

1. *M. Ashraf Rizvi* , Effective Communication Skills**,** Commonly used as supplementary/reference material for communication theory and soft skills.

Online Resources:

1. https://in.indeed.com/career-advice/resumes-cover-letters/communication-skills
2. https://www.skillsyouneed.com/ips/communication-skills.html
3. https://corporatefinanceinstitute.com/resources/management/communication/
4. https://www.coursera.org/articles/communication-skills

Course code:CSS-L	**Course name: Communication skills- Lab**
category: Program Core Course	
Credits: 01 **Teaching scheme:** P-2	**Evaluation scheme:** TW–50, PR–00
Pre-requisites: 1	
Laboratory Outcomes: At the end of the session, the students will be able to -	
CO1: Student will be able to apply basic theory and principles in day to day communication	
CO2: Student will be able to effectively apply non-verbal skills.	
CO3: Student will be able to competently use communication skills in interview and presentation.	
CO4: Student will be able to use appropriate words as per the context.	
CO5: Student will be able to apply written communication in day to day life.	

List of Experiments–

Sr.No.	Name of Experiment
1	Give self-Introduction.
2	State the importance of body language in communication.
3	Prepare effective techniques for successful interview .
4	Prepare guidelines for presentation.
5	Design a power point presentation on any topic.
6	Organize debate on given topic.
7	Prepare job application with resume.
8	Discuss some important personal attributes required for a team.
9	Write some tips for professional e-mail writing.
10	Write a case study on any given situation.

Text Books:

1. *Sanjay Kumar, Pushpa Lata* (Oxford University Press) ,Communication Skills ,Core textbook covering communication fundamentals and practice.
2. *Urmila Rai* (Himalaya Publications), Communication Skills Standard student textbook with explanations and exercises.
3. *E. Sureshkumar, P. Sreehari, J. Savithri,* English for Success(Foundation Books) ,Student-oriented book blending communication and English skill practice.

Reference Books:

- *M. Ashraf Rizvi* , Effective Communication Skills**,** Commonly used as supplementary/reference material for communication theory and soft skills.

Online Resources:

1. https://in.indeed.com/career-advice/resumes-cover-letters/communication-skills
2. https://www.skillsyouneed.com/ips/communication-skills.html
3. https://corporatefinanceinstitute.com/resources/management/communication
4. https://www.coursera.org/articles/communication-skills

Course code: 22IUCI0601L **Course name: Data Communication and Networking Lab** **Course category:**Program Core Course		
Credits: 01	**Teaching scheme:** P-2	**Evaluation scheme:** TW–30, PR–20
Pre-requisites: Understanding of computer hardware (CPU, memory, storage), Programming Fundamentals, Basic computer organization		
Laboratory Outcomes: At the end of the session, the students will be able to -		
LO1: Analyse the function of data communication and computer Network		
LO2: Select relevant transmission media and switching Techniques.		
LO3: Analyse the transmission errors with respect to IEEE standards		
LO4: Configure various networking devices		
LO5: Configure different TCP/IP services		

List of Experiments–

Sr.No.	Name of Experiment
1	Configure Peer to Peer Network with at least three hosts
2	Study of network IP
3	Connect the computers in Local Area Network.
4	Create desired standard network cable including cross cable and test by using cable tester
5	Connect computer using given topology
6	Connect computers using wireless media
7	Write a C program for CRC Error Detection
8	Create a network using Bluetooth
9	Share printer and folder in a network and transfer a file from one computer to another
10	Install operating system
11	Configure file server
12	Configure DHCP server

Text Books: 1. Data communication and networkin Author: Forouzan Behrouz
Reference Books: ComputerNetworksAuthor:TanenbaumAndrewS
Online Resourceshttps://www.tutorialspoint.com https://www.w3schools.com

Course code: 22IUCI0602L **Course name: Advanced Java Programming Lab** **Course category:** PCC		
Credits: 2	**Teaching scheme:** P-4	**Evaluation scheme:** TW–60, PR–40
Pre-requisites: Core Java, Object-Oriented Programming (OOP), Java AWT (GUI Programming Basics), Java Collections & Data Structures, Basic Networking Concepts, Basic Database/SQL Concepts		
Laboratory Outcomes: At the end of the session, the students will be able to -		
LO1: Create Java programs using GUI frameworks such as AWT or Swing.		
LO2: Apply knowledge of event handling mechanisms in AWT and Swing components		
LO3: Write Java code to handle various events in a program.		
LO4: Apply networking concepts to communicate over networks. Also, handle data base operations		
LO5: Utilize Java collections to manage and manipulate data efficiently.		
LO6: Create Java to handle HTTP requests and generate dynamic web content.		

List of Experiments–

Sr.No.	Name of Experiment
1	Write a program to demonstrate the use of AWT components like Label, Textfield, TextArea, Button, Checkbox, RadioButton, List and Choice, etc.
2	Write a program to show use of different Layout.
3	Write a program using AWT to create a menubar where menubar contains menu items such as File, Edit, View and create a submenu under the File menu: New and Open.
4	Write a program using swing to display a ScrollPane and JComboBox in an Applet with the items — English, Marathi, Hindi, Sanskrit.
5	Write a program to create a JTable.
6	Write a program to create a JTree.
7	Write a program to launch a JProgressBar.
8	Write a program to demonstrate status of key on Applet window such as Key_Pressed, KeyReleased, KeyUp, KeyDown.
9	Write a program to demonstrate various mouse events using MouseListener and MouseMotionListener interface.
10	Write a program to demonstrate the use of JTextField and JPasswordField using Listener Interface.
11	Write a program to demonstrate the use of InetAddress class and its factory methods.
12	Write a program to demonstrate the use of URL and URLConnection class and its methods.
13	Write a program to implement chat Server using ServerSocket and Socket class.
14	Write a program to insert and retrieve the data from database using JDBC.
15	Write a program to demonstrate Collections Framework classes and interfaces and use of Generics.
16	Write a program to demonstrate the use of HttpServlet as a parameterized Servlet.
17	Write a Servlet program to send username and password using HTML forms and authenticate the user.
18	Write a program to create Session using HttpSession class.

Text Books: 1.PatrickNaughton-HerbertScheldt,TheCompleteReferenceJava2(ThirdEdition)TMH 2.Deitel&Deitel: “HowtoProgramJAVA”, PearsonEducation
Reference Books: 1.E.Balagurusamy, ProgrammingwithJava, BPBPublication 2.Java2 BlackbookbyStevenHolzner 3.KathySieraa&BertBates,HeadFirstJava(2ndEdition),O’reilly 4.DarrelInce&AdamFreeman,programmingtheInternet withJAVA,Addison-Wesley 5.Dr.G.T.Thampy,ObjectOrientedProgramminginJava,DreamtechPress
Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: 22IUCI0603L **Course category:**PCC		**Course name: Software Testing Lab**
Credits: 01	**Teaching scheme:** P-2	**Evaluation scheme:** TW–50, PR–00
Pre-requisites: Basic Programming Knowledge, Software Engineering		
Laboratory Outcomes: At the end of the session, the students will be able to -		
LO1: Apply basic concepts of software Testing.		
LO2: Apply various software Testing Methods.		
LO3: Prepare Test cases for different type sand levels of Testing.		
LO4: Prepare Test plan for an Application.		
LO5: Identify bugs to create defect report of given application.		
LO6: Test software for Performance measures using automated testing tools		

List of Experiments–

Sr.No.	Name of Experiment
1	Designtestcases Forsimplecalculator application.(BB Testing)
2	Design test cases for railway reservation form
3	Design test cases fore-commerce(Flipkart ,Amazon)login Form
4	Design test cases for Web Pages Testing any Web Sites
5	Write program and design test cases for the following Control and decision making statement. 1)For...Loop 2) Switch...case 3)Do...While 4)lf...else
6	Prepare test plan for an identified Mobile application.
7	Design test plan and test cases for Notepad (MS Window based) Application.
8	Prepare defect report after executing test cases for library management system
9	Prepare defect report after executing test cases for Withdrawn of amount from ATM Machine.
10	Prepare defect report after executing test cases for any login form.
11	Design and run test cases for WordPad(MS Windows based). Using an Automated tool.
12	Design and run test cases for MSWord application using an Automation Tool.

Text Books:

1. Sriniwasan Desikan, Gopalswamy Ramesh, *Software Testing: Principles and Practices*, Pearson Education, Pearson India, 2005.
   ISBN: 978-81-7758-121-8.
2. M. G. Limaye, *Software Testing: Principles, Techniques and Tools*, Tata McGraw-Hill Education, New Delhi, 2007.
   ISBN: 978-0-07-013990-9.

Reference Books:

1. Naresh Chauhan, *Software Testing: Principles and Practices*, Oxford University Press, Noida.
2. Yogesh Singh, *Software Testing*, Cambridge University Press, Bengaluru.
   ISBN: 978-1-107-65278-1.

Online Resources:

1.http://www.selenium.com
2.http://en.wikipedia.org/wiki/Test_automation
3.http://en.wikipedia.org/wiki/Software_testing#Testing_tools
4.http://www.softwaretestingsoftware.com
5.www.nptel.ac.in/courses/
6.www.toolsqa.com

Course code: 22IUCI0604L **Course name: Database Management System Lab** **Course category:** PCC
Credits: 01 **Teaching scheme:** P-2 **Evaluation scheme:** TW–30, PR–20
Pre-requisites:1 Design and create databases, 2 SQL DDL commands 3. Develop and execute PL/SQL programs
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Re calls and understands fundamental concepts related to databases.
LO2: Apply normalization rules to ensure data integrity
LO3: Apply SQL commands to create and manage databases.
LO4: Apply PL/SQLcode to implement business logic and manipulate data with in the database.
LO5: Create stored procedures and functions to en capsulatere usable logic.
LO6: Implement security measures (e.g.,access controls, encryption) to protect the data base and ensure confidentiality.

List of Experiments–

Sr.No.	Name of Experiment
1	**Create a database** and perform: i) Assign primary key. ii) Modify the table structure - add column, change the data type of column. iii) Delete the column from table. iv) Insert, update, and delete the record from table. v) Retrieve data from the table according to condition given.
2	**Create and Execute DDL commands** using SQL. ii) Apply following Integrity constraints on table: iii) Primary key, Foreign key, Unique key constraint, Null, No Null, and Check constraint.
3	**Create and Execute DML commands using SQL.**
4	Write Queries using following operators: Arithmetic Operators Comparison Operators Logical Operators Set Operator Range Searching operators (Between) Pattern matching operators (Like)
5	Write Queries using following Functions: String Arithmetic Date and time Aggregate Functions
6	Execute Queries using the Select command with Where, Having Group by and order by clauses .Execute the queries for implementation of Inner And Outer Join
7	Execute Queries using the Select command with Where, Having Group by and order by clauses.
8	Execute the queries for implementation of Inner and Outer Join
9	Implement Views: i) Create different views. ii) Insert, modify, and delete records through views. iii) Delete the views.

10	Create and Execute Indexes, Sequences, and Synonyms in SQL**.**
11	Write a PL/SQL program **using if-then-else, for, while, and nested loop.**
12	Creation of procedures, database triggers, and cursor**.**
13	Executing DCL commands using SQL**:** i) Create users. ii) Grant privileges to user. iii) Revoke privileges from users.

Text Books:

1. 1.Database System Concepts McGraw Hill in ANSIC Korth McGraw Hill Education,New Delhi,2015
2. Introduction to Database Management Systems1SRD Group McGraw Hill Education,New Delhi, 2015

Reference Books:

1. Introduction to Relational databases & SQL programming Allen McGraw HillEducation, New Delhi, 2015
2. Complete Reference:Mysql Vikram Vaswani McGraw Hill Education,NewDelhi.2015

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: 22IUCI0605L **Course category:** PCC		**Course name: Operating System Lab**
Credits: 01	**Teaching scheme:** P-2	**Evaluation scheme:** TW–50, PR–20
Pre-requisites:1		
Laboratory Outcomes: At the end of the session, the students will be able to -		
CO1: Use operating system tools to perform various functions		
CO2: Apply scheduling algorithms to calculate turnaroundtimeandaveragewaiting time.		
CO3: Learn sharing of resources without interference using mutual exclusion.		
CO4: Calculate efficiency of different memory management techniques.		
CO5: Apply file management techniques.		
CO6: Improve Linux as a technical plat form		

List of Experiments–

Sr.No.	Name of Experiment
1	Download Install and configure Ubuntu LINUX Operating system on your machine.
2	Execute General purpose and process related command on your machine
3	Execute File and Directory manipulation related command on your machine
4	Write a shell script to find greatest among three numbers ,Fibonacci series using conditional statement.
5	Write a shell script to list all files in a directory.
6	Write a shell script to find whether a give files present. Write a shell script to check read, write permission of given file.
7	Write a C Program to calculate total waiting and Turnaround time of n process with FCFS algorithm.
8	Simulate algorithm for deadlock prevention
9	Simulate a page replacement algorithm.
10	Write a C Program to implement FIFO page replacement algorithm

Reference Books:

1. OperatingsystemConcepts-SilberschatzGalvin/JohnWileyandSons

2.Operating System– Godbole Achyut,TataMCGrow HillEducation,2015,ISBN:9780070591134

Online Resources: 1.NPTEL / SWAYAM lectures.

SEM-VII

Course code: CCI24PCL401 **Course name: Artificial Intelligence** **Course category:** PCC	
Credits: 3 **Teaching scheme:** L-3	**Evaluation scheme:** CA–60, ESE–40
Pre-requisites:	
Course Objectives:	
1.	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: To understand the fundamentals of Artificial Intelligence.	
CO2: To design smart system using different search strategies of Artificial Intelligence	
CO3: Demonstrate knowledge of reasoning and knowledge representation for solving real world Problems.	
CO4: Illustrate the construction of learning and expert system.	

Contents –

Unit	Content	Teaching hours
1	**Introduction to AI**. 1.1 Introduction: History & overview of Artificial Intelligence, Evolution of AI ,Applications of AI, 1.2 Classification of AI Systems with respect to the environment, Artificial Intelligence vs Machine Learning. 1.3 Types of AI Agent, Concept of Rationality, nature of environment, structure of agents. Turing Test in AI.	**12**
2	**Problem Solving**. 2.1 Uninformed Search, depth first search, breadth first search 2.2 Heuristic Search Techniques: Generate-and-Test; Hill Climbing; Properties of A* algorithm, Best first Search; Problem Reduction. Constraint Satisfaction problem: Interference in CSPs; Backtracking search for CSPs; Local Search for CSPs; structure of CSP Problem. 2.3 Beyond Classical Search: Local search algorithms and optimization problem, local search in continuous spaces, searching with nondeterministic action and partial observation, online search agent and unknown environments	**12**
3	**Knowledge and Reasoning** 3.1 Knowledge-Based Agent in Artificial intelligence: Architecture, Approaches to designing a knowledge-based agent. 3.2 knowledge representation: Techniques of knowledge representation, Propositional logic, Rules of Inference, First-Order Logic, Forward Chaining and backward chaining in AI. 3.3 Reasoning in Artificial intelligence: Types of Reasoning and Probabilistic reasoning, Uncertainty.	**10**

4	**Learning & Game** 4.1 Learning from examples: Overview of different forms of learning, Supervised learning, Unsupervised learning, 4.2 Learning Decision Trees, regression and classification with linear model, SVM. 4.3 Ensemble learning, Reinforcement learning. Artificial neural network 4.4. Search under adversarial circumstances. 4.5 Optimal decision in game, minimax algorithm, alpha-beta pruning, games with an element of chance imperfect real time decision, stochastic games, partially observable games, stat of art game program, alternative approaches	**11**

Text Books: 1. Russell, S. and Norvig, P. 2015. Artificial Intelligence - A Modern Approach, 3rd edition,Prentice Hall.
2. J. Gabriel, Artificial Intelligence: Artificial Intelligence for Humans (Artificial Intelligence, Machine Learning), Create Space Independent Publishing Platform, First edition, 2016

Reference Books: 1. Introduction to Artificial Intelligence & Expert Systems, Dan W Patterson, PHI. 2010 2. S Kaushik, Artificial Intelligence, Cengage Learning, 1st ed.2011.
2. Ric, E., Knight, K and Shankar, B. 2009. Artificial Intelligence, 3rd edition, Tata McGraw Hill.
3. Luger, G.F. 2008. Artificial Intelligence -Structures and Strategies for Complex Problem Solving, 6th edition, Pearson
4. Alpaydin, E. 2010. Introduction to Machine Learning.2nd edition, MIT.

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: CCI24PCL402 **Course category:** PCC		**Course name: Human Computer Interaction**
Credits: 2	**Teaching scheme:** L-2	**Evaluation scheme:** CA–60, ESE–40
Pre-requisites:		
Course Objectives:		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Explain importance of HCI study and principles of user-centered design (UCD) approach.		
CO2: Develop understanding of human factors in HCI design.		
CO3: Develop understanding of models, paradigms, and context of interactions.		
CO4: Design effective user-interfaces following a structured and organized UCD process.		
CO5: Evaluate usability of a user-interface design.		
CO6: Apply cognitive models for predicting human-computer-interactions		

Contents –

Unit	Content	Teaching hours
1	**Introduction** What is HCI? Disciplines involved in HCI, Why HCI study are important? The psychology of everyday things Donald A. Norman, Principles of HCI, User-centered Design. Measurable Human factors.	**7**
2	**Understanding The Human And Human Interaction** Input-output channels, Human memory, Human emotions, Individual differences, Psychology. Ergonomics, Human errors, Models of interaction, Paradigms of Interactions, Interaction styles, Interactivity, Context of interaction, User experience.	**8**
3	**HCI Models And Theories** User Profiles, categorization of users, Goal and task hierarchy model, Linguistic model, Physical and device models, GOMS, Norman's 7 stage model, Cognitive architectures, Hierarchical task analysis (HTA), Uses of task analysis, Diagrammatic dialog design notations.	**8**
4	**Design Process** Design Rules : Principles that support usability, Design standards, Design Guidelines, What is interaction design?, The software design process, User focus, Scenarios, Navigation Design, Screen Design, Prototyping techniques, Wire-Framing, Understanding the UI Layer and Its Execution Framework, Model-View-Controller(MVC) Framework.	**7**

Text Books:

1. Alan Dix (2008). Human Computer Interaction. Pearson Education. ISBN 978-81-317-1703-5.

2. Ben Shneiderman; Catherine Plaisant; Maxine Cohen; Steven Jacobs (29 August 2013).

3. Designing the User Interface: Strategies for Effective Human-Computer Interaction. Pearson Education Limited. ISBN 978-1-292-03701-1

Reference Books:

1. Gerard Jounghyun Kim (20 March 2015). Human–Computer Interaction: Fundamentals and Practice.CRC Press. ISBN 978-1-4822-3390-2.
2. Donald A. Norman (2013). The Design of Everyday Things Basic Books. ISBN 978-0-465-07299-6.
3. Jeff Johnson (17 December 2013). Designing with the Mind in Mind: Simple Guide to Understanding User Interface Design Guidelines. Elsevier. ISBN 978-0-12-411556-9.

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: CCI24PCL403	**Course name: Network and cyber security**	
Course category: PCC		
Credits: 2	**Teaching scheme:** L-2	**Evaluation scheme:** CA–60, ESE–40
Pre-requisites: Knowledge of python programing		
Course Objectives:		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Learn key concepts, technologies, and strategies to secure information systems		
CO2: Develop basic understanding of security, cryptography, system attacks and defenses against them		
CO3: Identify the security issues in the network and resolve it		

Contents –

Unit	Content	Teaching hours
1.	Introduction to Computer Networks : Overview of computer, networks Types of networks (LAN, WAN, MAN), Network topologies and protocols, OSI and TCP/IP models, Network devices and their functions	**7**
2.	Network Design and Implementation: Network design principles, Subnetting and addressing, Routing and switching, Wireless networks, Network troubleshooting and tools Internet and Web Technologies: Internet architecture and services, HTTP, HTTPS, DNS, Web application architecture, Security considerations in web development	**8**
3.	Internet- Security Protocols: Introduction, Secure Socket Layer(SSL),Transport Layer Security(TLS),Secure Hyper Text Transfer Protocol(SHTTP),Secure Electronic Transaction(SET), SSLversus SET,3-D Secure Protocol, Email Security, WAP Security, Security in GSM, Security in 3G, IEEE 802.11 Security.	**8**
4.	Network Security Fundamentals: Introduction to network security, Authentication and access control, Encryption and cryptography, Firewalls and intrusion detection systems, Virtual Private Networks (VPNs), Intrusion, Case Study: IP Spoofing attacks.	**7**

Text Books: 1. AtulKahate, Cryptography and Network Security, 3e, McGraw Hill Education 2. John W. Rittinghouse, William M.Hancock, “Cyber security Operations Handbook”, Elsevier Pub. 3. Roberta Bragg, Mark Rhodes-Ousley, Keith Strassberg, “The Complete reference – Network Security” , Tata McGraw Hill publication
Reference Books: 1. " Network Security Essentials" by William Stallings 2. Behrouz A. Forouzan, DebdeepMukhopadhyay,Cryptography and Network Security, McGraw Hill Education. 3. VivekSood, 'Cyber Law Simplified', McGraw Hill Education
Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: CCI24PCP401 **Course category:** PCC		**Course name: Artificial Intelligence Lab**
Credits: 01	**Teaching scheme:** P-2	**Evaluation scheme:** TW–30, PR–20
Pre-requisites:1		
Laboratory Outcomes: At the end of the session, the students will be able to -		
LO1: To understand the fundamentals of Artificial Intelligence.		
LO2: To design smart system using different search strategies of Artificial Intelligence		
LO3: Demonstrate knowledge of reasoning and knowledge representation for solving real world Problems.		
LO4: Illustrate the construction of learning and expert system.		

List of Experiments–

Sr.No.	Name of Experiment
1.	Study and understand simple reflex and Model Based Agent
2.	Program to implement Uninformed Search Technique: Breadth First Search
3.	Program to implement Uninformed Search Technique: Depth First Search
4.	Write a Program to solve 8 Queen Problem
5.	Program to implement Informed Search Technique: A* Algorithm
6.	Implement Forward Chaining Algorithm
7.	Implement backward Chaining Algorithm
8.	Implement Supervised & Unsupervised Learning using Python
9.	Implement Mini-Max algorithm for game playing (Alpha-Beta pruning)
10.	Chatbot in Python

Text Books:

1. Russell, S. and Norvig, P. 2015. Artificial Intelligence - A Modern Approach, 3rd edition,Prentice Hall.
2. J. Gabriel, Artificial Intelligence: Artificial Intelligence for Humans (Artificial Intelligence, Machine Learning), Create Space Independent Publishing Platform, First edition, 2016

Reference Books:

1. Introduction to Artificial Intelligence & Expert Systems, Dan W Patterson, PHI. 2010 2. S Kaushik, Artificial Intelligence, Cengage Learning, 1st ed.2011.
2. Ric, E., Knight, K and Shankar, B. 2009. Artificial Intelligence, 3rd edition, Tata Mc Graw Hill.
3. Luger, G.F. 2008. Artificial Intelligence -Structures and Strategies for Complex Problem Solving, 6th edition, Pearson
4. Alpaydin, E. 2010. Introduction to Machine Learning. 2nd edition, MIT.

Online Resources:

1. www.tutorialspoint.com//ArtificialIntelligence/
2. https://swayam.gov.in/

Course code: CCI24 PCP402 **Course name: Human Computer Interaction Lab** **Course category:** PCC		
Credits: 01	**Teaching scheme:** P-2	**Evaluation scheme:** TW–30, PR–20
Pre-requisites:		
Course Objectives:		
Laboratory Outcomes: At the end of the session, the students will be able to -		
LO1: CO1: Explain importance of HCI study and principles of user-centered design (UCD) approach.		
LO2: Develop understanding of human factors in HCI design.		
LO3: Develop understanding of models, paradigms, and context of interactions.		
LO4: Design effective user-interfaces following a structured and organized UCD process.		
LO5: Evaluate usability of a user-interface design.		
LO6: Apply cognitive models for predicting human-computer-interactions		

List of Experiments–

Sr.No.	Name of Experiment
1.	Identify and observe bad designs Students are expected to submit minimum of 3 to 5 photographs of bad designs in their surrounding or home or any product or neighborhood and create a report mentioning why is it bad? They can submit word/pdf file having photos and description, source of photos and place and mention why is it bad and discuss the outcome during lab session.
2.	Feedback and Constraint: Products or interfaces should offer useful feedback to understand the state and have constraints to avoid mistakes while using them. Students are expected to identify and analyze minimum of 5 Curriculum for Third Year of Information Technology (2019 Course), SavitribaiPhule Pune University TE (Information Technology) Syllabus (2019 Course) 40 interfaces or products offering feedback and constraint. Prepare a report clearly showcasing feedback and constraint and support it with minimum of 5 photographs taken in their surrounding or home or neighborhood. Discuss the outcome during lab session
3.	a) Prototype and wire frame: Students are expected to choose a problem statement and identify – Types of users going to use (age, experience, environmental conditions during use etc.)Minimum 3 scenarios of use Create paper-based prototypes for scenarios. Use any open-source tool to wire frame scenarios.
4.	a) CSS: Students are expected to design minimum of 5 web pages using CSS for the problem statement chosen in assignment no. 4. Apply CSS properties Border, margins, Padding, Navigation, dropdown list to page
5.	CMS tool: Develop website using any CMS tool which falls into one of the categories blog, social networking, News updates, Wikipedia, E-commerce store. Website must include home page, and at least 5 forms. Use WordPress/ Joomla/ Drupal /PHP/ CSS/Bootstrap/ JavaScript.
6.	Evaluation of Interface: Students are expected to evaluate minimum of two products / software interface against known HCI evaluation.
7.	Usability Testing and User Feedback Report a) Conduct a usability test on an existing digital product or application (e.g., a mobile app, website, or software interface). Gather real-user feedback to evaluate its user-

	friendliness and functionality.
8.	Redesign a poorly designed user interface based on HCI principles and improve the experience for the user.
9.	Evaluate an existing website or application for accessibility compliance and improve it based on accessibility guidelines
10.	Design an interaction flow for a new feature or product and map out the user journey from start to finish.

Text Books:
1. Alan Dix (2008). Human Computer Interaction. Pearson Education. ISBN 978-81-317-17035.

2. Ben Shneiderman; Catherine Plaisant; Maxine Cohen; Steven Jacobs (29 August 2013).

3. Designing The User Interface: Strategies for Effective Human-Computer Interaction. Pearson Education Limited. ISBN 978-1-292-03701-1.

Reference Books:
1. Gerard Jounghyun Kim (20 March 2015). Human–Computer Interaction: Fundamentals andPractice. CRC

Press. ISBN 978-1-4822-3390-2.

2. Donald A. Norman (2013). The Design of Everyday Things Basic Books. ISBN 978-0-465-07299-6.

3. Jeff Johnson (17 December 2013). Designing with the Mind in Mind: Simple Guide to Understanding User Interface Design Guidelines. Elsevier. ISBN 978-0-12-411556-9

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: CCI24 PCP403 **Course name: Network and Cyber Security Lab** **Course category:** PCC		
Credits: 01	**Teaching scheme:** P-2	**Evaluation scheme:** TW–30, PR–20
Pre-requisites:		
Course Objectives:		
Laboratory Outcomes: At the end of the session, the students will be able to -		
CO1: Learn key concepts, technologies, and strategies to secure information systems		
CO2: Develop basic understanding of security, cryptography, system attacks and defenses against them		
CO3: Identify the security issues in the network and resolve it.		

List of Experiments–

Sr.No.	Name of Experiment
1.	Study of different types of Network cables (Cat5,Cat6 & Fiber optics)and practically implement the cross-wired cable and straight wired cable using crimping tool with Fiber Optics Splicing (joining)
2.	Install and configure wired and wireless NIC and transfer files between system LAN and Wireless LAN
3.	Install and configure different Network Devices: Hub, Switch, Router etc.
4.	Connect the computers in LAN /Configure LAN Network for the LAB.
5.	Configure Host IP, Subnet mask and default gateway in a system in LAN(TCP/IP configuration)
6.	Study of IKE (Internet Key Exchange) protocol in detail
7.	Establish peer to peer network connection using two systems using switch and router in a
8.	Write case study on IIB hit by Ransomware Attack
9.	Configure internet connection and use IPCONFIG, PING/Tracer and Net stat utilities to debug
10.	the network issues.(also study basic networking commands)

Text Books:

1. AtulKahate, Cryptography and Network Security, 3e, McGraw Hill Education
2. John W. Rittinghouse, William M.Hancock, “Cyber security Operations Handbook”, Elsevier Pub.
3. Roberta Bragg, Mark Rhodes-Ousley, Keith Strassberg, “The Complete reference – Network Security” , Tata McGraw Hill publication

Reference Books:

1. Network Security Essentials " by William Stallings
2. Behrouz A. Forouzan, Debdeep Mukhopadhyay, Cryptography and Network Security, McGraw Hill Education.
3. VivekSood, Cyber Law Simplified;, McGraw Hill Education.

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: MGM56VEL102 **Course category:** PCC		**Course name: Constitution of India**
Credits: 2	**Teaching scheme:** L-2	**Evaluation scheme:** CA–30, ESE–20
Pre-requisites:		
Course Objectives:		
1.		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Understand and Explain the Significance of Indian Constitution as the Fundamental Law of the Land.		
CO2: Exercise Fundamental Rights in Proper Sense at the Same Time Identify Responsibilities in National Building.		
CO3: Analyse the Indian Political System, Electoral Process, Emergency Provisions and Amendment Procedure.		
CO4: Understand the Powers and Functions of the Union, State and Local Governments in detail		

Contents –

Unit	Content	Teaching hours
1.	**Historical Background and Philosophy of Indian Constitution** Societies before and after the Constitution Adoption, Introduction to the Indian constitution, Making of the Constitution. Role of the Constituent Assembly, Preamble and Nature of Indian Constitution, Salient Features of Indian Constitution.	**8**
2.	**Fundamental Rights & Duties** Right to Equality, Right to Freedom, Right against Exploitation, Right to Freedom of Religion. Cultural and Educational Rights, Right to Constitutional Remedies, Fundamental Duties.	**8**
3.	**Directive Principles of State Policy and Other Important Constitutional Provisions** Meaning and Significance of Directive Principles, Economic Principles, Political Principles. Emergency Provisions under Indian Constitution, Election Commission.	**8**
4.	**Organs of Government** Centre and State Legislature, Centre and State Executive, Centre and State Judiciary, Local Self Government.	**6**

Text Books:

1. D. D. Basu, Introduction to the Constitution of India, Lexis Nexis, 2013.
2. Pravin Kumar, Basic Mechanical Engineering, Pearson Education; Second Edition, 2018.
3. M. P. Jain, Outline of Indian Legal and Constitutional History, Lexis Nexis, 2014 .Constitutional Law of India - J.N.Pandey.

Reference Books:

1. Austin, The Constitution of India, Cornerstone of a Nation. Oxford University press 1966.

Online Resources: 1.https://onlinecourses.nptel.ac.in/

Course code: CCI24HSL404 Course **name: Business Management & Financial Accounting** **Course category:** PCC
Credits: 2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–60, ESE–40
Pre-requisites:
Course Objectives:
1.
Course Outcomes: At the end of the course, the students will be able to -
CO1: Analyse the Risk and Return of Alternative Sources of Financing.
CO2: Estimate Cash Flows from a Project, including Operating, Net Working Capital and Capital Spending.
CO3: Define Basic Terminology used in Finance and Accounts.
CO4: Prepare Financial Statements.

Contents –

Unit	Content	Teaching hours
1.	**Introduction to Business Management** Aims, Objective and Function of Business Management, Principles of Management. Concept of Business Finance, Goals & Objectives of Financial Management, Sources of Financing-Long Term Financing and Short Term Financing.	**7**
2.	**Business Capital Management** Concept of Business Working Capital, Significance, Types. Adequacy of Working Capital, Factors Affecting Working Capital Needs. Financing Approaches for Working Capital, Methods of Forecasting, Working Capital Requirements.	**8**
3.	**Basics of Financial Accounting** Meaning, Scope and Importance of Financial Accounting, Financial Accounting - Concepts and Conventions, Classification of Accounts. Rules and Principles Governing Double Entry Book-Keeping System, Meaning, Preparation of Journal, Ledger, Cash Book & Trial Balance.	**7**
4.	**Financial Statement Preparation and analysis** Preparation of Financial Statements. Profit & Loss Account, Balance Sheet. Ratio Analysis.	**8**

Text Books:

1. Financial Management by Khan & Jain, Text, Problem & Cases, Tata McGraw Hill Publication 5th Edition.
2. Tulsian Financial Management by Dr.P.C.Tulsian, S.Chand Publication 5th Edition.
3. Taxman's Financial Management by Ravi M. Kishore, Taxmann 2017 Edition.

Reference Books:

1. A Textbook of Financial, Cost & Management Accounting by Dr.P.Periasamy, Himalaya Publishing House.
2. Fundamentals of Financial Management by Bhabhtosh Banerjee, PHI publication, 2nd Edition.

Online Resources:1. https://onlinecourses.nptel.ac.in/

Course code: CCI24FPJ405	**Course name:** Field Project **Course category:** FP
Credits: 2	**Teaching scheme:P-4** **Evaluation scheme:** TW : 50, PR:–00
Pre-requisites:	
Course Objectives:	
1. To get familiar with client-server architecture.	
2. To explore recent web technologies**.**	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Design web pages using HTML, CSS and JavaScript.	
CO2: Build appropriate server-side scripts using PHP.	
CO3: Create dynamic web application using PHP and MySQL.	
CO4: Develop solution to real world problem using appropriate web technologies and content management system.	
Contents –	

Sr. No.	List of Experiments
1	Develop informative web pages using HTML5 (Use all possible formatting for example font, colour etc.). Web page should include various images, links within the page, links to other pages for navigation, new tabs.
2	Design an online registration form for any application.
3	Use Cascaded Style Sheets(CSS) to style the web pages designed. Make use of border, margins, padding, navigation, dropdown list etc.
4	Design an XML document to store information about a student in an engineering college affiliated to MGM University. The information must include RegistrationNo, Name, Name of Brach, Year of Joining, and e- mail id. Make up sample data for 3 students. Create a CSS style sheet and use it to display the document.
5	Design, Develop and Deploy new web application using Bootstrap in consultation of course instructor.
6	Write a Java Script to design a simple calculator to perform the following operations: sum, product, difference and division.
7	Write a JavaScript that calculates the squares and cubes of the numbers from 0 to 10 and outputs HTML text that displays the resulting values in an HTML table format.
8	Develop and demonstrate a HTML5 file that includes JavaScripts that uses functions for the following problems: Parameter: A string Output: The position in the string of the left-most vowel. Parameter: A number Output: The number with its digits in the reverse order.
9	Write a JavaScript code that displays text “TEXT- GROWING” with increasing font size in the interval of 100ms in RED COLOR, when the font size reaches 50pt it displays “TEXT-SHRINKING” in BLUE colour. Then the font size decreases to 5pt.
10	Write a PHP program to keep track of the number of visitors visiting the web page and to display this count of visitors, with proper headings.

11	Write a PHP program to display a digital clock which displays The current time of the server.
12	Write a PHP program to display the student records which are stored in the database.
13	Write the PHP programs to do the following: a. Implement simple calculator operations. b. Find the transpose of a matrix. c. Multiplication of two matrices. d. Addition of two matrices.
14	Develop web site using any CMS tool which falls into one of the categories blogs, social networking, News updates, Wikipedia, E commerce store. Website must include home page, and at least 3 pages.
15	Study Experiment/Project: Develop a web application project using the languages and concepts learnt in the exercises with good looks. You can use any web technologies and Frameworks and databases.

Text Books:
1. Ralph Moseley & M.T. Savaliya,―Developing Web Applications ‖, Wiley publications, 2007
2.Kogent Learning Solutions Inc,Web Technologies, Black book, Dreamtech Press.2009

Reference Books:
1. IvanBayross, Web Enabled Commercial Application Development Using HTML, JavaScript, DHTML and PHP, BPB Publications.2005, Third Edition
2.StevenM. Schafer," HTML, XHTML and CSS", Wiley India Edition., 5th Edition 2011

Online Resources:
1. MicrosoftAcademyCoursesonHTML5&CSS.3
2.Udemy and Coursera Courses on Web Technologies.
3.Spoken Tutorial Course on PHP and MySql.
4.W3Schools,Tutorials Point tutorials on HTML,CSS,JavaScript,PHP,Bootstrap.

SEM-VIII

Course code: CCI24PCL451 **Course category:** PCC		**Course name: Machine Learning**
Credits: 2	**Teaching scheme:** L-2	**Evaluation scheme:** CA–60, ESE–40
Pre-requisites:		
Course Objectives:		
1.		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Recognize the characteristics of machine learning that makes it useful to real-world problems.		
CO2: Apply the different supervised learning methods of support vector machine and tree based models.		
CO3: Use different linear methods for regression and classification with their optimization through different regularization techniques..		
CO4: Implement and assess the results of advanced machine learning algorithms..		

Contents –

Unit	Content	Teaching hours
1	**Introduction to ML** Introduction to Machine Learning: History of ML, Examples of Machine Learning Applications Learning Types ML Life cycle, AI & ML, dataset for ML Data Pre-processing, Training versusTesting, Positive and Negative Class, Cross-validation.	7
2	**Learning** Types of Learning: Supervised, Unsupervised and Semi-Supervised Learning. Supervised: Learning a Class from Examples, Types of supervised Machine learning Algorithms, Unsupervised: Types of Unsupervised Learning Algorithm, 2.4 Dimensionality Reduction: Introductionto Dimensionality Reduction, Subset Selection Introduction to Principal Component Analysis.	7
3	**Classification & Regression** Classification: Binary and Multiclass Classification: Assessing Classification Performance, Handling more than two classes, Multiclass Classification-One vs One, One vs Rest. Regression: Assessing performance of Regression Error measures, Overfitting and Underfitting, Catalysts for Overfitting, VC Dimensions.	8

4	**Advanced Machine Learning Techniques** Decision Trees and Random Forests, Support Vector Machines (SVM), Clustering algorithms: K-Means, Hierarchical clustering, Neural Networks and Deep Learning basics, Introduction to TensorFlow and PyTorch.	**8**

Text Books:

1. Peter Flach: Machine Learning: The Art and Science of Algorithms that Make Sense of Data, Cambridge University Press, Edition 2012.

Reference Books:

1“Introduction to Machine Learning”,EthemAlpaydin, PHI 2nd Edition-2013.
2."Introduction to Machine Learning with Python: A Guide for Data Scientists" by Andreas C. Müller and Sarah Guido.
3. “Elements of Statistical Learning" by Trevor Hastie, Robert Tibshirani, and Jerome Friedman.
4."Python Machine Learning" by Sebastian Raschka and VahidMirjalili
5."Reinforcement Learning: An Introduction" by Richard S. Sutton and Andrew G. Barto

Online Resources:

1. https://www.tutorialspoint.com/machine_learning/index.htm
2.https://www.geeksforgeeks.org/machine-learning/
3. https://onlinecourses.nptel.ac.in/

Course code: CCI24PCL452 **Course category:** PCC		**Course name: Cloud Computing**
Credits: 2	**Teaching scheme:** L-2	**Evaluation scheme:** CA–60, ESE–40
Pre-requisites: Basic knowledge of programming (preferably in Python or Java) and introductory statistics.		
Course Objectives:		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Maintain Cloud Based Application.		
CO2: Implement virtualization in Cloud Computing.		
CO3: Maintain Storage System in Cloud.		
CO4: Maintain Cloud Services.		

Contents –

Unit	Content	Teaching hours
1.	**Cloud Computing Fundamentals** Cloud Computing, Essential characteristics of cloud computing Cloud Deployment Model: Public cloud, Private cloud, Community cloud, Hybrid cloud Cloud Service Models: laaS, PaaS, SaaS Cloud Economics and Benefits Architecture of Cloud Computing Cloud Computing Infrastructure Cloud-Based Integrated Development Environment (IDE) to write, run, and debug code with a browser.	**7**
2.	**Cloud Security** Introduction, Virtualization Reference Model, Characteristics of virtualized environment Virtualization Types Technology Example: VMWare, Microsoft Hyper-V, KVM, Xen Advantages: Virtual Machine(VM), VM Migration, VM Consolidation, VM Management Disadvantages of Virtualization	**8**

3.	**Risk And Management** System Architecture, Virtualize Data Centre (VDC) Architecture, VDC Environment, server, storage, networking, desktop and application virtualization techniques and benefits. Block and file level storage virtualization, Virtual Provisioning, and automated storage tiering, Virtual storage area network(VSAN) and benefits, Cloud file systems: Google File System GFS and Hadoop Distributed File System HDFS**,**	**8**
4.	**Cloud Compliance and Disaster Recovery Strategies** Service Provider and users An architecture of federated cloud computing Service Level Agreement (SLA) management: Types of SLA, Life cycle of SLA. Service catalog, management and functional interfaces of services, Cloud portal and its functions Cloud Service life cycle phases: Service planning, service creation, service operation and service termination 4.7 Cloud resource management Ab-initio Resource Assignment Periodic Resource Optimization	**7**

Text Books:

1. Cloud Computing, Principals and Paradigms,Buyya Rajkumar, J. Broberg, A. Goscinski, A John Wilwy & Sons, Inc., Pubication, ISBN: 978-0-470-88799-8
2. Cloud Computing, Sharma Rishabh Wiley Publication, ISBN: 978-81-265-5306-8.
3. Mastering Cloud Computing, Buyya Rajkumar, Vecchiola Christian, Selvi S Thamarai, McGraw Hill Publication, ISBN 978-1-25-902995-0
4. Cloud Computing,Singh Shailendra. Oxford University Press, ISBN: 9780199477388

Reference Books:

1. J. Arundel and J. Domingus, *Cloud Native DevOps with Kubernetes*. Sebastopol, CA, USA: O'Reilly Media, 2019.
2. M. J. Kavis, *Architecting the Cloud: Design Decisions for Cloud Computing Service Models (SaaS, PaaS, and IaaS)*. Indianapolis, IN, USA: Wiley, 2014.

Online Resources:

1. https://www.geeksforgeeks.org/
2. https://onlinecourses.nptel.ac.in/

Course code: CCI24PCL453 Course name: Ethical Hacking and Information Security **Course category:** PCC
Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–60, ESE–40
Pre-requisites: Computer Network.
Course Objectives:
Course Outcomes: At the end of the course, the students will be able to -
1. Introduce the fundamentals of ethical hacking and information security, including hacker classifications, security models, and legal and ethical issues. 2. Provide knowledge of reconnaissance, footprinting, and network scanning techniques along with associated threats, tools, and counter measures. 3. Familiarize students with enumeration, sniffing, and password-cracking techniques and methods to secure systems and networks. 4. Create awareness about malware, cyber attacks, social engineering, IoT threats, and modern security challenges, along with effective prevention strategies. Course Outcomes: At the end of the course, the students will be able to -
CO1: Understand ethical hacking concepts, hacker types, legal issues, and information security models. **CO2:** Analyze footprinting, reconnaissance, and network scanning techniques with countermeasures. **CO3:** Apply enumeration, sniffing, and password-cracking techniques to identify vulnerabilities. **CO4**: Evaluate cyber attacks, malware, and social engineering threats and suggest defenses.

Contents –

Unit	Content	Teaching hours
1.	**Introduction to Ethical Hacking and Information Security** Introduction to Ethical Hacking, Hacker Classifications and their role. Legal and ethical consideration in hacking, Techniques of Hacking. Introduction to Information Security, Information Security Models: Computer Security, Network Security, Service Security, Application Security, Security Architecture Information Security Program: The Process of Information Security, Component Parts of Information Security Program, Risk Analysis and Ethical Hacking,Active and Passive Threats and Exploits	12

2.	**Foot printing, Reconnaissance and Scanning networks** What is foot printing? Active and passive foot printing, purpose of foot printing, Objectives of foot printing, foot printing threats, Types of foot printing, foot printing countermeasures. Scanning networks: Network scanning and its types, o objectives of network scanning, scanning live systems, scanning techniques-TCP Connect / Full Open Scan, Types of Stealth scans, port scanning countermeasures, IDS evasion techniques, Banner grabbing and its tools, vulnerability scanning, proxy servers, anonymizers, IP spoofing and its countermeasures.	12
3.	**Enumeration and sniffing, Password Cracking** Enumeration and Sniffing: What is Enumeration? Enumeration techniques, Enumeration types, Enumeration countermeasures, what is sniffing? Wire trapping and its types, packet sniffing, sniffing threats, how sniffers work?, sniffing methods-ARP spoofing and MAC flooding, active and passive sniffing, types of sniffing attacks, sniffing countermeasures, Sniffing detection techniques Password Hashes, Password Cracking types, Dictionary Attack, Brute Force Attacks, Cracking Passwords using John the Ripper, Other password Cracking tools, How passwords are stored in Linux,/etc/passwd and /etc/shadow, How passwords are stored in Windows, Testing SSH Password and Hardening of SSH, Password Cracking Countermeasures.	12
4.	**Trojans and other Attacks:** Worms, viruses, Trojans, Types of worms, viruses and worms, Preventing malware attacks, types of attacks: (DoS / DDoS), Waterhole attack, brute force, phishing and fake WAP, Eavesdropping, Man-in-the-middle, buffer overflow, DNS poisoning, ARP poisoning, Identity Theft, IoT Attacks, BOTs and BOTNETs, Steganography - text, image and audio and video, types of Social Engineering: Physical social engineering, Remote social engineering and hybrid social engineering.	9

Text Books:

1. EC-Council "Ethical Hacking and Countermeasures Attack Phases", Cengage Learning
2. James S. Tiller, "The Ethical Hack: A Framework for Business Value Penetration Testing", Auerbach Publications, CRC Press.
3. The Basics of Hacking and Penetration Testing, Patrick Engebretson 2nd edition Syngress.
4. Ethical Hacking A Comprehensive Beginner's Guide to Learn and Master Ethical Hacking, By Hilary Morrison, hein smith · 2018, Create Space Independent Publishing Platform

Reference Books:

1. EC-Council, "Ethical Hacking and Countermeasures Attack Phases", Cengage Learning.

2. Michael Simpson, Kent Backman, James Corley, "Hands-On Ethical Hacking and Network Defense", Cengage Learning.

3. Matt Walker, All-In-One-CEH-Certified-Ethical-Hacker-Exam Guide

C

Course code: CCI24PCP451 **Course category:** PCC		**Course name: Machine Learning Lab**
Credits: 01	**Teaching scheme:** P-2	**Evaluation scheme:** TW–30, PR–20
Pre-requisites:1		
Laboratory Outcomes: At the end of the session, the students will be able to -		
CO1: Recognize the characteristics of machine learning that makes it useful to real-world problems		
CO2: Apply the different supervised learning methods of support vector machine and tree based models.		
CO3: Use different linear methods for regression and classification with their optimization through different regularization techniques.		
CO4: Implement and assess the results of advanced machine learning algorithms.		

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1.	Write a python program to import and export data using Pandas library functions.	2
2.	Cleanse and preprocess a dataset, handling missing values and outliers.	2
3.	Use a dataset to implement and visualize a simple linear regression model.	2
4.	Evaluate machine learning models using accuracy, precision, recall, and F1 Score.	2
5.	Implement a decision tree and a random forest model on a dataset.	2
6.	Apply SVM to classify data points in a two-dimensional space.	2
7.	Implement K-Means clustering on a dataset.	2
8.	Build a basic neural network using a popular library like TensorFlow or PyTorch.	2
9.	Use a pre-trained model or train a basic model for image recognition.	2
10.	Case Study	2

Text Books:

1. Peter Flach: Machine Learning: The Art and Science of Algorithms that Make Sense of Data, Cambridge University Press, Edition 2012.

Reference Books:

1.“Introduction to Machine Learning”,EthemAlpaydin, PHI 2nd Edition-2013.
2."Introduction to Machine Learning with Python: A Guide for Data Scientists" by Andreas C. Müller and Sarah Guido.
3.“Elements of Statistical Learning" by Trevor Hastie, Robert Tibshirani, and Jerome Friedman.
4."Python Machine Learning" by Sebastian Raschka and VahidMirjalili
5."Reinforcement Learning: An Introduction" by Richard S. Sutton and Andrew G. Barto

Online Resources:

1. https://www.tutorialspoint.com/machine_learning/index.htm
2. https://www.geeksforgeeks.org/machine-learning/
3. https://onlinecourses.nptel.ac.in/

Course code: CCI24PCP452 **Course name: Cloud Computing Lab** **Course category:**PCC
Credits: 01 **Teaching scheme:** P-2Hrs/Week **Evaluation scheme:**CA–0, MSE–0, ESE–0, TW–30, PR–20
Pre-requisites: Basic knowledge of programming (preferably in Python or Java) and introductory statistics.
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Maintain Cloud Based Application.
LO2: Implement virtualization in Cloud Computing.
LO3: Maintain Storage System in Cloud.
LO4: Maintain Cloud Services.

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1.	Use Goggle Doc to make spreadsheet and notes	2
2.	Install/Configure cloud using JustCloud	2
3.	Use Cloud 9 to demonstrate use of different language.	2
4.	Create/Delete Virtual Machines using VMware (Private Cloud)	2
5.	Implement Storage Service on Cloud using OpenStack	2
6.	Use Open Stack for File Management	2
7.	Monitor cloud using Nagios Tool	2
8.	Create and Host Simple Web Application on Microsoft Azure/Google cloud/Any cloud platform(Part-I)	2
9.	Create and Host Simple Web Application on Microsoft Azure/Google cloud/Any cloud platform (Part-II)	2
10.	Implement Identity Management and Access Management using OpenStack	2

Reference Books:

1.J. Arundel and J. Domingus, *Cloud Native DevOps with Kubernetes*. Sebastopol, CA, USA: O'Reilly Media, 2019.

2.M. J. Kavis, *Architecting the Cloud: Design Decisions for Cloud Computing Service Models (SaaS, PaaS, and IaaS)*. Indianapolis, IN, USA: Wiley, 2014.

Online Resources:

1. https://www.geeksforgeeks.org/
2. https://onlinecourses.nptel.ac.in/

Course code: CCI24PCP453 Course name: Ethical Hacking and Information Security Lab Course category: PCC		
Credits: 01	**Teaching scheme:** P-2	**Evaluation scheme:** TW–30, PR–20
Pre-requisites: Fundamentals of Data Structures, Basic programming language.		
Laboratory Outcomes: At the end of the session, the students will be able to -		
LO1: Understand how to gather and review information related using different foot printing techniques		
LO2: Understand the different types of password attacks and assess the password hashes and Password strength .		
LO3: Understand Types of Virus, Trojans, Backdoor, and Key logger		
LO4: Understand website information like archived website and extract contents of a website.		

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1.	Foot printing and Reconnaissance: Performing foot printing using Google Hacking, website information, information about an archived website, to extract contents of a website, to trace any received email, to fetch DNS information.	2
2.	Scanning networks, Enumeration and sniffing: Use port scanning, Network scanning tools, IDS tool, sniffing tool and generate reports.	2
3.	Malware Threats: Worms, viruses, Trojans: Use Password cracking, Dictionary attack, Encrypt and decrypt passwords, DoS attack, ARP poisoning in windows, ipconfig, ping, net stat, trace route, Steganography tools.	2
4.	Developing and implementing malwares : Creating a simple key logger in python, creating a virus, creating a Trojan.	2
5.	Hacking web servers, web applications: Hacking a website by Remote File Inclusion, Disguise as Google Bot to view hidden content of a website, to use Kaspersky for Lifetime without Patch	2
6.	SQL injection and Session hijacking : SQL injection for website hacking, session hijacking.	2
7.	Using Cryptool to encrypt and decrypt password, implement encryption and decryption using Ceaser Cipher	2
8.	Pen testing : Penetration Testing using Metasploit and metasploitable,	2

Text Books:

1. EC-Council “Ethical Hacking and Countermeasures Attack Phases”, Cengage Learning
2. James S. Tiller, “The Ethical Hack: A Framework for Business Value Penetration Testing”, Auerbach Publications, CRC Press.
3. The Basics of Hacking and Penetration Testing, Patrick Engebretson 2nd edition Syngress.
4. Ethical Hacking A Comprehensive Beginner's Guide to Learn and Master Ethical Hacking, By Hilary Morrison, hein smith · 2018, Create Space Independent Publishing Platform

Reference Books:

1. EC-Council, "Ethical Hacking and Countermeasures Attack Phases", Cengage Learning.

2. Michael Simpson, Kent Backman, James Corley, "Hands-On Ethical Hacking and Network Defense", Cengage Learning.

3. Matt Walker, All-In-One-CEH-Certified-Ethical-Hacker-Exam Guide

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: CCI24VSP455 **Course name: UI UX Design Lab** **Course category:** VSEC
Credits: 02 **Teaching scheme:** P-4 **Evaluation scheme:** TW–60, PR–40
Pre-requisites:1. Knowledge of python programing, & Front End desing
Course Objectives: 1. Explain design thinking concept 2. Select appropriate visual design for given problem. 3. Create interactions using design tool 4. Create innovative design prototype for given applications.
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Use design tool to collect user requirements.
LO2: Use any Design tool to create a 'blank project'.
LO3: Create library/repository of created assets in the design tool.
LO4: Use components and navigations to design quiz like page in design tool.

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1.	Use Design tool for user requirement collection and analysis	2
2.	Use Design tool for user requirement collection and analysis of various interfaces such as kiosks	2
3.	Recreate a given user interface using any open source design tool	2
4.	Create grid system for the given screen using any design tool	2
5.	Design given user interface using various components such as auto-layouts in the design tool	2
6.	Use horizontal scrolling to create pages for given website/ App	2
7.	Use vertical scrolling for a given website/ App	2
8.	Recreate given website for UI design, color, images, interactions, menu	2
9.	Create navigations for the given website/ App	2
10.	Design a quiz for given user interface	2
11.	Create any two gamification effects for given user interface in given scenario	2
12.	Create gamification for task completion in website such as LMS/ retail website/ banking website	2
13.	Create any five micro animations for the given user interface in given scenario	2
14.	Create prototyping with different interactions – tab, click, hover, delay. for the given user interface	2
15.	Convert created prototype in HTML page(s)	2
16.	Mini Porject	2

Text Books:

1. Kilian Langenfeld, Design Thinking for Beginners

Reference Books:

1. Jesse James Garrett , The Elements of User Experience: User-Centered Design for the Web and Beyond , New Riders Publishing, 201 West 103 Street, Indianapolis, IN 46290 800-545-5914 ISBN:978-0-321-68368-7
2. Fabio Staiano, Designing and Prototyping Interfaces with Figma, Packt Publishing Ltd, Grosvenor House, 11 St Paul's Square, Birmingham, B3 1RB ISBN-10: 180056418X

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: CSI24HSL456	**Course name: Entrepreneurship Development**	
Course category: PCC		
Credits: 2	**Teaching scheme:** L-2	**Evaluation scheme:** CA–60, ESE–40
Pre-requisites:		
Course Objectives:		
1.		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: To Develop Skills related to Various Functional Areas of Management (Marketing Management, Financial Management, Operations Management, Personnel Management etc.).		
CO2: To Develop Skills related to Project Planning And Business Plan Development.		
CO3: To Determine and Compare Various Skills of Successful Entrepreneurs, Business Organizations and Business Laws.		
CO4: To Examine Essentials to Avoid Failure in Entrepreneurship		

Contents –

Unit	Content	Teaching hours
1.	**Introduction to Entrepreneurship Development** Entrepreneurship Development, Evolution of Entrepreneurship, Meaning, Definition and Concept of Enterprise. Concepts of Entrepreneurship, Role of Entrepreneurship in Economic Development, Factors Affecting Entrepreneurship. Entrepreneurial Development models and Theories, Entrepreneurial Trait Tests, Business Opportunity Identification.	**8**
2.	**Entrepreneur** Major Types of Entrepreneurship, Techno Entrepreneurship, Women Entrepreneurship, Social Entrepreneurship, Intrapreneurship (Corporate entrepreneurship), Rural Entrepreneurship, Family Business etc. The Entrepreneurial Decision Process, Entrepreneurial Environment – Political, Legal, Technological, Natural, Economic, Socio – Cultural etc.	**7**
3.	**Project Planning, Skills for Successful Entrepreneurs & Business Organizations and Business Laws:** Project Report, Project Appraisal, Setting Up an Industrial Unit – Procedure And Formalities in Setting Up an Industrial Unit; Business Plan Development. Communication Skills, Creativity and Problem-Solving, Innovation, Negotiation Skills, Risk Management. Types of Business Organizations – Sole, Proprietorship, Joint Hindu Family Business, Partnership, Limited Liability Partnership (LLP), Corporate Governance, Franchising, Business Laws in India to Start Business.	**8**
4.	**Concepts of Entrepreneurship Failure** Issues of Entrepreneurial failure, Reasons of Entrepreneurial Failure, Essentials to Avoid Failure in Entrepreneurship. Case Study of Failure in Entrepreneurship.	**7**

Text Books: 1. Dynamics of Entrepreneurship Development by Vasant Desai. 2. Entrepreneurship Development and small business management by Poornima M. Charantimath. 3. Dynamics of Entrepreneurship Development by Vasant Desai.
Reference Books: 1.Innovation and Entrepreneurship by Peter F. Drucker 2.Kathleen R Allen, Launching New Ventures, An Entrepreneurial Approach, Cengage Learning, 2016.Vijay Sathe, Corporate Entrepreneurship, Cambridge, 2009.
Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: MGM24VEL201 **Course name:** Energy Conservation & Management
Course category: VEC
Credits: 2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–30, ESE–20
Duration of ESE: 01 Hour
Pre-requisites: Industrial Automation
Course Objectives:
1. To introduce the fundamentals of energy conservation, Indian energy scenario, and national energy policies.
2. To develop understanding of energy auditing methodologies, instruments, and energy management systems.
3. To explain the relationship between energy use, environment, climate change, and sustainable development.
4.To identify and evaluate energy conservation opportunities in electrical and thermal systems.
Course Outcomes: At the completion of the course, students should be able to:
CO1: Explain energy conservation principles, Indian energy scenario, and provisions of the Energy Conservation Act and related policies.
CO2: Perform basic energy audits using standard procedures and instruments, and interpret energy audit reports.
CO3: Analyze the impact of energy usage on environment and climate change, and describe global and national mitigation mechanisms.
CO4: Identify and recommend suitable energy conservation measures for electrical and thermal systems

Contents –

Unit	Content	Teaching hours
1	**Fundamentals of Energy Conservation** **Energy Scenario:** Classification of Energy, Indian energy scenario, Sectorial energy consumption (domestic, industrial and other sectors), energy needs of growing economy, energy intensity, long term energy scenario, energy pricing, energy security, energy conservation and its importance, energy strategy for the future. **Energy Conservation Act 2001 and related policies:** Energy conservation Act 2001 and its features, notifications under the Act, Schemes of Bureau of Energy Efficiency (BEE) including Designated consumers, State Designated Agencies, Electricity Act 2003, Integrated energy policy, National action plan on climate change, ECSBC code for Building Construction 2024	08
2	**Energy Audit And Instruments** Definition, objectives, and types of energy audit (preliminary and detailed). Methodology of conducting energy audit. Energy audit report format. Instruments used in energy auditing: Lux meter, power analyser, data logger,	08

	flow meter, etc. Roll of BEE, Energy Auditors and managers. Management. Bureau of Energy Efficiency (BEE) and its role. Energy Conservation Act 2001 and its features. Star rating of appliances. ISO 50001: Energy Management System (basic overview). Government initiatives and programs (PAT, ECBC, UJALA, etc.) Case studies of energy audits in industries and commercial buildings.	
3	**Energy and environment, air pollution, climate change** United Nations Framework Convention on Climate Change (UNFCC), sustainable development, Kyoto Protocol, Conference of Parties (COP), Clean Development Mechanism (CDM), CDM Procedures case of CDM – Bachat Lamp Yojna and industry; Prototype Carbon Fund (PCF). Renewable Energy for Industrial development	06
4	**Energy Conservation Opportunities** Electrical: Buildings & Lighting Systems, Motors, Pumps, Transformers Power Transmission & Distribution Thermal: Boilers, Furnaces & Waste Heat Recovery Systems, Cogeneration Systems, HVAC, Cooling Towers & DG Systems	08

Text Books:

1. K. V. Sharma, P. Venkataseshaiah, Energy management and conservation I K International Publishing House Pvt. Ltd, ISBN-13:978-9381141298
2. . Chetan singh, solanki, Renewable energy technologies: a practical guide for beginners, PHI Learning Pvt. Ltd., 2008 ISBN 8120334345, 9788120334342.
3. Shukla, Shailendra K, Green building technologies, 2023, OM Publications New Delhi.
4. Sustainable Smart Homes and Buildings with Internet of Things, ramod Singh Rathore, Abhishek Kumar, Surbhi Bhatia, Arwa Mashat, Thippa Reddy Gadekallu, 2025 Scrivener Publishing LLC

Reference Books:

1. Sam Kubba, Handbook of Green Building Design and Construction, Butterworth-Heinemann; 1st edition (30 July 2012); CBSPD - NEW DELHI -110092 , ISBN13 :978-0123851284.
2. Bureau of Energy Efficiency (BEE), Energy Conservation & Sustainable Building Code (ECSBC) – India 2024.
3. J. Vijayalaxmi, Sustainable Building Performance Towards a Greener India, Springer.

SEM-IX

Course Code:CCI24PCL501 Course Name: Cyber and Digital Forensics **Course Category:** PCC
Credits: 2 **Teaching scheme:** L-2 **Evaluation Scheme:** CA:60, ESE:40
Duration of External Exam: 2 Hrs.
Pre-requisites: Basics knowledge of cyber security.
Course Objectives: 1. Illustrate forensic duplication and file system analysis 2. Identify the need of digital forensic and role of digital evidences 3. Compare the use of various tools for data recovery 4. Assess the network forensics to collect digital evidences
Course Outcomes: At the end of the course, the students will be able to – **CO1**: Understand the fundamentals of Ethical Hacking and Cyber Crime **CO2**: Analyze and apply Digital Forensics **CO3**: Implement a Computer Security Incident Response **CO4**: Perform Forensic Duplication and Imaging **CO5**: Analyze and Investigate Network Forensics

Contents:

Unit No.	Content	Teaching Hours
1.	**Ethical Hacking and Cyber Crime Fundamentals** Introduction to Hacking and Ethical Hacking,Difference between Hacking and Ethical Hacking,Steps of Ethical Hacking (Reconnaissance to Reporting,Tools used for Ethical Hacking (Overview),Introduction to Cyber Crime,Types and Categories of Cyber Crime, Prevention from Cyber Crime	8
2.	**Digital Forensics and Digital Evidence** Introduction to Digital Forensics and its Need,Rules and Principles of Digital Forensics,Types of Digital Forensics (Computer, Mobile, Network, Cloud)Digital Evidence: Types and Characteristics,Collection and Preservation of Digital Evidence,Challenges in Digital Evidence Handling	7
3.	**Incident Response and Forensic Duplication** Introduction to Computer Security Incidents,Goals and Importance of Incident Response,Incident response methodology,Introduction to Forensic Duplication,Forensic Image Formats(RAW,E01,AFF),Traditional Duplication and Live System	8

	Duplication,Forensic Duplication Tools	
4.	**Disk, Network Forensics and Forensic Tools** Disk and File System Analysis – Media Analysis Concepts,File System Abstraction Model, Partition Identification and Recovery,Virtual Machine Disk Images and Forensic Containers,Data Analysis Methodology, Investigating Applications and Malware Handling,Network Forensics – Traffic Analysis and Evidence Collection,Network Forensics: Routers Investigation, Password Cracking and Routers as Response Tools,Forensic Tools – Need, Types and Tasks Performed	7

Text Books:

1. Nilakshi Jain, Dhananjay Kalbande, Digital Forensic: The fascinating world of Digital Evidences, Wiley India Pvt Ltd.
2. Jason Luttgens, Matthew Pepe, Kevin Mandia, Incident Response and computer forensics, Tata McGraw Hill

Reference Books:

1. Cory Altheide, Harlan Carvey, Digital forensics with open-source tools, Syngress Publishing, Inc
2. Asey, E. (2011). Digital evidence and computer crime: Forensic science, computers and the internet. Academic Press

Online Resources: 1. https://onlinecourses.swayam2.ac.in/cec20_lb06

Course code: CCI24PCL502 Course name: Cyber Law, Ethics & Security Policies
Course category: PCC
Credits: 2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Cyber Security
Course Objectives: 1. To introduce students to the fundamental concepts of cyber law and the legal framework governing cyberspace at national and international levels 2. To make students aware of cyber-crimes, cyber offenses, and the legal procedures for investigation, adjudication, and punishment. 3. To develop an understanding of ethical issues in computing, including professional ethics, digital responsibility, and ethical decision-making in cyber environments. 4. To impart knowledge of data protection, privacy laws, intellectual property rights, and compliance requirements in the digital domain.
Course Outcomes: At the end of the course, the students will be able to -
CO1: Understand and explain fundamental concepts of cyber law, cyber space, and the legal framework governing digital activities.
CO2: Identify various types of cyber-crimes and analyse relevant laws, acts, and penalties applicable at national and international levels. problems
CO3: Apply ethical principles and professional codes of conduct to resolve ethical issues and dilemmas in computing and cyber environments.
CO4: Evaluate real-world cyber law cases, ethical violations, and security breaches to propose appropriate legal, ethical, and policy-based solutions.

Contents –

Unit	Content	Teaching hours
1.	**Introduction to Cyber Law** • Cyberspace: concept, nature, and characteristics • Need and scope of cyber law • Cyber law framework: national and international perspectives • Information Technology Act, 2000: objectives, features, and amendments • Digital signatures, electronic records, and electronic contracts • Authorities under IT Act: CCA, Certifying Authorities, Adjudicating Officers • Jurisdiction issues in cyberspace	**7**

2.	**Cyber Crimes and Legal Provisions** • Classification of cyber-crimes: against individuals, property, organizations, and government • Common cyber-crimes: hacking, phishing, identity theft, cyber stalking, cyber fraud • Cyber terrorism and cyber warfare • Offences and penalties under IT Act, 2000 • Investigation of cyber-crimes and digital evidence • Cyber-crime reporting mechanisms in India • Case studies on cyber law violations	8
3.	**Introduction to cyber ethics** • Ethical issues in computing and cyberspace • Professional ethics and codes of conduct (ACM / IEEE) • Intellectual Property Rights: copyright, patents, trademarks in cyberspace • Software piracy, plagiarism, and ethical hacking • Privacy issues and data protection • Social and ethical challenges of emerging technologies	**7**
4.	**Information Security and Security Policies** • Information security concepts: confidentiality, integrity, availability • Security threats, vulnerabilities, and attacks • Security governance and risk management • Organizational security policies: types and importance • Access control policies and authentication mechanisms • Data protection and privacy policies • Security standards and compliance	**8**

Text Books: 1. Cyber Security Governance, Policy, Ethics & Law – Kindle Edition by David Tuffely 2. Cyber Laws and IT Protection by Harish Chander & Gagandeep Kaur
Reference Books: 1. Cyber Law – An exhaustive section-wise Commentary on The Information Technology Act – Pavan Duggal 2. Introduction to Information Security and Cyber Laws – Surya Prakash Tripathi, Ritendra Goel & Praveen Kumar Shukla
Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: CCI24PCL503	**Course Name: Blockchain Architecture**
Course Category: PCC	
Credits: 2	**Teaching scheme:** L-2 Evaluation **scheme:** CA–60, ESE–40
Pre-requisites: Computer Networks, Cryptography	
Course Objectives:	
1. To understand the fundamentals of blockchain, its evolution, and the concept of decentralization. 2. To study blockchain architecture, versions, and the concept of smart decentralized ecosystems with cryptocurrencies. 3. To explore blockchain mechanisms like proof systems, hashing, and Bitcoin's transaction and payment structures. 4. To compare and design blockchain platforms and assess their applicability in different domains.	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Understand the basic structure, history, and types of blockchain systems and decentralization methods..	
CO2: Analyze blockchain architecture, smart contracts, and cryptocurrencies in various real-world use cases.	
CO3: Apply cryptographic principles and consensus algorithms in Bitcoin transactions and wallet mechanisms.	
CO4: Evaluate and design distributed applications using modern blockchain platforms.	

Contents –

Unit	Content	Teaching hours
1.	**Unit I: Introduction to Blockchain Technology** Discover Blockchain Technology: Blockchain, Growth of blockchain technology, Distributed systems, History of blockchain and Bitcoin, Types of blockchain. Decentralization: Methods of decentralization, Routes of decentralization, Blockchain and full ecosystem decentralization, Smart contracts, Decentralized organizations and platforms for decentralization	8
2.	**Unit II: Decentralization and Smart Ecosystem** Blockchain: Architecture, Versions, Variants, Use cases, Life use cases of blockchain, Blockchain vs shared Database, Introduction to cryptocurrencies, Types, Applications. Bitcoins: Introducing Bitcoin, Bitcoin digital keys and addresses, Transactions, Blockchain mining. Alternative Coins. Limitations of Bitcoin.	7

3.	**Unit III: Blockchain Architecture, Cryptocurrencies, and Bitcoin** Concept of Double Spending, Hashing, Proof of work. Proof of Stake. Bitcoin Network and payments, Bitcoin network, Wallets, Bitcoin payments, Innovation in Bitcoin, Bitcoin Clients and APIs.	8
4.	**Unit IV: Blockchain Platforms and Applications** Introduction to Blockchain Platforms: Ethereum, Hyperledger, IOTA, EOS, Multichain, Bigchain, etc. Advantages and Disadvantages, EthereumvsBitcoin, Design a new blockchain, Potential for disruption, Design a distributed application, Blockchain applications..	7

Text Books:

1) I. Bashir, Mastering Blockchain: Unlocking the Power of Cryptocurrencies, Smart Contracts, and Decentralized Applications, 4th ed., Packt Publishing, 2023.

2) A. D. Lewis, Blockchain and the Law: The Rule of Code, Harvard University Press, 2021

Reference Books:

1) M. Swan, Blockchain: Blueprint for a New Economy, O'Reilly Media, 2015.

2) R. Wattenhofer, The Science of the Blockchain, Inverted Forest Publishing, 2016.

3) D. Drescher, Blockchain Basics: A Non-Technical Introduction in 25 Steps, Apress, 2017.

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: CCI24PCL504	**Course Name: Big Data**
Course Category: PCC	
Credits: 2 **Teaching scheme:** L-2	**Evaluation scheme:** CA–60, ESE–40
Pre-requisites: Fundamentals of Programming, Basic Knowledge of Databases and SQL	
Course Objectives:	
1. Understand the characteristics, challenges, and technologies of Big Data. 2. Learn various Big Data storage and management techniques. 3. Explore tools like Hive and Pig for data analysis and processing. 4. Develop skills to implement Big Data processing solutions for real-world applications.	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Understand characteristics and technologies of Big Data..	
CO2: Apply relevant Big Data Storage techniques for a given application.	
CO3: Apply Big Data Technologies Hive and Pig to handle different nature of data.	
CO4: Analyze Big Data Processing Technique for a given application.	

Contents –

Unit	Content	Teaching hours
1.	**Unit - I Introduction to Big Data Technologies** 1.1 Introduction :Definition of Big Data, Characteristics of big data, types of big data 1.2 Applications of big data 1.3 Difference between big data and traditional data in terms of volume, velocity, variety, veracity, value, scalability 1.4 Big data technology-definition, types : operational and analytical 1.5 Classification of big data technologies based on its purpose: Data Storage, Data Mining, Data Analytics, Data Visualization	**7**
2.	**Unit - II Big Data Ecosystem and Architecture** 2.1 Data Storage Solutions: Hadoop architecture, HDFS features, HDFS architecture and operations 2.2 Map Reduce: The Map Task, The Reduce Task, grouping by key, Partitioner and combiners, Detailed Map-Reduce execution Workflow with basic word count, max-temperature finding example using python library 2.3 NoSQL databases: NoSQL data architecture , key value stores, family stores, graph stores and document stores, RDBMS vs NoSQL 2.4 MongoDB: Introduction to MongoDB, key features, Operation on database using MongoDB: CRUD	**8**

3.	**Unit - III Handling data and Query with Hive and pig** 3.1 Apache Hive: Hive architecture, warehouse directory and megastore, Hive data types, loading data into tables, Hive file formats, Hive built in functions, Querying data, Sorting, Aggregation 3.2 Apache Pig: Architecture, execution modes, built-in functions, Pig operations: group and join, combine and split, filter	**7**
4.	**Unit - IV Big Data Processing Tools** 4.1 Apache Spark: Need for Apache Spark, Apache Spark architecture, Spark components, Resilient Distributed Datasets(RDD) transformations 4.2 Spark SQL, Spark Session, Data Frame operations 4.3 Apache Kafka: Need, benefits, Kafka Messaging System: Point to point and Public subscribe messaging system, Kafka cluster architecture	**8**

Text Books:

1) **M. Vijayalakshmi, Radha Shankarmani** – *Big Data Analytics*, Wiley, 2nd Edition, 2017, ISBN: 9788126565757
2) **Tom White** – *Hadoop: The Definitive Guide,* 4th Edition, O'Reilly Media, 2015, ISBN: 9781491901632

Reference Books:

1) **Holden Karau, Andy Konwinski, Patrick Wendell, Matei Zaharia** – *Learning Spark: Lightning-Fast Data Analytics*, O'Reilly Media, 2015, ISBN: 9781449358624
2) **Pramod J. Sadalage, Martin Fowler** – *NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence*, Addison-Wesley, 2012, ISBN: 9780321826626
3) **Kieran Healy** – *Data Visualization: A Practical Introduction*, Princeton University Press, 2018, ISBN: 9780691181622

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Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: CCI24PCP501L **Course name:** Cyber and Digital Forensics Lab **Course Category:** PCC
Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** TW–30, PR–20
Duration of External Exam: 2 Hrs.
Pre-requisites: Computer Networking
Course Objectives:
1. Introduce fundamental concepts of network security, threats, and defense mechanisms.
2. Understand cryptographic techniques and their applications in securing communication.
3. Explore secure communication protocols and mechanisms for network authentication.
4. Implement security measures and analyse emerging trends in cybersecurity.
Laboratory Outcomes: At the end of the course, the students will be able to -
LO1: Understand and apply ethical hacking concepts to identify basic security threats, cyber crimes, and preventive measures in real-world systems..
LO2: Explain and use digital forensics principles to recognize, collect, and preserve digital evidence in a safe and legally acceptable manner.
LO3: Demonstrate incident response and forensic duplication techniques to handle security incidents and create forensic copies of digital data..
LO4: Analyze disk and network evidence using forensic tools to support cyber crime investigation and security analysis..

Contents –

Sr. No.	List of Experiments	Practical Hours
1	Collect basic information of a given website using simple tools. Find its IP address, domain details, and hosting information and record your observations.	2
2	Given different situations, identify whether they represent hacking or ethical hacking and justify your answer.	2
3	Analyze 5 real-life scenarios and classify each into appropriate types of cyber crime (phishing, identity theft, cyber stalking, hacking, online fraud)..	2
4	Demonstrate and explain all the steps of ethical hacking from Reconnaissance to Reporting using a flow diagram.	2
5	Explain the complete digital forensic investigation process (Identification, Collection, Preservation, Analysis, Reporting) with a simple example..	2
6	Given different digital items (email screenshot, image file, log file),	2

	identify which are digital evidence and mention their type.	
7	A system is infected with malware. Design an incident response plan showing how you will detect, contain, remove, and recover from the attack..	2
8	Given different forensic image formats (.img, .E01, .AFF), identify each and write one feature of every format..	2
9	Using a Wireshark screenshot or diagram, explain what a network packet is and how it can be used as digital evidence in an investigation.	2

Text Books:

1. Nilakshi Jain, Dhananjay Kalbande, Digital Forensic: The fascinating world of Digital Evidences, Wiley India Pvt Ltd.
2. Jason Luttgens, Matthew Pepe, Kevin Mandia, Incident Response and computer forensics, Tata McGraw Hill

Reference Books:

1. Cory Altheide, Harlan Carvey, Digital forensics with open-source tools, Syngress Publishing, Inc
2. Asey, E. (2011). Digital evidence and computer crime: Forensic science, computers and the internet. Academic Press

Online Resources: 1. https://onlinecourses.swayam2.ac.in/cec20_lb06

Course code: CCI24PCP503	**Course Name: Block chain Architecture Lab**	**Course category:** PCC
Credits: 01	**Teaching scheme:** P-2	**Evaluation scheme:** TW–30, PR–20

Pre-requisites: Computer Networks, Cryptography

Course Objectives:

1. To understand the fundamentals of blockchain, its evolution, and the concept of decentralization.
2. To study blockchain architecture, versions, and the concept of smart decentralized ecosystems with cryptocurrencies.
3. To explore blockchain mechanisms like proof systems, hashing, and Bitcoin's transaction and payment structures.
4. To compare and design blockchain platforms and assess their applicability in different domains.

Laboratory Outcomes: At the end of the session, the students will be able to -

LO1: Understand the basic structure, history, and types of blockchain systems and decentralization methods..

LO2: Analyze blockchain architecture, smart contracts, and cryptocurrencies in various real-world use cases.

LO3: Apply cryptographic principles and consensus algorithms in Bitcoin transactions and wallet mechanisms.

LO4: Evaluate and design distributed applications using modern blockchain platforms

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1.	To install and configure blockchain development tools.	2
2.	To understand blockchain structure and block linking using Python.	2
3.	To simulate decentralized nodes communicating in a blockchain network.	2
4.	To deploy a simple smart contract on Ethereum using Remix IDE.	2
5.	To perform ether transactions between accounts.	2
6.	To simulate a Bitcoin-like transaction and mining process.	2
7.	To demonstrate consensus algorithms.	2
8.	To create a customized blockchain prototype	2
9.	To develop and deploy a simple DApp using Solidity and web3.js.	2
10.	To explore and compare platforms like Ethereum, Hyperledger, and IOTA.	2

Text Books:

1) I. Bashir, Mastering Blockchain: Unlocking the Power of Cryptocurrencies, Smart Contracts, and Decentralized Applications, 4th ed., Packt Publishing, 2023.

2) A. D. Lewis, Blockchain and the Law: The Rule of Code, Harvard University Press, 2021

Reference Books:

1) M. Swan, Blockchain: Blueprint for a New Economy, O'Reilly Media, 2015.

2) R. Wattenhofer, The Science of the Blockchain, Inverted Forest Publishing, 2016.
3) D. Drescher, Blockchain Basics: A Non-Technical Introduction in 25 Steps, Apress, 2017.

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: CCI24PCP504 **Course Name: Big Data Lab** **Course category:** PCC
Credits: 01 **Teaching scheme:** P-2Hrs/Week **Evaluation Scheme:** TW–30, PR–20
Pre-requisites: Fundamentals of Programming**, Basic Knowledge of Databases and SQL**
Course Objectives: 1. Understand the characteristics, challenges, and technologies of Big Data. 2. Learn various Big Data storage and management techniques. 3. Explore tools like Hive and Pig for data analysis and processing. 4. Develop skills to implement Big Data processing solutions for real-world applications
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Understand the basic structure, history, and types of blockchain systems and decentralization methods..
LO2: Analyze blockchain architecture, smart contracts, and cryptocurrencies in various real-world use cases.
LO3: Apply cryptographic principles and consensus algorithms in Bitcoin transactions and wallet mechanisms.
LO4: Evaluate and design distributed applications using modern blockchain platforms.

List of Experiments–

Sr.No.	Name of Experiment	Practiacal Hours
1.	Write a python program for analyzing the characteristics of any large dataset case study	2
2.	Install Single-node Hadoop cluster	2
3.	Write HDFS commands for file operations	2
4.	Write MapReduce program for word count operation	2
5.	Write MapReduce program to sort array elements	2
6.	Hive installation and Queries	2
7.	Apache Pig: a)Install Apache Pig b)Use Pig operators	2
8.	Develop spark program to perform basic operations on data frames using python	2
9.	Write Program to perform RDD transformations	2
10.	Develop program to produce and consume message and build simple pipeline	2

Text Books:

1. M. Vijayalakshmi, Radha Shankarmani – *Big Data Analytics*, Wiley, 2nd Edition, 2017, ISBN: 9788126565757
2. Tom White – *Hadoop: The Definitive Guide*, 4th Edition, O'Reilly Media, 2015, ISBN: 9781491901632

Reference Books:

1. Holden Karau, Andy Konwinski, Patrick Wendell, Matei Zaharia – *Learning Spark: Lightning-Fast Data Analytics*, O'Reilly Media, 2015, ISBN: 9781449358624
2. Pramod J. Sadalage, Martin Fowler – *NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence*, Addison-Wesley, 2012, ISBN: 9780321826626
3. Kieran Healy – *Data Visualization: A Practical Introduction*, Princeton University Press, 2018, ISBN: 9780691181622

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Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: CSI24PEL505	**Course Name:** Basics of CC
Course category: PEC	
Credits: 2 **Teaching scheme: L-2**	**Evaluation scheme:** CA–60, ESE–40
Pre-requisites:1. Computer Network.	
Course Objectives:	
1. Understand the fundamentals and history of cloud computing.	
2. Explore cloud services like IaaS, PaaS, SaaS, and others.	
3. Discuss methods to enhance cloud data and infrastructure security.	
4. Introduce virtualization and types of hypervisors.	
5. Compare cloud support plans and cost models.	
Laboratory Outcomes: At the end of the session, the students will be able to -	
CO1: Explain mainframe, client-server, parallel, and distributed computing architectures.	
CO2: Illustrate cloud computing architectures and delivery models.	
CO3: Identify infrastructure security measures at network, host, and application levels.	
CO4: Explain virtualization concepts and types, including Type-I and Type-II hypervisors.	
CO 5: Apply knowledge of core services and pricing models of AWS, GCP, Azure and Databricks.	

Contents –

Unit	Content	Teaching hours
1	**Model computing** -Introduction to Mainframe architecture & Client-server architecture, Parallel & Distributed Computing, Cluster & Grid Computing, Definition and Evolution of Cloud Computing	4
2	**Cloud Computing and Cloud Services -** Cloud Computing architecture, cloud delivery models, cloud deployment models, Benefits and challenges, Service oriented architecture (SOA), Cloud Services: Infrastructure-as-a- Service (IaaS), Platform-as-a-Service (PaaS), Software-as-a-Service (SaaS), Identity-as-a-service (IDaaS), Storage-as-a-service.	8
3	**Cloud Security-** Cloud Security: Infrastructure Security – Network level security, Host level security, Application level Security. Data security and Storage - Data privacy and security Issues.	4
4	**Virtualization and SLA -** Virtualization: Introduction to virtualization, Hypervisor: Type-I & Type II, Types of Virtualization, Pros and cons of virtualization, Virtualization applications in enterprises: Server virtualization, Desktop and Application Virtualization, Storage and Network Virtualization SLA Management, SLA Management in Cloud.	8
5	**Cloud Platforms in Industry-** Amazon Web Services- Compute Services, Storage Services, Communication Services and Additional Services, GCP, Billing and pricing & Support plan, Pricing model of AWS & GCP, Different support levels offered by AWS and GCP, Google AppEngine-Architecture and Core Concepts, Application Life-Cycle, cost model. Microsoft Azure- Azure Core Concepts, SQL Azure	6

Text Books: 1. Barrie Sosinsky, “Cloud computing Bible” Wiley India Pvt Ltd, 2011. 2. Rajkumar Buyya, Christian Vecchiola, S. ThamaraiSelvi ,”Mastering Cloud Computing”, McGraw Hill,2013
Reference Books: 1. Hughues- Hallett Gleason, Calculus Single and Multivariable, 4th edition, Wiley. 2. B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, NewDelhi.
Online Resources: 1. NPTEL / SWAYAM lectures.

Course code: CCI24PEP505	**Course Name:** Basics of CC Lab
Course category: PEC	
Credits: 1 **Teaching scheme: P-2**	**Evaluation scheme:** TW–30, PR–20
Pre-requisites:1. Computer Network.	
Course Objectives:	
1. Understand the fundamentals and history of cloud computing.	
2. Explore cloud services like IaaS, PaaS, SaaS, and others.	
3. Discuss methods to enhance cloud data and infrastructure security.	
4. Introduce virtualization and types of hypervisors.	
5. Compare cloud support plans and cost models.	
Laboratory Outcomes: At the end of the session, the students will be able to -	
LO1: Explain mainframe, client-server, parallel, and distributed computing architectures.	
LO2: Illustrate and compare cloud computing architectures and delivery models.	
LO3: Identify infrastructure security measures at network, host, and application levels.	
LO4: Explain virtualization concepts and types, including Type-I and Type-II hypervisors.	
LO5: Apply knowledge of core services and pricing models of AWS, GCP, Azure and Databricks.	

Contents –

Sr. No.	List of Experiments	Practical Hours
1.	Implementing Basic Concepts in Cloud Computing.	2
2.	Creating a Warehouse Application in SalesForce.com.	2
3.	Create an Application using Apex Programming in SalesForce.com.	2
4.	Implementation of SOAP Web services in C#/JAVA Applications.	2
5.	Implementation of Virtualization. Install Oracle Virtual box and create VMs.	2
6.	Install a C compiler in the virtual machine and execute a sample program.	2
7.	Setup account for AWS and GCP	2
8.	Create a Cloud Storage bucket using Amazon Simple Storage Service (Amazon S3).	2
9.	Working with AWS CLI	2
10.	Creating Azure Container Instances	2
11.	Case Study of Google App Engine	2

Text Books: 1. Barrie Sosinsky, “Cloud computing Bible” Wiley India Pvt Ltd ,2011. 2. Rajkumar Buyya, Christian Vecchiola, S. ThamaraiSelvi ,”Mastering Cloud Computing”, McGraw Hill,2013
Reference Books: 1. Hughues- Hallett Gleason, Calculus Single and Multivariable, 4th edition, Wiley. 2. B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, NewDelhi.
Online Resources: 1. NPTEL / SWAYAM lectures.

Course Code: CSI24PEL506 **Course Name:** Dev-Ops-I **Course Category:** PEC	
Credits: 2 **Teaching scheme:** L-2 **Evaluation Scheme:** CA:60, ESE:40	
Duration of External Exam: 2 Hrs	
Pre-requisites: Software Development Life Cycle (SDLC), Version Control Systems, Basic Programming Knowledge.	
Course Objectives: 1. Introduce software development concepts, SDLC models, and software quality practices. 2. Understand DevOps concepts and the role of version control in modern software development. 3. Develop practical skills in using Git for source code and version management. 4. Apply advanced Git features for effective branch, merge, and release management. 5. Familiarize students with current DevOps trends, tools, and industry practices.	
Course Outcomes: At the end of the course, the students will be able to – **CO1**: Describe Study of various Software Development Methodologies and Cloud Computing. **CO2:** Understand the basic Linux commands with respect to Devops and basic of version control. **CO3:** Understand the concept of Devops commands and workflow in a systematic way. **CO4:** Apply the advanced concept of DevOps with real time applications. **CO5:** understand the concept of DevOps Methodology and tools in a systematic way.	

Contents:

Unit No.	Content	Teaching Hours
1.	**Introduction to Software Development**: What is Software Development, Software Development Life Cycle, Traditional Models for SDLC, Elaborate software development life cycle: Waterfall model, iterative waterfall model, Prototype model, evolutionary model, spiral model, agile model, software testing and debugging, software verification and validation, Comparison of different life cycle models, User Interface Design, Coupling and Cohesion, Pham-Nordmann-Zhang Model (PNZ model), Schick-Wolverton software reliability model.	8
2.	**Introduction to DevOps and software version control:** What is DevOps, Industry Importance of DevOps, DevOps Lifecycle, Continuous Development, Continuous Testing, Configuration Management, Continuous Integration, Continuous Monitoring of software throughout its development life cycle, Understanding basics of version control, Control Concepts of different types of Version Control Systems	8
3.	**Basic Git:** Introduction to Git, Git lifecycle- Working Directory, Staging Area, Local Repository, Remotel Repository Common Git commands, Git Workflow, Working with Remote Repository, Version controlling using Git	4
4.	**Advanced Git:** Source code management with Git, Comparison	4

	commands, Branch management strategies, and merge strategies, Rebasing, Interactive rebase Stashing and Tagging.	
5.	**DevOps Trends:** DevOps Market Trends, DevOps Engineer Skills, DevOps Delivery Pipeline, DevOps Ecosystem, Role of a DevOps Engineer, Devops Tools: Git, Docker, Selenium, Maven, Jenkins, Puppet, Ansible, Kubernetes, Nagios.	6

Text Books:

1. "The Phoenix Project: A Novel About IT, DevOps, and Helping Your Business Win" Gene Kim, Kevin Behr, George Spafford.
2. Linux Pocket Guide: Essential Commands By Daniel J. Barrett, O'reilly.
3. "Learning Git" Anna Slater.

Reference Books:

1."The DevOps Toolkit: Kubernetes" by Viktor Farcic.
2. "Jenkins: The Definitive Guide: Continuous Integration for the Masses" by John Ferguson Smart.
3. Devops: A Software Architect's Perspective (Sei Series In Software Engineering) By Len Bass, Ingo Weber, Liming Zhu, Addison-Wesley.

Online Resources:

1. https://www.guvi.in/blog/best-websites-to-learn-devops/
2. https://www.geeksforgeeks.org/devops-tutorial/
3. GitHub Learning Lab.

Course Code: CCI24PEP506 **Course Name:** Dev-Ops-I Lab
Course Category: PEC
Credits: 1**Teaching Scheme:** P-2 **Evaluation Scheme:** TW–30, PR–20
Duration of External Exam: 3 Hrs.
Pre-requisites: Software Development Life Cycle (SDLC), Version Control Systems, Basic Programming Knowledge.
Course Objectives:
1. To provide hands-on experience with the core principles and tools used in DevOps, including version control, continuous integration, continuous delivery, and containerization.
2. To introduce students to the practical application of DevOps practices through real-world tasks, utilizing tools like Git, Jenkins, Docker, and Kubernetes for automating development workflows and deployment processes.
Laboratory Outcomes: At the end of the course, the students will be able to –
LO1. Describe various software development methodologies and understand cloud computing concepts.
LO2. Demonstrate proficiency in basic Linux commands necessary for DevOps environments.
LO3. Understand and implement DevOps commands and follow systematic workflows for building.
LO4. Apply advanced DevOps concepts to **real-world scenarios,** demonstrating how DevOps tools and practices improve software reliability, scalability, and security.
LO5. Understand and apply DevOps methodologies and use related **tools like Git, Jenkins, Docker, Kubernetes, Ansible,** and others in a systematic way to automate software development and deployment lifecycle.

Contents –

Sr. No.	List of Experiments	Practical Hours
1.	Explore DevOps architecture	2
2.	Create a Git repository for a simple project, and then practice common commands such as git init, git clone, git commit, git push, and git pull. Tools: Git, GitHub (or any Git hosting service).	2
3.	Installation of Oracle VM Virtual Box and create Virtual Machine Practical.	2
4.	Installation of Linux, Implementation of basic Linux commands: chmod, grep, wget, chown, find, cat, echo, ifconfig, cp,ping,kill, tail, rm, rmdir, cd, mkdir, vi, mv.	2
5.	Installation of packages using RPM and YUM	2

6.	Installation of Git, Implementing common Git Command	2
7.	Repository creation in Git, Git Branch, source code management with Git	2
8.	Installation of Maven and Work	2
9.	Continuous Testing with Selenium	2
10.	Build a simple DevOps pipeline using Jenkins that performs the following tasks: 1. Pulls code from a Git repository. 2. Builds the code (using Maven or similar). 3. Runs automated tests (using Selenium). 4. Builds a Docker image of the application. 5. Deploys the Docker container to a test environment. Tools: Jenkins, Git, Maven, Docker, Selenium.	2
11.	Mini project	2

Text Books:

1."The Phoenix Project: A Novel About IT, DevOps, and Helping Your Business Win" Gene Kim, Kevin Behr, George Spafford.
2. Linux Pocket Guide: Essential Commands By Daniel J. Barrett, O'reilly.
3."Learning Git" Anna Slater.

Reference Books:

1."The DevOps Toolkit: Kubernetes" by Viktor Farcic.
2. "Jenkins: The Definitive Guide: Continuous Integration for the Masses" by John Ferguson Smart.
3. Devops: A Software Architect's Perspective (Sei Series In Software Engineering) By Len Bass, Ingo Weber, Liming Zhu, Addison-Wesley.

SEM-X

Course code: CCI24PCL551 **Course name: Natural Language Processing** **Course category: PCC**
Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–60, ESE–40,
Pre-requisites: Knowledge of programing language, Data Structures
Course Objectives: 1 To understand natural language processing concept. 2 To get acquainted with word level , syntax analysis & language model. 3 To understand semantic analysis of NLP & model. 4 To design systems that uses NLP techniques
Course Outcomes: At the end of the course, the students will be able to -
CO1: Have a broad understanding of the field of natural language processing.
CO2: Get knowledge of morphology, language model & POS
CO3: Understand concept of Lexical Semantics
CO4 : Understand NLP techniques to design real world NLP applications

Contents –

Unit	Content	Teaching hours
1.	**Introduction :** History of NLP, Study of Human Language, NLP pipeline, levels of NLP , Knowledge in language processing , Ambiguity in Natural language , stages in NLP, challenges of NLP , Applications of NLP, Linguistic Resources. **Text Normalization** : Tokenization, sentence segmentation, Stopwords, stemming, lemmatization. Regular Expressions & Automata :Regex fundamentals, Regular sets, grammars, Finite automata & relationships among RE, RG, FA	**12**
2.	**Word Level Analysis :** Morphology analysis –survey of English Morphology, Inflectional morphology & Derivational morphology, Lemmatization, finite state transducers (FST) , Morphological parsing , N gram language model. **Syntax Analysis :** Part-Of-Speech tagging(POS)- Tag set for English (Penn Treebank) , Rule based POS tagging, Stochastic POS tagging, Issues –Multiple tags & words, Unknown words. Introduction to CFG, Sequence labelling: Hidden Markov Model (HMM)	**12**
3.	**Semantic Analysis :** Lexical Semantics, Attachment for fragment of English-sentences, noun phrases, Verb phrases, prepositional phrases, Relations among lexemes & their senses –Homonymy, Polysemy, Synonymy, Hyponymy, WordNet, Robust Word Sense Disambiguation (WSD) ,Dictionary based approach	**12**

4.	**Pragmatic analysis :** Discourse –model & structure, reference resolution, reference phenomenon , syntactic & semantic constraints on co reference **Applications of NLP :** Sentiment Analysis, information Extraction ,Machine Translation , Text Summarization ,Grammar and Text Analysis , Modern NLP Applications : Virtual Assistants ,Chatbots and Customer Service , Speech Recognition ,Spam Filters , Predictive Text , Ethics & fairness in NLP systems	**9**

Text Books:

1. Daniel Jurafsky, James H. Martin ―Speech and Language Processing , Prentice Hall,
2. Christopher D.Manning and Hinrich Schutze, ― Foundations of Statistical Natural Language Processing ―, MIT Press.

Reference Books:

1. Siddiqui and Tiwary U.S., Natural Language Processing and Information Retrieval, Oxford University Press
2. Daniel M Bikel and Imed Zitouni ― Multilingual natural language processing applications‖ Pearson
3. Alexander Clark (Editor), Chris Fox (Editor), Shalom Lappin (Editor) ― The Handbook of Computational Linguistics and Natural Language Processing
4. Steven Bird, Ewan Klein, Natural Language Processing with Python, O'Reilly

Online Resources: 1.NPTEL / SWAYAM lectures, Tutorials Point

Course code: CCI24PCL552	**Course Name: Blockchain Security**
Course Category: PCC	
Credits: 2 **Teaching scheme:** L-2	**Evaluation scheme:** CA–60, ESE–40
Pre-requisites: Computer Networks, Cryptography, Blockchain Fundamentals	
Course Objectives:	
1. Introduce the fundamental principles and models of blockchain security. 2. Analyze network-level threats and defense mechanisms in blockchain environments. 3. Evaluate security aspects of consensus mechanisms and their vulnerabilities. 4. Develop skills in assessing and securing smart contracts and blockchain applications.	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Understand blockchain security principles, objectives, and architecture models.	
CO2: Analyze network-level vulnerabilities and corresponding defense mechanisms in blockchain systems	
CO3: Evaluate consensus protocols and their resilience against attacks.	
CO4: Design and implement security mechanisms for smart contracts and blockchain applications	

- **Contents –**

Unit	Content	Teaching hours
1.	**Overview of blockchain security** Introduction to Blockchain Security, Security Objectives and Principles, Blockchain Security Model, Cryptographic Foundations, Identity and Authentication, Key and Wallet Security, Access Control in Blockchain, Node Security and Privacy Mechanisms	**8**
2.	**Network-Level Vulnerabilities and Attacks** Blockchain Network Architecture, Security Issues in P2P Protocols, Eclipse Attacks, Routing Attacks, Denial of Service (DoS) and DDoS Attacks, Timejacking and Network Delay Exploits, Defense Mechanisms at Network Layer	**7**
3.	**Security of consensus protocols** Overview of Consensus Protocols, Proof of Work (PoW) Security, Proof of Stake (PoS) and Delegated PoS, Raft and PBFT Consensus, 51% Attacks, Selfish Mining Attack, Long-Range and Nothing-at-Stake Attacks, Consensus Security Enhancements	**8**
4.	**Smart Contract and Application-Level Security** Introduction to Smart Contract Security, Bitcoin Script Security, Ethereum Smart Contract Security, Hyperledger Fabric Chaincode Security, Static and Dynamic Analysis of Smart Contracts, Formal Verification and Security Auditing, Case Studies and Future Trends	**7**

Text Books:

1) M. Swan, *Blockchain: Blueprint for a New Economy*, O'Reilly Media, 2015.

2) I. Bashir, *Mastering Blockchain: Unlocking the Power of Cryptocurrencies, Smart Contracts, and Decentralized Applications*, 4th ed., Packt Publishing, 2023.

3) A. Narayanan, J. Bonneau, E. Felten, A. Miller, and S. Goldfeder, *Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction*, Princeton University Press, 2016.

Reference Books:

1) D. Drescher, *Blockchain Basics: A Non-Technical Introduction in 25 Steps*, Apress, 2017.

2)R. Wattenhofer, *The Science of the Blockchain*, Inverted Forest Publishing, 2016.

3) L. Luu, D. Chu, H. Olickel, P. Saxena, and A. Hobor, "Making Smart Contracts Smarter," *Proc. ACM SIGSAC Conf. Computer and Communications Security (CCS)*, pp. 254–269, 2016..

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: CCI24PCP551 **Course name: Natural Language Processing Lab** **Course Category: PCC**
Credits: 01 **Teaching scheme:** P-2 **Evaluation scheme:** TW–30, PR–20
Pre-requisites:1. Knowledge of python programing, Data Structure
Course Objectives: 1. To study available libraries & dataset related to NLP 2. To get acquainted with designing skill of fundamental NLP operations. 3. Design any application of NLP
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: know available libraries & dataset related to NLP
LO2: design and implement fundamental NLP operations
LO3: design an application of NLP

List of Experiments–

Sr. No.	Name of Experiment	Practical Hours
1.	Study & installation of different libraries in Python for NLP	2
2.	Study different datasets for NLP	2
3.	Study & implementation of read , write & append data in file.	2
4.	Study & implementation of tokenization concept (word , sentence) .	2
5.	Implement concept to find synonyms and antonyms of the word.	2
6.	Implement concept of stop word removing from document .	2
7	Implement concept of unigram, bigram , trigram .	2
7.	Implement concept of POS	2
8.	Implement concept stemming & lemmatization	2
9.	Implement concept of named entity recognition (NER)	2
10.	Write a complete NLP task using cleaning & pre processing of	2
11.	Implement any one application of NLP .	2

Text Books:
1 Machine Learning in data science using Python - Dr. R Nageshwara Rao (dreamtech press).
2. Steven Bird, Ewan Klein, Natural Language Processing with Python, O'Reilly

Reference Books:
1. Christopher D.Manning and Hinrich Schutze, ― Foundations of Statistical Natural Language Processing ―, MIT Press.
2. Daniel Jurafsky, James H. Martin ―Speech and Language Processing , Prentice Hall

Online Resources: 1.NPTEL / SWAYAM , scribd , w3school, geeksforgeeks

ourse code: CCI24PCP552 **Course Name: Blockchain Security Lab** **Course category:** PCC
Credits: 01 **Teaching scheme:** P-2 **Evaluation scheme:**, TW–30, PR–20
Pre-requisites: Computer Networks, Cryptography, Blockchain Fundamentals
Course Objectives: 1. Introduce the fundamental principles and models of blockchain security. 2. Analyze network-level threats and defense mechanisms in blockchain environments. 3. Evaluate security aspects of consensus mechanisms and their vulnerabilities. 4. Develop skills in assessing and securing smart contracts and blockchain applications
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Understand blockchain security principles, objectives, and architecture models.
LO2: Analyze network-level vulnerabilities and corresponding defense mechanisms in blockchain systems
LO3: Evaluate consensus protocols and their resilience against attacks.
LO4: Design and implement security mechanisms for smart contracts and blockchain applications

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1.	To explore blockchain security components and architecture	2
2.	To perform hashing using SHA-256 for blockchain data integrity	2
3.	To create and manage Ethereum wallets securely	2
4.	To analyze blockchain P2P traffic for vulnerabilities	2
5.	To simulate a simple DoS attack on a local blockchain node.	2
6.	To understand how Proof-of-Work ensures security in blockchain	2
7.	To simulate Proof-of-Stake consensus.	2
8.	To detect vulnerabilities in a smart contract.	2
9.	To verify smart contract correctness using formal tools.	2
10.	To deploy a secure DApp integrating best security practices.	2

Text Books:

1) M. Swan, *Blockchain: Blueprint for a New Economy*, O'Reilly Media, 2015.

2) I. Bashir, *Mastering Blockchain: Unlocking the Power of Cryptocurrencies, Smart Contracts, and Decentralized Applications*, 4th ed., Packt Publishing, 2023.

3) A. Narayanan, J. Bonneau, E. Felten, A. Miller, and S. Goldfeder, *Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction*, Princeton University Press, 2016.

Reference Books:

1) D. Drescher, *Blockchain Basics: A Non-Technical Introduction in 25 Steps*, Apress, 2017.

2)R. Wattenhofer, *The Science of the Blockchain*, Inverted Forest Publishing, 2016.

3) L. Luu, D. Chu, H. Olickel, P. Saxena, and A. Hobor, "Making Smart Contracts Smarter," *Proc. ACM SIGSAC Conf. Computer and Communications Security (CCS)*, pp. 254–269, 2016..

Online Resources: 1.NPTEL / SWAYAM lectures.

Course Code:CCI24PCL553 **Course Name: Internet of Things** **Course Category:** PCC
Credits: 2 **Teaching scheme:** L-2 **Evaluation Scheme:** CA:60, ESE:40
Duration of External Exam: 2 Hrs
Pre-requisites: Basic electronics, Networking
Course Objectives: 1. Identify the key components and architecture of IoT systems. 2. Define the various levels of IoT architecture 3. Define sensor networks and explain their importance in the IoT ecosystem. 4. Discuss the advantages and limitations of each access technology.
Course Outcomes: At the end of the course, the students will be able to - **CO1:** Describe the IoT Characteristics and Conceptual Framework. **CO2:** Differentiate between the levels of the IoT architectures. **CO3:** Interpret sensor network and its components. **CO4:** Analyze the IoT access technologies. **CO5:** Illustrate various protocols at network layer and application layer for IoT.

Contents:

Unit No.	Content	Teaching Hours
1.	**Introduction to IoT :** Introduction to IoT- Defining IoT, Characteristics of IoT, Conceptual Framework of IoT, Physical design of IoT, Logical design of IoT, Functional blocks of IoT, Communication models & APIs, Basics of networking Communication protocol, wireless sensor networks. Convergence of IT and OT , IoT Challenges, IoT protocol vs Web Protocol stack	6
2.	**IoT Network Architecture and Design :** Drivers Behind New Network Architectures : Scale,Security,Constrained Devices and Networks ,Data,Legacy Device Support Architecture : The IoT World Forum (IoTWF) Standardized Architecture :Layer 1-7, IT and OT Responsibilities in the IoT Reference Model,Additional IoT Reference Models, A Simplified IoT Architecture, The Core IoT Functional Stack ::Layer 1-3 , Analytics Versus Control Applications , Data Versus Network Analytics Data Analytics Versus Business Benefits , Smart Services, IoT Data Management and Compute Stack :Fog Computing, Edge Computing ,The Hierarchy of Edge, Fog and Cloud	8
3.	**Smart & Connecting Objects in IoT** Sensors, Actuators, and Smart Objects , Sensors , Actuators, Micro-Electro-Mechanical Systems (MEMS) Smart Objects: A Definition , Trends in Smart Objects, Sensor Networks , Wireless Sensor Networks (WSNs) , Communication Protocols for WSN,RFID ,NFC Communications Criteria : Range , Frequency Bands , Power Consumption ,	8

	Topology , Constrained Devices , Constrained-Node Networks , Data Rate and Throughput , Latency and Determinism , Overhead and Payload IoT Access Technologies : Standardization and Alliances , Physical Layer , MAC Layer ,Topology ,Security and Conclusion of IEEE 802.15.4	
4.	**IoT Network Layer and Application protocols** The Business Case for IP , The Key Advantages of Internet Protocol ,Adoption or Adaptation of the Internet Protocol ,The Need for Optimization ,Constrained Nodes , Constrained Networks IP Versions , Optimizing IP for IoT The Transport Layer , IoT Application Transport Methods, Generic Web-Based Protocols , 4.3 IoT Application Layer Protocols , CoAP, MQTT, AMQP	8

Text Books:

1. Arsheep Bahga (Author), Vijay Madisetti, Internet Of Things: A Hands-On Approach Paperback, Universities Press, Reprint 2020

2. David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton, Jerome Henry, IoT Fundamentals Networking Technologies, Protocols, and Use Cases for the Internet of Things CISCO.

Reference Books:

1. Pethuru Raj, Anupama C. Raman, The Internet of Things: Enabling Technologies, Platforms, and Use Cases by , CRC Press.

2. Raj Kamal, Internet of Things, Architecture and Design Principles, McGraw Hill Education, Reprint 2018.

Online Resources:

1NPTEL: Sudip Misra, IIT Khargpur, Introduction to IoT: Part-1, https://nptel.ac.in/courses/106/105/106105166/

Course Code: CCI24PCP553 Course Name: Internet of Things Lab **Course Category:** PCC
Credits: 1**Teaching Scheme:** P-2 **Evaluation Scheme:** TW–30, PR–20
Duration of External Exam: 3 Hrs.
Pre-requisites: Basic electronics, Networking
Course Objectives:
1. Identify the key components and architecture of IoT systems.
2. Define the various levels of IoT architecture
3. Define sensor networks and explain their importance in the IoT ecosystem.
4. Discuss the advantages and limitations of each access technology.
Laboratory Outcomes:
At the end of the course, the students will be able to -
LO1: Describe the IoT Characteristics and Conceptual Framework.
LO2: Differentiate between the levels of the IoT architectures.
LO3: Interpret sensor network and its components.
LO4: Analyze the IoT access technologies.
LO5: Illustrate various protocols at network layer and application layer for IoT.

Contents –

Sr. No.	List of Experiments	Practical Hours
1.	To implement the interfacing of different IoT sensors with Raspberry Pi pico/Arduino/ModeMCU	2
2.	To implement interfacing of actuators based on the data collected using IoT sensors. (like led switch ON/OFF, stepper motor)	2
3.	To demonstrate Contiki OS for RPL (like Create 2 border router and 10 REST clients, Access order router from other network (Simulator))	2
4.	To demonstrate the working of 6LoWPAN in Contiki OS (simulator)	2
5.	Write a program on Raspberry Pi to push and retrieve the data from cloud like thing speak/ things board/ AWS/ Azure etc.	2
6.	To implement IoT Data processing using Pandas.	2
7.	Write a program on Arduino / Raspberry Pi that subscribes to an MQTT broker for temperature data and prints it.	2
8.	Write a program to create a TCP Server on Arduino/Raspberry Pi and respond with humidity data to TCP client when requested.	2
9.	Write a program for ESP8266 DHT11/DHT22 Temperature and Humidity Web Server with Arduino IDE	2
10.	Write a program to Control Your ESP8266 From Anywhere in the World	2

Text Books:

1. Arsheep Bahga (Author), Vijay Madisetti, Internet Of Things: A Hands-On Approach Paperback, Universities Press, Reprint 2020

2. David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton, Jerome Henry, IoT Fundamentals Networking Technologies, Protocols, and Use Cases for the Internet of Things CISCO.

Reference Books:

1. Pethuru Raj, Anupama C. Raman, The Internet of Things: Enabling Technologies, Platforms, and Use Cases by , CRC Press.

2. Raj Kamal, Internet of Things, Architecture and Design Principles, McGraw Hill Education, Reprint 2018.

Online Resources:

1NPTEL: Sudip Misra, IIT Khargpur, Introduction to IoT: Part-1, https://nptel.ac.in/courses/106/105/106105166/

Course code: CCI24PEL554	**Course name:** CC Using AWS	**Course category:** PEC
Credits: 2	**Teaching scheme:** L-2	**Evaluation scheme:** CA–60, ESE–40
Duration of External Exam: 2 Hrs		
Pre-requisites: Basics of CC.		
Course Objectives:		
1. Host a basic application (e.g., Employee Directory) on AWS.		
2. Understand the Amazon EC2 instance lifecycle and compute services.		
3. Explain Docker and Kubernetes and their architectures.		
4. Understand S3 storage components, including buckets, objects, and keys.		
5. Configure IAM users, groups, and password policies.		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Identify the components of AWS Global Infrastructure.		
CO2: Describe the Amazon EC2 instance lifecycle and container services.		
CO3: Define containerization concepts and tools like Docker and Kubernetes.		
CO4: Explain S3 components, storage classes, and versioning.		
CO5: Create and manage IAM policies, groups, and permissions**.**		

Contents –

Unit	Content	Teaching hours
1	**AWS Essentials -**AWS Global Infrastructure, Interacting with AWS, Security and the AWS Shared Responsibility Model, Protect the AWS Root User, Introduction to AWS Identity and Access Management, Role Based Access in AWS, Hosting the Employee Directory Application on AWS	6
2	**Introduction to Amazon Elastic Compute Cloud-**Compute as a Service on AWS, Amazon EC2 Instance Lifecycle, Container Services on AWS,What is Serverless?, Introduction to AWS Lambda, Networking on AWS, Introduction to Amazon VPC, Amazon VPC Routing, Secure Your Network with Amazon VPC Security.	6
3	**Introduction to Containerisation -**What is Docker & Kubernetes?, Architecture of Docker & Kubernetes,AWS Elastic container services, Amazon Elastic Kubernetes service, Google Kubernetes engine.	6
4	**Cloud Storage -**S3 Storage :Buckets, Objects, and Keys, S3 Data Consistency, S3 Storage Classes,S3 Management , Versioning, S3 Bucket Security, Amazon S3 Glacier Archive Storage, S3 Glacier Vaults and Archives , Shared File Systems at AWS	6
5	**Security Services -**Creating IAM Users and Groups, Defining a Password Policy, Enabling MFA Protection, Creating IAM Policies, Creating Permission Boundaries, Exploring AWS Security Tools, Exploring CloudTrail Events, Setting up AWS Organizations and RAM	6

Text Books:
1. Danilo Poccia , "AWS Lambda in Action: Event-driven Serverless Applications", Manning Publication, 2016
2. Mark Wilkins , "Learning Amazon Web Services (AWS): A Hands-On Guide to the Fundamentals of AWS Cloud", Addison-Wesley Professional,2019 Computing", McGraw Hill,2013
3. Andreas Wittig and Michael Wittig, " Amazon Web Services in Action", Manning Publications, 2015

Reference Books:
1. Raoul Alongi , "AWS: The Most Complete Guide to Amazon Web Services from Beginner to Advanced Level", Kindle Edition, 2019
2. Ben Piper (Author), David Clinton , "AWS Certified Cloud Practitioner Study Guide:CLF-C01 Exam 1st Edition", Sybex, 2019

Course Code: CCI24PEL555	**Course Name:** DevOps-II	**Course Category:** PEC
Credits: 2	**Teaching scheme:** L-2	**Evaluation Scheme:** CA:60, ESE:40
Duration of External Exam: 2 Hrs.		
Pre-requisites: Software Development Basics, Version Control (Git),Operating Systems & Networking		
Course Objectives: **1.** Use version control tools for collaborative development.. **2.** Implement Continuous Integration and Continuous Deployment.. **3**. Apply containerization and orchestration in DevOps. **4.** Practice monitoring, logging, and security in DevOps. 5. Apply cloud and IaC practices to achieve DevOps Zenith.		
Course Outcomes: At the end of the course, the students will be able to –		
CO1: Apply version control and collaboration using Git.		
CO2: Design and implement CI/CD pipelines.		
CO3: Deploy applications using Docker and Kubernetes.		
CO4: Implement monitoring, logging, and DevSecOps practices		
CO5: Build scalable and automated cloud infrastructure using IaC and CI/CD.		

Contents:

Unit No.	Content	Teaching Hours
1.	**Version Control and Collaboration: Version Control Systems (VCS):** Introduction to Git and, GitHub/GitLab/Bitbucket, **Git Basics:** Cloning, branching, committing, merging, resolving conflicts, **Collaboration with Git**: Forking, pull requests, and code reviews, **Branching Strategies**: Git Flow, feature branching, trunk-based development, **Team Collaboration**: Integrating with CI/CD tools, using version control for DevOps	6
2.	**Continuous Integration and Continuous Deployment (CI/CD): CI/CD Overview:** Introduction to Continuous Integration and Continuous Deployment/Delivery, **CI Tools**: Jenkins, GitLab CI, Travis CI, CircleCI, **Setting up a CI Pipeline**: Automating builds and tests with CI tools, **CD Pipelines:** Automating the deployment process to various environments (staging, production), **Test Automation:** Integrating unit, integration, and UI tests into the CI/CD pipeline	6
3.	**Containerization and Virtualization:** Introduction to Containers: Differences between containers and virtual machines, Docker Basics: Docker containers, images, Docker file, and Docker Compose, Containerization Workflow: Creating and managing containers, building Docker images, Kubernetes Basics: Overview of Kubernetes for container orchestration, Using Containers in DevOps: Benefits of containers in CI/CD, micro services, and scalability.	6
4.	**Monitoring, Logging, and Security in DevOps:** Monitoring in DevOps: Importance of application and infrastructure monitoring (Prometheus, Grafana), Log Management: Tools like ELK Stack (Elasticsearch, Logstash, Kibana), Fluentd, Alerting and Metrics: Setting up alerts, performance monitoring, and health checks, DevSecOps: Integrating security practices into the DevOps lifecycle, Security Tools: Vulnerability scanning, secrets management (HashiCorp Vault)	6

5.	**Cloud Infrastructure & DevOps Zenith:** Cloud Platforms in DevOps: Role of AWS, Azure, and GCP in modern infrastructure, Infrastructure as Code (IaC): Concept, benefits, and automation using Terraform and Cloud Formation, Configuration Management: Basics of Ansible for consistent and repeatable deployments, **DevOps Zenith Concept:** Achieving the highest maturity level in DevOps through full automation, IaC-driven infrastructure, reliable CI/CD pipelines, monitoring, security integration, and scalability, Cloud Automation & Reliability: Autoscaling, high availability, and cost-optimized infrastructure as key indicators of DevOps Zenith.	6

Text Books:

1. "The Phoenix Project: A Novel About IT, DevOps, and Helping Your Business Win" by Gene Kim, Kevin Behr, and George Safford.
2. "Accelerate: The Science of Lean Software and DevOps: Building and Scaling High Performing Technology Organizations" by Nicole Forsgren, Jez Humble, and Gene Kim.
3. "DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations" by Gene Kim, Patrick Debois, John Willis, and Jez Humble.

Reference Books:

1. Version Control with Git" by Jon Loeliger and Matthew McCullough.
2. "Pro Git" by Scott Chacon and Ben Straub
3. "Docker: Up & Running" by Karl Matthias and Sean P. Kane
4. Kubernetes Up & Running: Dive into the Future of Infrastructure" by Kelsey Hightower, Brendan Burns, and Joe Beda.

Online Resources:

1. Virtual Labs
2. GitHub Repositories

Course code: CCI24PEL556		**Course name:** Image Processing
Course Category: PEC		
Credits: 03	**Teaching scheme:** L-3	**Evaluation scheme:** CA–60, ESE–40
Pre-requisites: **1.** Any one programming skill with		
Course Objectives:		
1. Demonstrate basic image operations for pre-processing		
2. Introduce filtering techniques for image smoothing and sharpening		
3. Introduce image segmentation techniques for object detection		
4. Demonstrate restoration techniques for noisy images.		
5. Explore advanced concepts like object recognition and feature extraction		
Course Outcomes: At the end of the session, the students will be able to –		
CO1: Explain image sampling, quantization, and basic operations.		
CO2: Apply smoothing and sharpening filters to improve image quality		
CO3: Explain segmentation techniques like thresholding and edge detection		
CO4: Apply DCT and Wavelet Transform for image compression.		
CO5: Analyse the use of machine learning in image recognition applications		

Content

Unit	Content	Teaching Hours
1.	**Introduction to Image Processing -** Definition and Applications of Image Processing, Components of an Image Processing System, Types of Digital Images: Binary, Grey scale, and Color, Image Sampling and Quantization, Basic Operations: Image Addition, Subtraction, Multiplication, and Division	8
2.	**Image Enhancement in Spatial and Frequency Domains -** Basics of Image Enhancement , Spatial Domain Techniques: Point Processing, Histogram Equalization, and Filtering, Frequency Domain Techniques: Fourier Transform and Filtering, Smoothing and Sharpening Filters, Noise Removal Techniques	10
3.	**Image Segmentation and Morphological Processing** - Thresholding Techniques for Image Segmentation, Edge Detection: Sobel, Prewitt, and Canny Operators, Region-Based Segmentation: Region Growing and Splitting, Morphological Operations: Dilation, Erosion, Opening, and Closing, Morphological Applications in Image Analysis	9
4.	**Image Compression and Restoration** - Basics of Image Compression: Lossy vs. Lossless, Compression Techniques: JPEG, PNG, and GIF, Transform Coding: Discrete Cosine Transform (DCT) and Wavelet Transform, Image Restoration Techniques: Noise Modelling and Inverse Filtering, Applications of Image Compression in Multimedia	9
5.	**Advanced Topics in Image Processing**- Object Recognition and Pattern Matching, Introduction to Machine Learning in Image Processing, Feature Extraction: SIFT, SURF, and HOG, Image Processing in AI Applications: Facial Recognition, Medical Imaging, Introduction to OpenCV for Image Processing	9

Text Books: 1. Fundamentals of Digital Image Processing by Anil K. Jain, Prentice Hall 2. Image Processing, Analysis, and Machine Vision by Milan Sonka, Vaclav Hlavac, and Roger Boyle, 4th Edition,Cengage Learning
Reference Books: 1. Digital Image Processing by Rafael C. Gonzalez and Richard E. Woods, 4th edition, Pearson Education

Course code: CCI24PEL557	**Course name:** Quantum Computing
Course category: PEC	
Credits: 03 **Teaching scheme:** L-3 **Evaluation scheme:** CA: 60, ESE: 40	
Pre-requisites: **1.** Any one learn programming skill	
CourseObjectives:	
1. Relate the differences between the classical and quantum system.	
2. Apply the concepts of complex numbers to quantum computing.	
3. Examine the various algorithms using quantum programming	
4. Develop various quantum applications.	
Course Outcomes: At the end of the session, the students will be able to -	
CO1: Understand the basics of quantum computing.	
CO2: Understand the background of Quantum Mechanics.	
CO3: Analyze the computation models.	
CO4: Model the circuits using quantum computation.	
CO5: Understand the quantum operations such as noise and error–correction.	

Content

Unit	Content	Teaching Hours
1.	**Basic Quantum Theory**: Global Perspectives – Quantum Bits – Quantum Computation – Quantum Algorithms – Experimental Quantum Information Processing – Quantum Information.	9
2.	**Algorithms:** Deutsch's Algoirthm, Deutsch-Jozsa Algoirthm Simon's Periodicity Algorithm, Grover's Search Algorithm, Shor's Factoring Algorithm	9
3.	**Programming Languages:** Quantum Assembly Programming, Toward Higher-Level Quantum Programming, Quantum Computation Before Quantum Computers	9
4.	**Theoretical Computer Science:** Deterministic and Nondeterministic Computations, Probabilistic Computations, Quantum Computations **Cryptography:** Quantum Key Exchange Algorithms- BB84, B92 and EPR Protocols, Quantum Teleportation	9
5.	**Information Theory:** Classical Information and Shannon Entropy, Quantum Information and von Neumann Entropy, Classical and Quantum Data Compression, Error Handling Codes	9

Text Books:

1. Noson S Yanofsky and Mirco A. Mannucci, Quantum Computing for Computer Scientists, Cambridge University Press
2. Vishal Sahni, Quantum Computing, Tata McGraw-Hill Edition

Reference Books:

1. N. David Mermin, "Quantum Computer Science: An Introduction", Cambridge University Press, 2007.
2.Bernhardt, Quantum Computing for Everyone, The MIT Press

Course code: CCI24PEP554 **Course name:** CC Using AWS Lab		
Course category: PEC		
Credits: 1	**Teaching scheme:** P-2	**Evaluation scheme:** TW–30, PR–20
Duration of External Exam: 2 Hrs		
Pre-requisites: Basics of CC.		
Course Objectives:		
1. Host a basic application (e.g., Employee Directory) on AWS.		
2. Understand the Amazon EC2 instance lifecycle and compute services.		
3. Explain Docker and Kubernetes and their architectures.		
4. Understand S3 storage components, including buckets, objects, and keys.		
5. Configure IAM users, groups, and password policies.		
Laboratory Outcomes:At the end of the course, the students will be able to -		
LO1: Identify the components of AWS Global Infrastructure.		
LO2: Describe the Amazon EC2 instance lifecycle and container services.		
LO3: Define containerization concepts and tools like Docker and Kubernetes.		
LO4: Explain S3 components, storage classes, and versioning.		
LO5: Create and manage IAM policies, groups, and permissions.		

Contents –

Sr. No.	List of Experiments	Practical Hours
1.	Create and configure a Cloud Storage bucket using Amazon S3 to store and manage object.	2
2.	Operate AWS services through the AWS CLI by executing basic commands for resource management.	2
3.	Design and deploy a Custom VPC with subnets, route tables, and internet gateways for secure networking.	2
4.	Configure Security Groups to control inbound and outbound traffic for EC2 instances.	2
5.	Provision Reserved Instances and demonstrate cost optimization strategies in AWS.	2
6.	Install and verify the CloudWatch Agent on EC2 instances to enable monitoring.	2
7.	Create and test a CloudWatch Alarm to trigger notifications based on performance metrics.	2
8.	Design and implement DynamoDB tables with appropriate keys and attributes for NoSQL storage.	2

9.	Build and query a data warehouse using Amazon Redshift and perform serverless querying with Amazon Athena on S3 data.	2
10.	Create and manage IAM Users and Groups with appropriate policies for access control.	2
11.	Deploy EC2 instances using AWS CloudFormation templates to automate infrastructure provisioning.	2
12	Provision and manage a Google Compute Engine instance using Terraform for Infrastructure-as-Code deployment.	2

Text Books:
1. Danilo Poccia , "AWS Lambda in Action: Event-driven Serverless Applications", Manning Publication, 2016
2. Mark Wilkins , "Learning Amazon Web Services (AWS): A Hands-On Guide to the Fundamentals of AWS Cloud", Addison-Wesley Professional,2019 Computing", McGraw Hill,2013
3. Andreas Wittig and Michael Wittig, " Amazon Web Services in Action", Manning Publications, 2015

Reference Books:
1. Raoul Alongi , "AWS: The Most Complete Guide to Amazon Web Services from Beginner to Advanced Level", Kindle Edition, 2019
2. Ben Piper (Author), David Clinton , "AWS Certified Cloud Practitioner Study Guide:CLF-C01 Exam 1st Edition", Sybex, 2019

Course Code: CCI24PEP555	**Course Name:** Dev-Ops-II Lab	
Course Category: PEC		
Credits: 1	**Teaching Scheme:** P-2	**Evaluation Scheme:** TW–30, PR–20
Duration of External Exam: 3 Hrs.		
Pre-requisites: Programming Skills, Cryptographic Fundamentals, Tools & Software Knowledge		
Course Objectives:		
1. Understand DevOps principles, culture, and lifecycle		
2. Apply Git for version control and collaborative development.		
3. Design and implement Continuous Integration and Continuous Deployment pipelines.		
4. Use containerization and orchestration tools such as Docker and Kubernetes.		
5. Implement monitoring, logging, and real-world DevOps case studies		
Laboratory Outcomes: At the end of the course, the students will be able to –		
LO1. Use Git for version control, collaboration, and code review.		
LO2. Build and manage CI/CD pipelines for automated delivery.		
LO3. Deploy applications using Docker, Docker Compose, and Kubernetes.		
LO4. Apply monitoring and logging using Prometheus, Grafana, and ELK Stack.		
LO5. Analyse and implement DevOps solutions through case studies.		

Contents –

Sr. No.	List of Experiments	Practical Hours
1.	Collaborating Using Git - Fork, Pull Requests, and Code Reviews	2
2.	Setting up a Continuous Integration (CI) Pipeline	2
3.	Explore Continuous Deployment (CD) Pipeline	2
4.	Apply Docker Basics - Containerization	2
5.	Understand Docker Compose - Multi-Container Application	2
6.	Introduction to Kubernetes - Basic Cluster Setup	2
7.	Monitoring with Prometheus and Grafana	2
8.	Apply Log Management with ELK Stack	2
9.	Create cloud infrastructure using Terraform, enable autoscaling and verify resource creation.	2
10.	Case Study/Implementation	2
11	Mini project	2

Text Books:

1. "The Phoenix Project: A Novel About IT, DevOps, and Helping Your Business Win" by Gene Kim, Kevin Behr, and George Safford.
2. "Accelerate: The Science of Lean Software and DevOps: Building and Scaling High Performing Technology Organizations" by Nicole Forsgren, Jez Humble, and Gene Kim.
3. "DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations" by Gene Kim, Patrick Debois, John Willis, and Jez Humble..

Reference Books:

1. Version Control with Git" by Jon Loeliger and Matthew McCullough.
2. "Pro Git" by Scott Chacon and Ben Straub
3. "Docker: Up & Running" by Karl Matthias and Sean P. Kane
4. Kubernetes Up & Running: Dive into the Future of Infrastructure" by Kelsey Hightower, Brendan Burns, and Joe Beda.

Online Resources:

1. https://git-scm.com/book/en/v2
2. https://docs.gitlab.com/ee/ci/

.**Course code:** CCI24PEP556 **Course name:** Image Processing Lab **Course category:** PEC
Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** TW–30, PR–20
Pre-requisites: Mathematical fundamentals like matrices, basic linear algebra.
Course Objectives: 1**.** Demonstrate basic image operations for pre-processing. 2. Introduce filtering techniques for image smoothing and sharpening. 3. Introduce image segmentation techniques for object detection 4. Demonstrate restoration techniques for noisy images 5. Explore advanced concepts like object recognition and feature extraction
Laboratory Outcomes: At the end of the course, the students will be able to - **LO1:** Explain image sampling, quantization, and basic operations. **LO2:** Apply smoothing and sharpening filters to improve image quality. **LO3:** Explain segmentation techniques like thresholding and edge detection. **LO4:** Apply DCT and Wavelet Transform for image compression. **LO5:** Analyse the use of machine learning in image recognition applications**.**

Contents –

Sr. No.	List of Experiments	Practical Hours
1.	Load and display images in Python/Matlab, convert between color spaces (RGB to grayscale).	2
2.	Implement histogram equalization to enhance image contrast.	2
3.	Apply smoothing (e.g., Gaussian filter) and sharpening filters (e.g., Laplacian filter).	2
4.	Perform image enhancement in the frequency domain using Fourier Transform.	2
5.	Implement Sobel, Prewitt, and Canny operators for edge detection.	2
6.	Apply global and adaptive thresholding for image segmentation.	2
7.	Demonstrate dilation, erosion, opening, and closing using binary images.	2
8.	Implement JPEG compression using DCT and analyze compression ratios.	2
9.	Extract features using SIFT and HOG for object recognition.	2
10.	Build a simple object recognition system using OpenCV and a machine learning model.	2

Text Books:

1. Fundamentals of Digital Image Processing by Anil K. Jain, Prentice Hall

2. Image Processing, Analysis, and Machine Vision by Milan Sonka, Vaclav Hlavac, and Roger Boyle, 4th Edition,Cengage Learning
Reference Books: 1. Digital Image Processing by Rafael C. Gonzalez and Richard E. Woods, 4th edition, Pearson Education 2. Computer Vision: Algorithms and Applications by Richard Szeliski, Springer
Online Resources: 1. NPTEL / SWAYAM lectures.

Course code: CCI24PEP557 **Course name:** Quantum Computing Lab		
Course category: PEC		
Credits: 1	**Teaching scheme:** P-2	**Evaluation scheme:** TW–30, PR–20
Pre-requisites: Basics of linear algebra and quantum mechanics is required		
Course Objectives:		
1. Relate the Differences between the classical and quantum System.		
2. Apply the concept of complex numbers to quantum computing		
3. Examine the Various algorithms using quantum programming		
4. Develop various quantum applications.		
Laboratory Outcomes: At the end of the course, the students will be able to -		
LO1: Understand the basics of quantum computing.		
LO2: Understand the background of Quantum Mechanics.		
LO3: Analyze the computation models.		
LO4: Model the circuits using quantum computation.		
LO5: Understand the quantum operations such as noise and error–correction.		

Contents –

Sr. No.	List of Experiments	Practical Hours
1.	Write a program to convert a complex number from Cartesian to polar and vice versa.	2
2.	Write a program to check whether a given square matrix is Hermitian or not, Unitary or not.	2
3.	Write a program that accepts two matrices and constructs their tensor product.	2
4.	Write a program to perform the little marble experiment	2
5.	Write a program that simulates the first quantum system. The user should be able to specify how many points the particle can occupy	2
6.	Design and implement a quantum computer emulator in the language of your choice.	2
7.	Study Algorithms-BB84, B92 and EPR Protocols	2
8.	Study Quantum Teleportation	2
9.	Write a program that lets the user choose how many 02 letters the source alphabet has, and then enter the probability distribution. The program should visualize it, and compute its Shannon entropy	2
10.	Write a program that lets the user enter two qubits and their corresponding probabilities	2

Text Books: 1. Noson S Yanofsky and Mirco A. Mannucci, Quantum Computing for Computer Scientists, Cambridge University Press 2. Vishal Sahni, Quantum Computing, Tata McGraw-Hill Edition
Reference Books: 1. N. David Mermin, "Quantum Computer Science: An Introduction", Cambridge University Press, 2007. 2. Chris Bernhardt, Quantum Computing for Everyone, The MIT Press
Online Resources: 1. NPTEL / SWAYAM lectures.

Course code: CCI24VSP558 Course Name: Mobile Application Development Lab **Course category:** Vocational Skill and Skill Enhancement Course
Credits: 02 **Teaching scheme:** P-2Hrs/Week **Evaluation Scheme:** CA–0, MSE–0, ESE–0, TW–60, PR–40
Pre-requisites: Basic Programming Knowledge**, Basic Knowledge of Databases and SQL**
Course Objectives:
1. To understand and interpret the architecture, features, and core components of the Android operating system.
2. To configure and utilize the Android development environment effectively for building and testing applications.
3. To design and develop Android applications using various user interface components and layout techniques.
4. To create database-driven Android applications by integrating data storage, retrieval, and management mechanisms.
5. To develop advanced Android applications that implement security features and manage permissions for safe and controlled access to system resources.
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Interpret the features of android operating system.
LO2: Use after configuring Android development environment**.**
LO3: Develop android applications using UI components and layouts
LO4: Create database driven Android applications.
LO5: Develop advanced Android applications that require relevant permissions for security.

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1.	Install Android IDE and create Android virtual device	2
2.	Develop a program to display “Hello World” onscreen	2
3.	Explore the directory structure in Android IDE	2
4.	Develop android application using View Text and Edit Text.	2
5.	Develop a program to implement Button, Image Button and Toggle Button	2
6.	Develop a program to design Checkbox and Radio button.	2
7.	Develop a program to implement Progress Bar	2
8.	Develop a program to create a login form using the above UI controls	2
9.	Write program to implement Linear layout and Constraint layout to create any registration form with Custom Toast Alert	2
10.	Develop a program to implement Grid View, Image View, Scroll View, List View for any management system like library management/hotel management	2
11	Develop a simple calculator which uses grid layout and GUI concepts	2
12	Develop a splash screen in android	2
13	Design and develop any convertor application like temperature convertor	2

	/currency convertor/ volume convertor	
14	Design and develop a simple countdown timer	2
15	Develop a program to implement Date Picker in application	2
16	Develop a program to implement Time Picker in application	2
17	Develop a program to create two simple activities for Login application	2
18	Develop a program to implement new Activity using explicit intent and implicit intent to open any other website	2
19	Develop a program to implement services like bluetooth/wifi	2
20	Develop a program to implement a broadcast receiver to switch between different modes like Airplane mode/Silent Mode/Loud Mode	2

Text Books:

1.Dixit, Prasanna Kumar – *Android*, Vikas Publication, New Delhi 2014, ISBN: 9789325977884
2.Hortan, John– *Android Programming for Beginners*, Packet Publications, 2015, ISBN : 978-1-78588-326-2

Reference Books:

1. Pradeep Kothari**–** *NAndroid Application Development*,Kogent Learning Solutions ISBN :9789351194095.
2.Maclean David , KomatineniSatya, Allen Grant**–** *Apress Publications, 2015, ISBN :978-1-4302-4680-0*

Online Resources:

1) https://onlinecourses.swayam2.ac.in/nou21_ge41/preview.
2) **https://developer.android.com/training/data-storage/sqlite**